Senior Thesis

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```
rhfs.original <- read.csv("RHFS_2024_Public_Variables.csv")

# Turning Categorical Variables into Factors
rhfs.original <- rhfs.original |>
  mutate(
    across(
      c(NUMCAT_R, OUTCOME, PROPCON, PROPTOWN,
        COMPLEX, NUMBLD_R, YRNEWBLG_R, YRPROPACQ_R, APPRWAIVER,
        CASHDWN, HOWACQ, HOWMVAL, OFFSTPARKAHS, UNITSTAT,
        ADAGENCY, ADAGENT, ADNEWSPAPER, ADONLINE, ADOTHONLINE,
        ADVERTISE, ADWEBSITE, MLS, MNGMNT, NUMOTHPROP,
        NUMOTHUNITS, OWNENT, OWNOTHPROP, COMMSPACE, CONCSS,
        ICABLE, IELEC, IGAS, IPARKING, IPOOL,
        ISEWER, ITRASH, IWATER, IWIFI, LEASLNG,
        LNGTRMPRV, NORNTAPPFEE, PAYAPP, PAYCASH, PAYCHK,
        PAYCREDIT, PAYONLINE, PAYOTH, PAYTHIRD, RCONTROLAHS,
        RECCOMMREN, RNTAPPFEEWAIV, SECDEPAMT, SENIOR, WEEKLNG,
        OPTAX_R, PROPANS, ACCEPT, BENEOTH, ECONDEVLP,
        ELDERLY, GOVBELOW, GOVGRANT, HISTORIC, LOWINAREA,
        LOWINCTEN, OTHRCV, RSUBSIDY, STLOWINC, YRMORT_R,
        DEBT, DEBT1, GREENMORT, LOCTYP, MORTGET,
        PMTFREQ, REGPAY, RPIPMI, RPIPRIN, RPIPRPINS,
        RPIPRPTAX, WHYNOMRT, SAMPSTRATA, MORTAMT_R, OUTBAL_R, RNTAPPFEE_R,
        YRPROPREHAB_R),
      as.factor
    )
   )

# Renaming some factor levels for later convenience
rhfs.original <- rhfs.original |>
  mutate(
    NUMCAT_R = fct_recode(
      NUMCAT_R,
      "1 Unit" = "0",
      "2-4 Units" = "1",
      "5-24 Units" = "2",
      "25-49 Units" = "3",
      "50+ Units" = "4"
    ),
    OUTCOME = fct_recode(
      OUTCOME,
      "Interview Compl; self response" = "1",
```

```
      "Assisted living limited interview; self response" = "3",
      "Conversion from partial to full interview" = "5",
      "Interview Compl; field interview" = "201",
      "Assisted living limited interview; field interview" = "203",
      "Sufficient partial; field interview" = "204",
      "Military limited field interview" = "206"
    )
  )

# Rearranging variables greatly improves View(rhfs)
questionnaire_order <- c(
  # 1c-1e and property definition
  "PROPTOWN", "PROPCON", "COMPLEX",
  "NUMBLD_R", "NUMUNITS_R", "NUMCAT_R",

  # 2-4e
  "UNITSTAT", "YRNEWBLG_R", "HOWACQ", "YRPROPACQ_R", "PURPRICE_R",
  "CASHDWN", "CASHDWNAMT", "MRKTVAL_R", "HOWMVAL", "APPRWAIVER",

  # 5-6: occupancy, rents, and rent payment methods
  "BROOMS", "TOTRENTVAC", "TRENOC_R", "TRENVA_R", "TOWNOC_R",
  "TEMPOC_R", "TOTHVA_R", "LOWRENT", "HIRENT", "RENT",
  "RENTNO_R", "RENTLT249_R", "RENT274_R", "RENT299_R", "RENT349_R",
  "RENT399_R", "RENT449_R", "RENT499_R", "RENT549_R", "RENT599_R",
  "RENT649_R", "RENT699_R", "RENT749_R", "RENT799_R", "RENT899_R",
  "RENT999_R", "RENT1099_R", "RENT1249_R", "RENT1499_R", "RENT1999_R",
  "RENT2499_R", "RENT2999_R", "RENT3499_R", "RENT3999_R", "RENT4499_R",
  "RENT4999_R", "RENT5000_R", "BED0_R", "BED1_R", "BED2_R", "BED3_R",
  "PAYONLINE", "PAYTHIRD", "PAYAPP", "PAYCREDIT", "PAYCHK", "PAYCASH", "PAYOTH",

  # 6a-11c
  "LEASLNG", "LNGTRMPRV", "WEEKLNG", "WEEKLNGNUM", "WEEKLNGPCT",
  "RNTAPPFEE_R", "NORNTAPPFEE", "RNTAPPFEEWAIV", "PCTFEEWAIV_R", "SECDEPAMT",
  "CONCSS", "CONCSSNUM", "SENIOR", "RCONTROL", "RCONTROLAHS",
  "OFFSTPARK", "OFFSTPARKAHS", "IELEC", "IGAS", "IWATER", "ISEWER", "ITRASH",
  "ICABLE", "IWIFI", "IPOOL", "IPARKING", "PROPANS",

  # 12a-15e
  "TOTCOLL", "COMMSPACE", "RECCOMMREN", "COMMRENT", "MNGMNT", "HRSMNGMNT",
  "ADVERTISE", "ADWEBSITE", "ADONLINE", "ADOTHONLINE", "ADNEWSPAPER",
  "ADAGENCY", "ADAGENT", "MLS",

  # 16: operating expenses
  "OPPAY", "OPTAX_R", "OPINSUR", "OPELGAS", "OPWTRSWR", "OPREP", "OPMNG",
  "OPPROTH", "OPSECUR", "OPLAND", "OPOTHEX", "OPEX_R",

  # 17-18c: capital expenses and ownership
  "CAPEX_R", "CPHEATAMT", "CPRFAMT", "CPWINAMT", "CPDOORAMT", "CPFLRAMT",
  "CPEXTAMT", "CPPLBAMT", "CPELEAMT", "CPKITAMT", "CPBATAMT", "CPACCAMT",
  "CPOTHAMT", "CPRESAMT_R", "OWNENT", "OWNOTHPROP", "NUMOTHPROP", "NUMOTHUNITS",

  # 19-23: subsidies, tax credits, and tax reductions
  "HUDUNITS", "GOVBELOW", "RSUBSIDY", "ELDERLY", "GOVGRANT", "BENEOTH",
```

```
  "LIHTCUNITS", "STLOWINC", "HISTORIC", "LOWINCTEN", "LOWINAREA", "ECONDEVLP", "OTHRCV",

  # 24-37: debt/mortgage section and HUD voucher history
  "DEBT", "WHYNOMRT", "DEBTNUM", "DEBT1", "YRMORT_R", "MORTAMT_R", "OUTBAL_R",
  "APPVAL_R", "GREENMORT", "MORTGET", "LOCTYP", "REGPAY", "PMTAMT_R", "PMTFREQ",
  "RPIPRIN", "RPIPRPINS", "RPIPMI", "RPIPRPTAX", "ACCEPT"
)

rhfs.original <- rhfs.original |>
  relocate(
    CONTROLPUF, OUTCOME, SAMPSTRATA, SURVYR,
    all_of(questionnaire_order)
  ) |>
  # Keep the full-sample and replicate weights together at the far right.
  relocate(
    WEIGHT, starts_with("REPWGT"), .after = last_col()
  )
```

```
# Makes View(rhfs) much more productive.
rhfs <- rhfs.original |>
  select(-starts_with("REPWGT"), -WEIGHT,
         -SURVYR, # is always 2024
         # J flag variables mark values that were manually reviewed; note that
         # they may become relevant if outliers are under scrutiny.
         -starts_with("J"),
         # A continuous variable coded as a factor without documentation; may
         # be useful for pure prediction
         # -PCTFEEWAIV_R,
         # Poorly documented variable with 99.5% of its values missing
         -WEEKLNGPCT
         )

# NAs in the dataset
missingvalues <- function(df) {
  NAs <- sum(sapply(df, \(x) sum(x == -8))) # Not applicable
  NRs <- sum(sapply(df, \(x) sum(x == -9))) # Not reported

  tibble(
    NAs = NAs, # Not applicable
    NRs = NRs, # Not reported
    perc.missing = (NAs + NRs) / prod(dim(df)) # % of missing values in the df
  )
}
missingvalues(rhfs)
```

```
## # A tibble: 1 x 3
##      NAs    NRs perc.missing
##    <int>  <int>        <dbl>
## 1 170275 144234        0.399
```

```
# Missing values by interview type
rhfs |>
  summarize(
```

```
    # pick(everything()) extracts a data frame of all non-grouping columns
    missingvalues(pick(everything())),
    .by = OUTCOME
  )
```

```
##                                              OUTCOME   NAs   NRs perc.missing
## 1                  Interview Compl; field interview 97135 73188    0.3911694
## 2               Sufficient partial; field interview 29948 36570    0.5148054
## 3                    Interview Compl; self response 37769 21028    0.3212635
## 4         Conversion from partial to full interview  3442  7740    0.4512510
## 5                  Military limited field interview   297   839    0.7131199
## 6 Assisted living limited interview; field interview  1286  3761    0.7128531
## 7   Assisted living limited interview; self response   398  1108    0.7090395
```

```
# Handling known Trivially Missing values
# Certain values in the columns below can only exist if REGPAY is 1, so -9 is
# wrongly assigned to the corresponding REGPAY values.
trivially.missing.regpay <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    REGPAY == -9 &
      # In a given observation, if any of these columns contain a value other than
      # -8 or -9 return True.
      if_any(c(PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, REGPAY, PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX) |>
  pull(id)

rhfs$REGPAY[trivially.missing.regpay] <- 1


trivially.missing.ownothprop <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    OWNOTHPROP == -9 &
      if_any(c(NUMOTHPROP, NUMOTHUNITS), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, NUMOTHPROP, NUMOTHUNITS) |>
  pull(id)

rhfs$OWNOTHPROP[trivially.missing.ownothprop] <- 1


trivially.missing.rntappfeewaiv <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    RNTAPPFEEWAIV == -9 &
      if_any(c(PCTFEEWAIV_R), \(x) !x %in% c(-8, -9))
    ) |>
  select(id, PCTFEEWAIV_R) |>
  pull(id)

rhfs$RNTAPPFEEWAIV[trivially.missing.rntappfeewaiv] <- 1
```

```r
# Total rental receipts as a percent of property value (TRENTRCTPV_R) is
# calculated with (TOTCOLL + COMMRENT) / MRKTVAL. If COMMSPACE = 2, then
# COMMRENT = 0, so TRENTRCTPV_R = TOTCOLL / MRKTVAL as derived below.
trivially.missing.trentrctpv_r <- rhfs |>
  mutate(id = row_number()) |>
  select(id, TRENTRCTPV_R, COMMSPACE, COMMRENT, RECCOMMREN, TOTCOLL, MRKTVAL_R) |>
  filter(TRENTRCTPV_R == -8 & COMMSPACE == 2 &
      if_all(c(TOTCOLL, MRKTVAL_R), \(x) !x %in% c(-8, -9))
    ) |>
  pull(id)

rhfs$TRENTRCTPV_R[trivially.missing.trentrctpv_r] <-
  rhfs$TOTCOLL[trivially.missing.trentrctpv_r] /
  rhfs$MRKTVAL_R[trivially.missing.trentrctpv_r]


# When CASHDWN is 2, CASHDWNAMT is indeed always -9
rhfs |>
  mutate(id = row_number()) |>
  select(id, CASHDWN, CASHDWNAMT) |>
  filter(CASHDWN == 2 & CASHDWNAMT != -9)
```

```
## [1] id         CASHDWN    CASHDWNAMT
## <0 rows> (or 0-length row.names)
```

```r
# If a CASHDWN payment wasn't made, then the CASHDWNAMT should be 0
rhfs$CASHDWNAMT <- ifelse(rhfs$CASHDWN == 2, 0, rhfs$CASHDWNAMT)


# Same idea as with CASHDWN/CASHDWNAMT
rhfs |>
  mutate(id = row_number()) |>
  select(id, WEEKLNG, WEEKLNGNUM) |>
  filter(WEEKLNG == 2 & WEEKLNGNUM != -9)
```

```
##     id WEEKLNG WEEKLNGNUM
## 1 2173       2          0
## 2 3507       2          0
## 3 4197       2          0
```

```r
# If no units are offered for daily/weekly rental, then WEEKLNGNUM (the count
# of those that are offered) should be 0
rhfs$WEEKLNGNUM <- ifelse(rhfs$WEEKLNG == 2, 0, rhfs$WEEKLNGNUM)


# WEEKLNG and WEEKLNGNUM (daily/weekly rental availability variables) are -8
# when NUMUNITS != 1. For 7 of these properties, LEASLNG = 3 (lease length for
# single-units is weekly) implies that WEEKLNG and WEEKLNGNUM are 1.
weekly_targets <- rhfs |>
  mutate(row = row_number()) |>
  filter(NUMUNITS_R == 1, LEASLNG == "3") |>
```

```
  select(row, LEASLNG, WEEKLNG, WEEKLNGNUM)

rhfs <- rhfs |>
  mutate(
    WEEKLNG = replace(WEEKLNG, weekly_targets$row, "1"),
    WEEKLNGNUM = replace(WEEKLNGNUM, weekly_targets$row, 1)
  )


# Viewing targets for imputation
rhfs |>
  mutate(id = row_number()) |>
  select(COMMSPACE, COMMRENT, RECCOMMREN) |>
  filter(COMMSPACE == 2)
```

```
##      COMMSPACE COMMRENT RECCOMMREN
## 1            2       -9         -9
## 2            2       -9         -9
## 3            2       -9         -9
## 4            2       -9         -9
## 5            2       -9         -9
## 6            2       -9         -9
## 7            2       -9         -9
## 8            2       -9         -9
## 9            2       -9         -9
## 10           2       -9         -9
## 11           2       -9         -9
## 12           2       -9         -9
## 13           2       -9         -9
## 14           2       -9         -9
## 15           2       -9         -9
## 16           2       -9         -9
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## 18           2       -9         -9
## 19           2       -9         -9
## 20           2       -9         -9
## 21           2       -9         -9
## 22           2       -9         -9
## 23           2       -9         -9
## 24           2       -9         -9
## 25           2       -9         -9
## 26           2       -9         -9
## 27           2       -9         -9
## 28           2       -9         -9
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## 38          2       -9        -9
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## 3467        2       -8        -9
## 3468        2       -8        -9
## 3469        2       -8        -9
## 3470        2       -8        -9
## 3471        2       -8        -9
## 3472        2       -8        -9
## 3473        2       -8        -9
## 3474        2       -8        -9
## 3475        2       -8        -9
## 3476        2       -8        -9
## 3477        2       -8        -9
## 3478        2       -8        -9
## 3479        2       -8        -9
## 3480        2       -8        -9
## 3481        2       -8        -9
## 3482        2       -8        -9
## 3483        2       -8        -9
## 3484        2       -8        -9
## 3485        2       -8        -9
## 3486        2       -8        -9
## 3487        2       -8        -9
## 3488        2       -8        -9
## 3489        2       -8        -9
## 3490        2       -8        -9
## 3491        2       -8        -9
## 3492        2       -8        -9
## 3493        2       -8        -9
```

```
## 3494         2       -8         -9
## 3495         2       -8         -9
## 3496         2       -8         -9
## 3497         2       -8         -9
## 3498         2       -8         -9
## 3499         2       -8         -9
## 3500         2       -8         -9
## 3501         2       -8         -9
## 3502         2       -8         -9
## 3503         2       -8         -9
## 3504         2       -8         -9
## 3505         2       -8         -9
## 3506         2       -8         -9
## 3507         2       -8         -9
## 3508         2       -8         -9
## 3509         2       -8         -9
## 3510         2       -8         -9
## 3511         2       -8         -9
## 3512         2       -8         -9
## 3513         2       -8         -9
## 3514         2       -8         -9
## 3515         2       -8         -9
## 3516         2       -8         -9
## 3517         2       -8         -9
## 3518         2       -8         -9
## 3519         2       -8         -9
## 3520         2       -8         -9
## 3521         2       -8         -9
## 3522         2       -8         -9
## 3523         2       -8         -9
## 3524         2       -8         -9
## 3525         2       -8         -9
## 3526         2       -8         -9
## 3527         2       -8         -9
## 3528         2       -8         -9
## 3529         2       -8         -9
## 3530         2       -8         -9
## 3531         2       -8         -9
## 3532         2       -8         -9
## 3533         2       -8         -9
## 3534         2       -8         -9
## 3535         2       -8         -9
## 3536         2       -8         -9
## 3537         2       -8         -9
## 3538         2       -8         -9
## 3539         2       -8         -9
## 3540         2       -8         -9
## 3541         2       -8         -8
## 3542         2       -8         -9
## 3543         2       -8         -9
## 3544         2       -8         -9
## 3545         2       -8         -9
## 3546         2       -8         -9
## 3547         2       -8         -9
```

```
## 3548        2       -8        -9
## 3549        2       -8        -9
## 3550        2       -8        -9
## 3551        2       -8        -9
## 3552        2       -8        -8
## 3553        2       -8        -9
## 3554        2       -8        -9
## 3555        2       -8        -9
## 3556        2       -8        -9
## 3557        2       -8        -9
## 3558        2       -8        -9
## 3559        2       -8        -9
## 3560        2       -8        -9
## 3561        2       -8        -9
## 3562        2       -8        -9
## 3563        2       -8        -9
## 3564        2       -8        -9
## 3565        2       -8        -9
## 3566        2       -8        -9
## 3567        2       -8        -9
## 3568        2       -8        -9
## 3569        2       -8        -9
## 3570        2       -8        -9
## 3571        2       -8        -9
## 3572        2       -8        -9
## 3573        2       -8        -9
## 3574        2       -8        -9
## 3575        2       -8        -9
## 3576        2       -8        -9
## 3577        2       -8        -9
## 3578        2       -8        -9
## 3579        2       -8        -9
## 3580        2       -8        -9
## 3581        2       -8        -9
## 3582        2       -8        -9
## 3583        2       -8        -9
## 3584        2       -8        -9
## 3585        2       -8        -9
## 3586        2       -8        -9
## 3587        2       -8        -9
## 3588        2       -8        -9
## 3589        2       -8        -9
## 3590        2       -8        -9
## 3591        2       -8        -9
## 3592        2       -8        -9
## 3593        2       -8        -9
## 3594        2       -8        -9
## 3595        2       -8        -9
## 3596        2       -8        -9
## 3597        2       -8        -9
## 3598        2       -8        -9
## 3599        2       -8        -9
## 3600        2       -8        -9
## 3601        2       -8        -9
```

```
## 3602         2       -8         -9
## 3603         2       -8         -9
## 3604         2       -8         -9
## 3605         2       -8         -9
## 3606         2       -8         -9
## 3607         2       -8         -9
## 3608         2       -8         -9
## 3609         2       -8         -9
## 3610         2       -8         -9
## 3611         2       -8         -9
## 3612         2       -8         -9
## 3613         2       -8         -9
## 3614         2       -8         -9
## 3615         2       -8         -9
## 3616         2       -8         -9
## 3617         2       -8         -9
## 3618         2       -8         -9
## 3619         2       -8         -9
## 3620         2       -8         -9
## 3621         2       -8         -9
## 3622         2       -8         -9
## 3623         2       -8         -9
## 3624         2       -8         -9
## 3625         2       -8         -9
## 3626         2       -8         -9
## 3627         2       -8         -9
## 3628         2       -8         -9
## 3629         2       -8         -9
## 3630         2       -8         -9
## 3631         2       -8         -9
## 3632         2       -8         -9
## 3633         2       -8         -8
## 3634         2       -8         -8
## 3635         2       -8         -9
## 3636         2       -8         -9
## 3637         2       -8         -9
## 3638         2       -8         -9
## 3639         2       -8         -9
## 3640         2       -8         -9
## 3641         2       -8         -9
## 3642         2       -8         -9
## 3643         2       -8         -9
## 3644         2       -8         -9
## 3645         2       -8         -9
## 3646         2       -8         -9
## 3647         2       -8         -9
## 3648         2       -8         -9
## 3649         2       -8         -9
## 3650         2       -8         -9
## 3651         2       -8         -9
## 3652         2       -8         -9
## 3653         2       -8         -9
## 3654         2       -8         -9
## 3655         2       -8         -9
```

```
## 3656         2       -8         -9
## 3657         2       -8         -9
## 3658         2       -8         -9
## 3659         2       -8         -9
## 3660         2       -8         -9
## 3661         2       -8         -9
## 3662         2       -8         -9
## 3663         2       -8         -9
## 3664         2       -8         -9
## 3665         2       -8         -9
## 3666         2       -8         -9
## 3667         2       -8         -9
## 3668         2       -8         -9
## 3669         2       -8         -9
## 3670         2       -8         -9
## 3671         2       -8         -9
## 3672         2       -8         -9
## 3673         2       -8         -9
## 3674         2       -8         -9
## 3675         2       -8         -9
## 3676         2       -8         -9
## 3677         2       -8         -9
## 3678         2       -8         -9
## 3679         2       -8         -9
## 3680         2       -8         -9
## 3681         2       -8         -9
## 3682         2       -8         -9
## 3683         2       -8         -9
## 3684         2       -8         -9
## 3685         2       -8         -9
## 3686         2       -8         -9
## 3687         2       -8         -9
## 3688         2       -8         -9
## 3689         2       -8         -9
## 3690         2       -8         -9
## 3691         2       -8         -9
## 3692         2       -8         -9
## 3693         2       -8         -9
## 3694         2       -8         -9
## 3695         2       -8         -9
## 3696         2       -8         -9
## 3697         2       -8         -9
## 3698         2       -8         -9
## 3699         2       -8         -9
## 3700         2       -8         -9
## 3701         2       -8         -9
## 3702         2       -8         -9
## 3703         2       -8         -9
## 3704         2       -8         -9
## 3705         2       -8         -9
## 3706         2       -8         -9
## 3707         2       -8         -9
## 3708         2       -8         -9
## 3709         2       -8         -9
```

```
## 3710        2       -8        -9
## 3711        2       -8        -9
## 3712        2       -8        -9
## 3713        2       -8        -9
## 3714        2       -8        -9
## 3715        2       -8        -9
## 3716        2       -8        -9
## 3717        2       -8        -9
## 3718        2       -8        -9
## 3719        2       -8        -9
## 3720        2       -8        -9
## 3721        2       -8        -8
## 3722        2       -8        -9
## 3723        2       -8        -9
## 3724        2       -8        -9
## 3725        2       -8        -9
## 3726        2       -8        -9
## 3727        2       -8        -9
## 3728        2       -8        -9
## 3729        2       -8        -9
## 3730        2       -8        -9
## 3731        2       -8        -9
## 3732        2       -8        -9
## 3733        2       -8        -9
## 3734        2       -8        -9
## 3735        2       -8        -9
## 3736        2       -8        -9
## 3737        2       -8        -9
## 3738        2       -8        -9
## 3739        2       -8        -9
## 3740        2       -8        -9
## 3741        2       -8        -9
## 3742        2       -8        -9
## 3743        2       -8        -9
## 3744        2       -8        -9
## 3745        2       -8        -9
## 3746        2       -8        -9
## 3747        2       -8        -9
## 3748        2       -8        -9
## 3749        2       -8        -9
## 3750        2       -8        -9
## 3751        2       -8        -9
## 3752        2       -8        -9
## 3753        2       -8        -9
## 3754        2       -8        -9
## 3755        2       -8        -9
## 3756        2       -8        -9
## 3757        2       -8        -9
## 3758        2       -8        -9
## 3759        2       -8        -9
## 3760        2       -8        -9
## 3761        2       -8        -9
## 3762        2       -8        -9
## 3763        2       -8        -9
```

```
## 3764         2       -8         -9
## 3765         2       -8         -9
## 3766         2       -8         -9
## 3767         2       -8         -9
## 3768         2       -8         -9
## 3769         2       -8         -9
## 3770         2       -8         -9
## 3771         2       -8         -9
## 3772         2       -8         -9
## 3773         2       -8         -9
## 3774         2       -8         -9
## 3775         2       -8         -9
## 3776         2       -8         -9
## 3777         2       -8         -9
## 3778         2       -8         -9
## 3779         2       -8         -9
## 3780         2       -8         -9
## 3781         2       -8         -9
## 3782         2       -8         -9
## 3783         2       -8         -9
## 3784         2       -8         -9
## 3785         2       -8         -9
## 3786         2       -8         -8
## 3787         2       -8         -9
## 3788         2       -8         -9
## 3789         2       -8         -9
## 3790         2       -8         -9
## 3791         2       -8         -9
## 3792         2       -8         -9
## 3793         2       -8         -9
## 3794         2       -8         -9
## 3795         2       -8         -9
## 3796         2       -8         -9
## 3797         2       -8         -9
## 3798         2       -8         -8
## 3799         2       -8         -9
## 3800         2       -8         -9
## 3801         2       -8         -9
## 3802         2       -8         -9
## 3803         2       -8         -9
## 3804         2       -8         -9
## 3805         2       -8         -9
## 3806         2       -8         -9
## 3807         2       -8         -9
## 3808         2       -8         -9
## 3809         2       -8         -9
## 3810         2       -8         -9
## 3811         2       -8         -9
## 3812         2       -8         -9
## 3813         2       -8         -9
## 3814         2       -8         -9
## 3815         2       -8         -9
## 3816         2       -8         -9
## 3817         2       -8         -9
```

```r
# If there is no commercial space on this property, then the property did not
# receive any rental income from a commercial use of the property.
rhfs$RECCOMMREN[rhfs$COMMSPACE == 2] <- "2"
rhfs$COMMRENT <- ifelse(rhfs$COMMSPACE == 2, 0, rhfs$COMMRENT)


rhfs |>
  mutate(id = row_number()) |>
  select(RECCOMMREN, COMMRENT) |>
  filter(RECCOMMREN == 2) |>
  summarize(
    n(),
    .by = COMMRENT
  )
```

```
##   COMMRENT  n()
## 1        0 3874
## 2       -9    9
```

```r
# If commercial rent was not received, then commercial rent should be 0
rhfs$COMMRENT <- ifelse(rhfs$RECCOMMREN == 2, 0, rhfs$COMMRENT)


# More informative than "-8"
rhfs <- rhfs |>
  mutate(
    UNITSTAT = fct_recode(
      UNITSTAT,
      "Not a single unit" = "-8"
    )
  )


# TOTRENTVAC's (total units for rent or vacant-for rent) 377 missing values are
# due to them being outside its universe of NUMUNITS >= 2. UNITSTAT (status of
# single unit) indicates that all of these values are rented or vacant for rent,
# pointing towards the idea that the 377 missing values should have a TOTRENTVAC
# of 1.
table(rhfs$TOTRENTVAC)
```

```
##
##   -9    1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
##  377   56  170   88  142   39   84   22   75   19   47   16   54   14   22   26
##   16   17   18   19   20   21   22   23   24   25   26   27   28   29   30   31
##   46    5   24   17   41   20   14   23   56   15   10   15   29   15   28   19
##   32   33   34   35   36   37   38   39   40   41   42   43   44   45   46   47
##   30    8   11   25   16   13    8   17   45   12   11    9   17   13   11   11
##   48   49   50   51   52   53   54   55   56   57   58   59   60   61   62   63
##   45   14   33    3    9    5    5    8   20    6    3   11   31    6   10    7
##   64   65   66   67   68   69   70   71   72   73   74   75   76   77   78   79
##   15    6    7    5   14    8   11    6   24    4    5   10    9    3    8    8
##   80   81   82   83   84   85   86   87   88   89   90   91   92   93   94   95
##   25    3    3   12   12    7    7    5   17    7   13    3    4    3   11    6
```

```
##   96   97   98   99  100  101  102  103  104  105  106  107  108  109  110  111
##   14    2   12   14   39    7    3    8   10    6    6    4   11    3   10    5
##  112  113  114  115  116  117  118  119  120  121  122  123  124  125  126  127
##   12    7    5    5    6    8    7   11   33    3    6    7    9    3    4    5
##  128  129  130  131  132  133  134  135  136  137  138  139  140  141  142  143
##   12    4   11    5    6    4    5    4   13    5    4    9   13    7    6    5
##  144  145  146  147  148  149  150  151  152  153  154  155  156  157  159  160
##   19    4    5    2   14    7   19    7   10    1    8    5    4    2    4   22
##  161  162  163  164  165  166  167  168  169  170  171  172  173  174  175  176
##    4    4    4    7    1    6    3   12    4    7    9    3    3    4    3   12
##  177  178  179  180  181  182  183  184  185  186  187  188  189  190  191  192
##    4    4    7   18    2    3    6    8    4    6    4   11    3    4    2   11
##  193  194  195  196  197  198  199  200  201  202  203  204  205  206  207  208
##    8    4    2   11    3   12    5   26   10    7    7    6    6    5    5   16
##  209  210  211  212  213  214  215  216  217  218  219  220  221  222  223  224
##    4   11    2    7    3    4    2   18    2    3    6    6    4    5    3    9
##  225  226  227  228  229  230  231  232  233  234  235  236  237  238  239  240
##    2    6    5    7    1    4    2    6    2    4    4    4    3    4    2   27
##  241  242  244  245  246  247  248  249  250  251  252  253  254  255  256  257
##    1    3    6    2    2    4    8    6    8    5   12    1    6    3    6    3
##  258  259  260  261  262  263  264  265  266  267  268  269  270  271  272  273
##    3    5    9    2    6    4    9    2    4    4    9    2    8    1    9    5
##  274  275  276  277  278  279  280  281  282  283  284  285  286  287  288  289
##    2    5    6    3    9    4   10    2    3    3    3    5    3    3   13    1
##  290  291  292  293  294  295  296  297  298  299  300  301  303  304  305  306
##    2    5    7    3    1    2    9    3    2    3   22    6    5   11    1    2
##  307  308  309  310  311  312  313  314  315  316  317  318  319  320  321  322
##    2    5    1    5    7   10    1    2    3    5    4    5    4   12    1    3
##  324  325  326  327  328  329  330  331  332  333  334  335  336  338  339  340
##    8    5    2    1    5    3    1    4    5    2    3    2   11    2    3    5
##  341  344  345  346  347  348  349  350  351  352  353  354  355  356  357  358
##    2    2    4    5    2    7    4    3    6    5    1    2    1    4    4    2
##  359  360  362  363  364  365  366  367  368  369  370  371  372  373  374  375
##    1   12    4    1    4    2    6    2    8    6   11    8   90    1    2    1
##  376  377  378  379  380  382  383  384  385  386  387  388  390  391  392  393
##    4    2    2    2    1    1    1    3    4    2    1    6    1    1    1    1
##  394  395  396  397  399  400  401  402  404  405  406  408  409  411  412  413
##    2    3    3    1    3    5    1    1    1    2    2    6    1    1    1    1
##  415  416  417  418  419  420  421  422  424  425  426  428  429  430  431  433
##    1    3    1    2    1    1    1    2    1    1    1    2    2    3    1    3
##  434  436  437  440  441  442  444  445  446  447  448  450  452  453  454  456
##    1    2    1    3    1    3    3    1    2    1    3    2    1    1    1    4
##  457  460  461  462  463  464  466  467  469  470  472  476  477  478  480  482
##    1    2    1    1    3    1    1    2    2    1    3    2    2    2    3    1
##  483  484  486  487  488  489  491  492  494  495  496  497  498  500  502  504
##    2    2    2    2    1    1    2    3    1    2    5    1    6    1    3    2
##  506  508  509  510  512  513  514  515  516  517  519  520  521  522  523  524
##    2    2    3    2    1    1    2    1    1    1    1    2    2    1    1    2
##  526  529  531  533  535  540  541  544  545  546  547  550  552  553  557  558
##    2    3    1    1    1    1    2    1    3    2   11    1    1    1    2    2
##  563  565  568  569  574  575  576  579  581  584  587  588  596  597  598  600
##    1    3    1    1    1    1    1    1    1    1    1    3    1    1    1    4
##  604  608  611  614  618  621  622  624  627  630  636  640  642  645  648  653
##    1    1    3    1    1    1    2    1    1    1    1    4    1    1    1    1
```

```
##  657  660  664  667  668  669  675  680  684  694  696  698  700  705  706  708
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    2    1
##  709  715  718  720  734  739  741  743  744  752  771  775  779  781  788  789
##    1    1    1    1    1    1    1    1    1    1    1    1    1    1    2    1
##  801  802  810  833  839  845  858  862  880  891  900  922 1000 1043 1097 1136
##    2    1    1    1    1    1    1    1    1    1    1    1    1    1    1    1
## 1137 1215 1236 1241 1274 1293 1296 1322 1324 1350 1354 1416 1444 1474 1530 1730
##    1    1    1    1    2    2    1    1    1    1    1    1    1    1    1    1
## 1817
##    1
```

```
table(rhfs$UNITSTAT)
```

```
##
## Not a single unit                 1                 2
##              4048               360                17
```

```
rhfs$TOTRENTVAC <- ifelse(rhfs$TOTRENTVAC == -9, 1, rhfs$TOTRENTVAC)


# Reverse engineering TOTRENTVAC indicates that it is constructed by NUMUNITS -
# TOWNOC_R - TEMPOC_R - TOTHVA_R (accurate 99.66% of the time). To achieve this
# accuracy, values in TOWNOC_R, TEMPOC_R, and TOTHVA_R are assigned to 0.
rhfs$TOWNOC_R[rhfs$TOWNOC_R %in% c(-9, -8)] <- 0
rhfs$TEMPOC_R[rhfs$TEMPOC_R %in% c(-9, -8)] <- 0
rhfs$TOTHVA_R[rhfs$TOTHVA_R %in% c(-9, -8)] <- 0


# When all components of OPEX_R are explicitly 0, then OPEX_R is 0. The
# corresponding OPEXPU_R (OPEX_R / NUMUNITS_R) are therefore also 0.
trivially.missing.opex_r <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    OPEX_R == -9 &
      if_all(c(OPPAY, OPINSUR, OPELGAS, OPWTRSWR, OPREP, OPMNG, OPPROTH,
               OPSECUR, OPLAND, OPOTHEX), \(x) x == 0)
    ) |>
  select(id, OPEX_R, OPEXPU_R, OPPAY, OPINSUR, OPELGAS, OPWTRSWR, OPREP, OPMNG,
         OPPROTH, OPSECUR, OPLAND, OPOTHEX) |>
  pull(id)

rhfs$OPEX_R[trivially.missing.opex_r] <- 0
rhfs$OPEXPU_R[trivially.missing.opex_r] <- 0


# When TRENOC_R is -8, then UNITSTAT is 1 (rented) or 2 (vacant, for rent).
rhfs |>
  select(TRENOC_R, UNITSTAT) |>
  filter(TRENOC_R == -8) |>
  summarize(
    n(),
    .by = UNITSTAT
  )
```

```
##   UNITSTAT n()
## 1        1 360
## 2        2  17
```

```
# TRENOC_R's (number of renter-occupied units) 377 -8s are due to them being
# outside of its universe of NUMUNITS >= 2. They can be derived from UNITSTAT.
rhfs$TRENOC_R <- ifelse(
  # If UNITSTAT is 1, then TRENOC_R is now 1.
  rhfs$UNITSTAT == "1", 1,
  # If UNITSTAT is 2, then TRENOC_R is now 0.
  ifelse(rhfs$UNITSTAT == "2", 0,
         # Otherwise, leave unchanged.
         rhfs$TRENOC_R))


# When TRENVA_R is -8, then UNITSTAT is 1 (rented) or 2 (vacant, for rent).
rhfs |>
  select(TRENVA_R, UNITSTAT) |>
  filter(TRENVA_R == -8) |>
  summarize(
    n(),
    .by = UNITSTAT
  )
```

```
##   UNITSTAT n()
## 1        1 360
## 2        2  17
```

```
# TRENVA_R's (number of vacant for rent units) 377 -8s are due to them being
# outside of its universe of NUMUNITS >= 2. They can be derived from UNITSTAT.
rhfs$TRENVA_R <- ifelse(
  # If UNITSTAT is 1, then TRENVA_R is now 0.
  rhfs$UNITSTAT == "1", 0,
  # If UNITSTAT is 2, then TRENVA_R is now 1.
  ifelse(rhfs$UNITSTAT == "2", 1,
         # Otherwise, leave unchanged.
         rhfs$TRENVA_R))


# BROOMS is observed for all but one single-unit property
rhfs |>
  select(BROOMS, BED0_R, BED1_R, BED2_R, BED3_R) |>
    filter(BED0_R == -8)
```

```
##     BROOMS BED0_R BED1_R BED2_R BED3_R
## 1        2     -8     -8     -8     -8
## 2        1     -8     -8     -8     -8
## 3        2     -8     -8     -8     -8
## 4        3     -8     -8     -8     -8
## 5        1     -8     -8     -8     -8
## 6       -9     -8     -8     -8     -8
## 7        1     -8     -8     -8     -8
## 8        2     -8     -8     -8     -8
```

```
## 9        1     -8     -8     -8     -8
## 10       3     -8     -8     -8     -8
## 11       3     -8     -8     -8     -8
## 12       1     -8     -8     -8     -8
## 13       3     -8     -8     -8     -8
## 14       2     -8     -8     -8     -8
## 15       1     -8     -8     -8     -8
## 16       2     -8     -8     -8     -8
## 17       3     -8     -8     -8     -8
## 18       2     -8     -8     -8     -8
## 19       1     -8     -8     -8     -8
## 20       4     -8     -8     -8     -8
## 21       3     -8     -8     -8     -8
## 22       4     -8     -8     -8     -8
## 23       2     -8     -8     -8     -8
## 24       4     -8     -8     -8     -8
## 25       2     -8     -8     -8     -8
## 26       3     -8     -8     -8     -8
## 27       2     -8     -8     -8     -8
## 28       3     -8     -8     -8     -8
## 29       2     -8     -8     -8     -8
## 30       2     -8     -8     -8     -8
## 31       3     -8     -8     -8     -8
## 32       3     -8     -8     -8     -8
## 33       0     -8     -8     -8     -8
## 34       2     -8     -8     -8     -8
## 35       3     -8     -8     -8     -8
## 36       1     -8     -8     -8     -8
## 37       2     -8     -8     -8     -8
## 38       3     -8     -8     -8     -8
## 39       2     -8     -8     -8     -8
## 40       2     -8     -8     -8     -8
## 41       1     -8     -8     -8     -8
## 42       2     -8     -8     -8     -8
## 43       3     -8     -8     -8     -8
## 44       3     -8     -8     -8     -8
## 45       3     -8     -8     -8     -8
## 46       3     -8     -8     -8     -8
## 47       2     -8     -8     -8     -8
## 48       2     -8     -8     -8     -8
## 49       2     -8     -8     -8     -8
## 50       1     -8     -8     -8     -8
## 51       3     -8     -8     -8     -8
## 52       2     -8     -8     -8     -8
## 53       3     -8     -8     -8     -8
## 54       1     -8     -8     -8     -8
## 55       3     -8     -8     -8     -8
## 56       2     -8     -8     -8     -8
## 57       3     -8     -8     -8     -8
## 58       3     -8     -8     -8     -8
## 59       1     -8     -8     -8     -8
## 60       3     -8     -8     -8     -8
## 61       3     -8     -8     -8     -8
## 62       3     -8     -8     -8     -8
```

```
## 63       2     -8     -8     -8     -8
## 64       3     -8     -8     -8     -8
## 65       3     -8     -8     -8     -8
## 66       2     -8     -8     -8     -8
## 67       3     -8     -8     -8     -8
## 68       1     -8     -8     -8     -8
## 69       2     -8     -8     -8     -8
## 70       2     -8     -8     -8     -8
## 71       2     -8     -8     -8     -8
## 72       3     -8     -8     -8     -8
## 73       1     -8     -8     -8     -8
## 74       2     -8     -8     -8     -8
## 75       2     -8     -8     -8     -8
## 76       3     -8     -8     -8     -8
## 77       3     -8     -8     -8     -8
## 78       1     -8     -8     -8     -8
## 79       2     -8     -8     -8     -8
## 80       4     -8     -8     -8     -8
## 81       2     -8     -8     -8     -8
## 82       3     -8     -8     -8     -8
## 83       2     -8     -8     -8     -8
## 84       1     -8     -8     -8     -8
## 85       1     -8     -8     -8     -8
## 86       3     -8     -8     -8     -8
## 87       3     -8     -8     -8     -8
## 88       3     -8     -8     -8     -8
## 89       2     -8     -8     -8     -8
## 90       4     -8     -8     -8     -8
## 91       5     -8     -8     -8     -8
## 92       3     -8     -8     -8     -8
## 93       1     -8     -8     -8     -8
## 94       1     -8     -8     -8     -8
## 95       2     -8     -8     -8     -8
## 96       3     -8     -8     -8     -8
## 97       3     -8     -8     -8     -8
## 98       2     -8     -8     -8     -8
## 99       5     -8     -8     -8     -8
## 100      2     -8     -8     -8     -8
## 101      4     -8     -8     -8     -8
## 102      5     -8     -8     -8     -8
## 103      2     -8     -8     -8     -8
## 104      4     -8     -8     -8     -8
## 105      1     -8     -8     -8     -8
## 106      1     -8     -8     -8     -8
## 107      3     -8     -8     -8     -8
## 108      3     -8     -8     -8     -8
## 109      3     -8     -8     -8     -8
## 110      1     -8     -8     -8     -8
## 111      3     -8     -8     -8     -8
## 112      1     -8     -8     -8     -8
## 113      3     -8     -8     -8     -8
## 114      3     -8     -8     -8     -8
## 115      2     -8     -8     -8     -8
## 116      3     -8     -8     -8     -8
```

```
## 117      3     -8     -8     -8     -8
## 118      2     -8     -8     -8     -8
## 119      3     -8     -8     -8     -8
## 120      3     -8     -8     -8     -8
## 121      1     -8     -8     -8     -8
## 122      3     -8     -8     -8     -8
## 123      1     -8     -8     -8     -8
## 124      1     -8     -8     -8     -8
## 125      1     -8     -8     -8     -8
## 126      2     -8     -8     -8     -8
## 127      2     -8     -8     -8     -8
## 128      2     -8     -8     -8     -8
## 129      2     -8     -8     -8     -8
## 130      0     -8     -8     -8     -8
## 131      1     -8     -8     -8     -8
## 132      2     -8     -8     -8     -8
## 133      2     -8     -8     -8     -8
## 134      2     -8     -8     -8     -8
## 135      3     -8     -8     -8     -8
## 136      1     -8     -8     -8     -8
## 137      2     -8     -8     -8     -8
## 138      3     -8     -8     -8     -8
## 139      3     -8     -8     -8     -8
## 140      1     -8     -8     -8     -8
## 141      2     -8     -8     -8     -8
## 142      2     -8     -8     -8     -8
## 143      2     -8     -8     -8     -8
## 144      2     -8     -8     -8     -8
## 145      1     -8     -8     -8     -8
## 146      3     -8     -8     -8     -8
## 147      4     -8     -8     -8     -8
## 148      4     -8     -8     -8     -8
## 149      2     -8     -8     -8     -8
## 150      2     -8     -8     -8     -8
## 151      1     -8     -8     -8     -8
## 152      1     -8     -8     -8     -8
## 153      2     -8     -8     -8     -8
## 154      2     -8     -8     -8     -8
## 155      2     -8     -8     -8     -8
## 156      1     -8     -8     -8     -8
## 157      3     -8     -8     -8     -8
## 158      3     -8     -8     -8     -8
## 159      2     -8     -8     -8     -8
## 160      2     -8     -8     -8     -8
## 161      2     -8     -8     -8     -8
## 162      3     -8     -8     -8     -8
## 163      3     -8     -8     -8     -8
## 164      0     -8     -8     -8     -8
## 165      2     -8     -8     -8     -8
## 166      2     -8     -8     -8     -8
## 167      2     -8     -8     -8     -8
## 168      1     -8     -8     -8     -8
## 169      2     -8     -8     -8     -8
## 170      3     -8     -8     -8     -8
```

```
## 171      2     -8     -8     -8     -8
## 172      2     -8     -8     -8     -8
## 173      2     -8     -8     -8     -8
## 174      2     -8     -8     -8     -8
## 175      3     -8     -8     -8     -8
## 176      3     -8     -8     -8     -8
## 177      3     -8     -8     -8     -8
## 178      2     -8     -8     -8     -8
## 179      1     -8     -8     -8     -8
## 180      3     -8     -8     -8     -8
## 181      2     -8     -8     -8     -8
## 182      3     -8     -8     -8     -8
## 183      2     -8     -8     -8     -8
## 184      3     -8     -8     -8     -8
## 185      1     -8     -8     -8     -8
## 186      2     -8     -8     -8     -8
## 187      2     -8     -8     -8     -8
## 188      2     -8     -8     -8     -8
## 189      1     -8     -8     -8     -8
## 190      2     -8     -8     -8     -8
## 191      2     -8     -8     -8     -8
## 192      2     -8     -8     -8     -8
## 193      4     -8     -8     -8     -8
## 194      3     -8     -8     -8     -8
## 195      2     -8     -8     -8     -8
## 196      2     -8     -8     -8     -8
## 197      2     -8     -8     -8     -8
## 198      1     -8     -8     -8     -8
## 199      2     -8     -8     -8     -8
## 200      2     -8     -8     -8     -8
## 201      2     -8     -8     -8     -8
## 202      3     -8     -8     -8     -8
## 203      3     -8     -8     -8     -8
## 204      2     -8     -8     -8     -8
## 205      4     -8     -8     -8     -8
## 206      2     -8     -8     -8     -8
## 207      2     -8     -8     -8     -8
## 208      1     -8     -8     -8     -8
## 209      3     -8     -8     -8     -8
## 210      2     -8     -8     -8     -8
## 211      2     -8     -8     -8     -8
## 212      2     -8     -8     -8     -8
## 213      3     -8     -8     -8     -8
## 214      1     -8     -8     -8     -8
## 215      3     -8     -8     -8     -8
## 216      3     -8     -8     -8     -8
## 217      3     -8     -8     -8     -8
## 218      1     -8     -8     -8     -8
## 219      2     -8     -8     -8     -8
## 220      3     -8     -8     -8     -8
## 221      1     -8     -8     -8     -8
## 222      2     -8     -8     -8     -8
## 223      0     -8     -8     -8     -8
## 224      2     -8     -8     -8     -8
```

```
## 225      2     -8     -8     -8     -8
## 226      3     -8     -8     -8     -8
## 227      3     -8     -8     -8     -8
## 228      2     -8     -8     -8     -8
## 229      3     -8     -8     -8     -8
## 230      2     -8     -8     -8     -8
## 231      2     -8     -8     -8     -8
## 232      2     -8     -8     -8     -8
## 233      2     -8     -8     -8     -8
## 234      1     -8     -8     -8     -8
## 235      2     -8     -8     -8     -8
## 236      3     -8     -8     -8     -8
## 237      2     -8     -8     -8     -8
## 238      2     -8     -8     -8     -8
## 239      2     -8     -8     -8     -8
## 240      4     -8     -8     -8     -8
## 241      2     -8     -8     -8     -8
## 242      2     -8     -8     -8     -8
## 243      2     -8     -8     -8     -8
## 244      3     -8     -8     -8     -8
## 245      3     -8     -8     -8     -8
## 246      2     -8     -8     -8     -8
## 247      1     -8     -8     -8     -8
## 248      3     -8     -8     -8     -8
## 249      1     -8     -8     -8     -8
## 250      3     -8     -8     -8     -8
## 251      2     -8     -8     -8     -8
## 252      2     -8     -8     -8     -8
## 253      2     -8     -8     -8     -8
## 254      1     -8     -8     -8     -8
## 255      3     -8     -8     -8     -8
## 256      4     -8     -8     -8     -8
## 257      3     -8     -8     -8     -8
## 258      2     -8     -8     -8     -8
## 259      3     -8     -8     -8     -8
## 260      4     -8     -8     -8     -8
## 261      1     -8     -8     -8     -8
## 262      1     -8     -8     -8     -8
## 263      2     -8     -8     -8     -8
## 264      2     -8     -8     -8     -8
## 265      3     -8     -8     -8     -8
## 266      3     -8     -8     -8     -8
## 267      3     -8     -8     -8     -8
## 268      1     -8     -8     -8     -8
## 269      4     -8     -8     -8     -8
## 270      3     -8     -8     -8     -8
## 271      1     -8     -8     -8     -8
## 272      2     -8     -8     -8     -8
## 273      2     -8     -8     -8     -8
## 274      3     -8     -8     -8     -8
## 275      2     -8     -8     -8     -8
## 276      2     -8     -8     -8     -8
## 277      2     -8     -8     -8     -8
## 278      2     -8     -8     -8     -8
```

```
## 279      2     -8     -8     -8     -8
## 280      3     -8     -8     -8     -8
## 281      2     -8     -8     -8     -8
## 282      2     -8     -8     -8     -8
## 283      3     -8     -8     -8     -8
## 284      2     -8     -8     -8     -8
## 285      2     -8     -8     -8     -8
## 286      3     -8     -8     -8     -8
## 287      3     -8     -8     -8     -8
## 288      3     -8     -8     -8     -8
## 289      4     -8     -8     -8     -8
## 290      2     -8     -8     -8     -8
## 291      1     -8     -8     -8     -8
## 292      3     -8     -8     -8     -8
## 293      1     -8     -8     -8     -8
## 294      2     -8     -8     -8     -8
## 295      3     -8     -8     -8     -8
## 296      2     -8     -8     -8     -8
## 297      1     -8     -8     -8     -8
## 298      3     -8     -8     -8     -8
## 299      1     -8     -8     -8     -8
## 300      2     -8     -8     -8     -8
## 301      3     -8     -8     -8     -8
## 302      2     -8     -8     -8     -8
## 303      2     -8     -8     -8     -8
## 304      3     -8     -8     -8     -8
## 305      2     -8     -8     -8     -8
## 306      1     -8     -8     -8     -8
## 307      3     -8     -8     -8     -8
## 308      2     -8     -8     -8     -8
## 309      1     -8     -8     -8     -8
## 310      2     -8     -8     -8     -8
## 311      1     -8     -8     -8     -8
## 312      3     -8     -8     -8     -8
## 313      2     -8     -8     -8     -8
## 314      2     -8     -8     -8     -8
## 315      1     -8     -8     -8     -8
## 316      2     -8     -8     -8     -8
## 317      3     -8     -8     -8     -8
## 318      3     -8     -8     -8     -8
## 319      3     -8     -8     -8     -8
## 320      3     -8     -8     -8     -8
## 321      2     -8     -8     -8     -8
## 322      3     -8     -8     -8     -8
## 323      2     -8     -8     -8     -8
## 324      3     -8     -8     -8     -8
## 325      1     -8     -8     -8     -8
## 326      2     -8     -8     -8     -8
## 327      3     -8     -8     -8     -8
## 328      2     -8     -8     -8     -8
## 329      5     -8     -8     -8     -8
## 330      3     -8     -8     -8     -8
## 331      2     -8     -8     -8     -8
## 332      2     -8     -8     -8     -8
```

```
## 333      2     -8     -8     -8     -8
## 334      3     -8     -8     -8     -8
## 335      2     -8     -8     -8     -8
## 336      2     -8     -8     -8     -8
## 337      4     -8     -8     -8     -8
## 338      1     -8     -8     -8     -8
## 339      2     -8     -8     -8     -8
## 340      2     -8     -8     -8     -8
## 341      1     -8     -8     -8     -8
## 342      4     -8     -8     -8     -8
## 343      4     -8     -8     -8     -8
## 344      2     -8     -8     -8     -8
## 345      2     -8     -8     -8     -8
## 346      2     -8     -8     -8     -8
## 347      2     -8     -8     -8     -8
## 348      3     -8     -8     -8     -8
## 349      3     -8     -8     -8     -8
## 350      4     -8     -8     -8     -8
## 351      2     -8     -8     -8     -8
## 352      2     -8     -8     -8     -8
## 353      3     -8     -8     -8     -8
## 354      3     -8     -8     -8     -8
## 355      4     -8     -8     -8     -8
## 356      4     -8     -8     -8     -8
## 357      2     -8     -8     -8     -8
## 358      2     -8     -8     -8     -8
## 359      4     -8     -8     -8     -8
## 360      2     -8     -8     -8     -8
## 361      2     -8     -8     -8     -8
## 362      3     -8     -8     -8     -8
## 363      2     -8     -8     -8     -8
## 364      3     -8     -8     -8     -8
## 365      2     -8     -8     -8     -8
## 366      2     -8     -8     -8     -8
## 367      2     -8     -8     -8     -8
## 368      4     -8     -8     -8     -8
## 369      3     -8     -8     -8     -8
## 370      2     -8     -8     -8     -8
## 371      2     -8     -8     -8     -8
## 372      2     -8     -8     -8     -8
## 373      3     -8     -8     -8     -8
## 374      2     -8     -8     -8     -8
## 375      3     -8     -8     -8     -8
## 376      4     -8     -8     -8     -8
## 377      4     -8     -8     -8     -8
```

```
# BED0_R, BED1_R, BED2_R, and BED3_R's 377 -8s are due to them being
# outside of their universe of NUMUNITS >= 2. They can be derived from BROOMS.
rhfs <- rhfs |>
  mutate(
    single = NUMUNITS_R == 1 & BROOMS >= 0,
    BED0_R = if_else(single, as.integer(BROOMS == 0), BED0_R),
    BED1_R = if_else(single, as.integer(BROOMS == 1), BED1_R),
    BED2_R = if_else(single, as.integer(BROOMS == 2), BED2_R),
```

```
    BED3_R = if_else(single, as.integer(BROOMS >= 3), BED3_R)
  ) |>
  select(-single)


# RENT's 27 rent-band variables are only applicable to properties with
# NUMUNITS >= 2. For single-unit properties, RENT provides the monthly rent,
# which can be used to determine the corresponding rent bands. Because RENT is
# rounded, some observations' possible original rents span multiple rent bands,
# preventing a unique assignment. Cells that must be 0 or 1 are imputed;
# ambiguous cells and properties with unreported RENT remain unchanged. RENT
# values topcoded as 5644 are assigned to RENT5000_R.

# Lower endpoints of the 27 rent bands, in ascending order.
rent_starts <- c(
  RENTNO_R = 0,
  RENTLT249_R = 1,
  RENT274_R = 250,
  RENT299_R = 275,
  RENT349_R = 300,
  RENT399_R = 350,
  RENT449_R = 400,
  RENT499_R = 450,
  RENT549_R = 500,
  RENT599_R = 550,
  RENT649_R = 600,
  RENT699_R = 650,
  RENT749_R = 700,
  RENT799_R = 750,
  RENT899_R = 800,
  RENT999_R = 900,
  RENT1099_R = 1000,
  RENT1249_R = 1100,
  RENT1499_R = 1250,
  RENT1999_R = 1500,
  RENT2499_R = 2000,
  RENT2999_R = 2500,
  RENT3499_R = 3000,
  RENT3999_R = 3500,
  RENT4499_R = 4000,
  RENT4999_R = 4500,
  RENT5000_R = 5000
)

# The corresponding upper endpoints.
rent_ends <- set_names(
  c(unname(rent_starts[-1]) - 1, Inf),
  names(rent_starts)
)

rent_cols <- names(rent_starts)

# Snapshot for comparison after the imputations.
```

```r
# Useful for checks that are not explicitly included in this .Rmd.
rent_before <- rhfs |>
  select(NUMUNITS_R, RENT, all_of(rent_cols))

rhfs <- rhfs |>
  mutate(
    .rent_low = case_when(
      NUMUNITS_R != 1 | RENT < 0 ~ NA_real_,
      RENT == 5644 ~ 5000,
      RENT == 1000 ~ 975,
      RENT < 1000 ~ pmax(RENT - 25, 0),
      TRUE ~ RENT - 50
    ),
    .rent_high = case_when(
      is.na(.rent_low) ~ NA_real_,
      RENT == 5644 ~ Inf,
      RENT < 1000 ~ RENT + 24,
      TRUE ~ RENT + 49
    ),
    across(
      all_of(rent_cols),
      \(x) case_when(
        .rent_low >= rent_starts[[cur_column()]] &
          .rent_high <= rent_ends[[cur_column()]] ~ 1L,
        .rent_high < rent_starts[[cur_column()]] |
          .rent_low > rent_ends[[cur_column()]] ~ 0L,
        TRUE ~ x
      )
    )
  ) |>
  select(-.rent_low, -.rent_high)


# If DEBT is 2 (no mortgage or similar debt), then DEBTNUM (number of mortgages
# or similar debts) is 0. This expands DEBTNUM's universe.
rhfs$DEBTNUM <- ifelse(rhfs$DEBT == 2, 0, rhfs$DEBTNUM)


# RCONTROL's 377 -8s are due to these values being outside of its universe of
# NUMUNITS >= 2. 311 of them can be derived from RCONTROLAHS.
rhfs <- rhfs |>
  mutate(
    RCONTROL = case_when(
      NUMUNITS_R == 1 & RCONTROL == -8 & RCONTROLAHS == "1" ~ 1,
      NUMUNITS_R == 1 & RCONTROL == -8 & RCONTROLAHS == "2" ~ 0,
      TRUE ~ RCONTROL
    )
  )


# LOWRENT and HIRENT are -8 for NUMUNITS >= 2. For 14 vacant-for-rent properties
# with reported, non-topcoded RENT (the topcoded value is RENT = 5644), both
# LOWRENT and HIRENT equal RENT.
```

```
rhfs |>
  select(UNITSTAT, LOWRENT, HIRENT, RENT) |>
  filter(UNITSTAT == 2)
```

```
##    UNITSTAT LOWRENT HIRENT RENT
## 1         2      -8     -8 1200
## 2         2      -8     -8 2000
## 3         2      -8     -8 2000
## 4         2      -8     -8  750
## 5         2      -8     -8 5644
## 6         2      -8     -8 1700
## 7         2      -8     -8 3000
## 8         2      -8     -8  750
## 9         2      -8     -8 1200
## 10        2      -8     -8 1500
## 11        2      -8     -8 1700
## 12        2      -8     -8 2000
## 13        2      -8     -8 3000
## 14        2      -8     -8 2000
## 15        2      -8     -8 1600
## 16        2      -8     -8   -9
## 17        2      -8     -8   -9
```

```
rhfs <- rhfs |>
  mutate(
    across(
      c(LOWRENT, HIRENT),
      \(x) if_else(
        NUMUNITS_R == 1 & UNITSTAT == "2" &
          RENT >= 0 & RENT != 5644 & x == -8,
        RENT, x
      )
    )
  )


# Viewing the targets for imputation
rhfs |>
  select(OPEX_R, OPPAY, OPINSUR, OPELGAS, OPWTRSWR, OPREP, OPMNG,
         OPPROTH, OPSECUR, OPLAND, OPOTHEX)
```

```
##        OPEX_R   OPPAY OPINSUR OPELGAS OPWTRSWR   OPREP   OPMNG OPPROTH OPSECUR
## 1          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 5          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 6          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 7          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 8          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 9          -9      -9      -9      -9       -9      -9      -9      -9      -9
## 10         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 11         -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 12       -9     -8     -8     -8     -9     -8     -8     -8     -8
## 13     3000     -8     -9     -9     -9   3000     -8     -8     -8
## 14       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 15       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 16       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 17       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 18       -9     -8     -8     -9     -9     -8     -8     -8     -8
## 19       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 20     4500     -8     -9     -9     -9   3100   1400     -8     -8
## 21       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 22     4500     -9   2000     -9     -9   2000     -9    500     -9
## 23      350     -8     -8      0      0    350     -8     -8     -8
## 24       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 25       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 26     2000   1000     -9      0      0      0   1000      0      0
## 27       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 28       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 29    12400     -8   2400     -8     -8  10000     -8     -8     -8
## 30    15108     -9     -9     -9     -9   7000     -9   3533     -9
## 31       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 32     8733     -8     -8     -8   1200   4000     -8   3533     -8
## 33       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 34       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 35       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 36    14500     -8     -8     -8   5200   1800   6000   1500     -8
## 37       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 38       -9     -8     -9     -9     -9     -9     -9     -9     -9
## 39       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 40      680      0    480     -8     -8    200     -8     -8     -8
## 41       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 42     3200     -8     -9     -8     -9   3000     -8     -8     -8
## 43     5000     -8     -8     -8     -9   5000     -8     -8     -8
## 44       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 45       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 46     4300     -8    800     -8     -8   3500     -8     -8     -8
## 47     2950      0    950      0      0   1800      0    200      0
## 48     5000     -8     -8     -8     -9   5000     -8     -8     -8
## 49     9633   3600   1500     -8     -8    500     -8   3533     -8
## 50     5060     -8     -8     -9     -9     -8    960     -8     -8
## 51       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 52       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 53    10000   2000   1500    600   1400   2500     -8   1000     -8
## 54      990     -8     -9     -8     -9      0    960     30      0
## 55       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 56       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 57     4300     -9   1500     -9     -9   1000   1600    200     -9
## 58      850      0    450     -8     -8    200     -8    200     -8
## 59       -9     -8     -9     -9     -9     -9     -9     -9     -9
## 60     3400     -8     -8     -8     -9   2500     -8     -8     -8
## 61       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 62     3000     -8   1000     -8     -8   2000     -8     -8     -8
## 63     2150     -8    760     50     80    490     -8    180     -8
## 64       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 65       -9     -8     -8     -8     -8     -8     -8     -8     -8
```

```
## 66      19100      -8     800      -8     2000   11000      -8     800      -8
## 67       3200      -9     950     400       -8    1000      -8      -8      -8
## 68         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 69         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 70       8380      -8       0       0        0    6000    1500      -8     180
## 71       6520    1200     930     200      670    2400     660     160       0
## 72       3570      -8     730      -8       -8    1600    1000     240      -8
## 73         -9      -8      -9      -9       -9      -9      -9      -9      -8
## 74       5810      -9     210      -9       -9    3000    2100      -9      -9
## 75       9705      -8     700       0      190     250      -8     200      -8
## 76       1150      -8     150      -8       -8    1000      -8      -8      -8
## 77       6000      -8    1500      -8       -8    4500      -8      -8      -8
## 78       1350      -8     250     600       -9      -9      -8     500      -8
## 79         -9      -8      -8      -8       -8      -8      -8      -8      -8
## 80       4010      -8     650      -8       -8    2600      -8      -8      -8
## 81      15624      -8    5524      -8       -9    8900      -8      -8      -8
## 82       4700      -8    3700      -8       -8      -8      -8      -8      -8
## 83       5400       0    2300       0        0     600     600       0       0
## 84         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 85       4400      -8    2400      -9       -9    1000      -8      -8      -8
## 86        500      -9      -9      -9       -9     500      -9      -9      -9
## 87       2520       0     790     200        0     760       0       0      -9
## 88         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 89         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 90      18567    5667    1500     200     1200   10000       0       0       0
## 91         -9      -8      -8      -8       -8      -8      -8      -8      -8
## 92         -9      -8      -8      -8       -8      -8      -8      -8      -8
## 93       8200      -8    3000      -8     5200      -8      -8      -8      -8
## 94         -9      -9      -9      -9       -9      -9      -9      -9      -9
## 95       5000      -8    2500      -8     1000    1000      -8      -8      -8
## 96         -9      -8      -8      -8       -8      -8      -8      -8      -8
## 97      24828      -9    2000      -9      250    8200      -9     450    2953
## 98       8000      -8     500      -9       -9    5000    2500      -8      -8
## 99      12700    1000    1300      -8       -8    8000      -8      -8      -8
## 100      3400       0    1400       0        0       0       0    2000       0
## 101     21599       0    5524       0     2200    4100    5200       0       0
## 102     14073       0    3800       0        0    2400    4200      -8    2953
## 103        -9       0      -9       0        0       0      -9       0       0
## 104      1500      -8    1000      -8       -8     500      -8      -8      -8
## 105        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 106        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 107        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 108        -9      -8      -8      -8       -8      -8      -8      -8      -8
## 109       600      -8      -9      -8       -8      -8      -8      -8      -8
## 110      1490       0     890       0        0     600       0       0       0
## 111      3600       0    3000       0        0     600       0       0       0
## 112        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 113        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 114      3500       0     800       0        0       0       0       0       0
## 115      4980       0    2500      -8       -8    2000      -8     100      -8
## 116        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 117      5460       0     510     480        0    2400       0    1400      -9
## 118        -9      -9      -9      -9       -9      -9      -9      -9      -9
## 119      5160      -8     660      -8       -8    4500      -8      -8      -9
```

```
## 120   19174    -8  1100  1400   600 15574    -8    -8    -8
## 121    6700     0  1800  1800    -9   500  2600    -8    -8
## 122    5610     0   410    -8    -8  4000  1200     0     0
## 123      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 124   10300    -8  1900    -8    -8    -8  1800    -8    -8
## 125    1600     0  1000    -8   600    -8    -8    -8    -8
## 126      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 127      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 128      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 129      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 130      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 131    4800     0     0     0     0    -9    -9     0     0
## 132      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 133    4230     0  1900     0   780   350     0     0     0
## 134    1700     0  1000     0   700     0     0     0     0
## 135    2000     0   850     0     0  1000     0   150     0
## 136    7480    -8   360    -8    -9   230    -8   290    -8
## 137      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 138      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 139    2700    -8    -8    -8    -8  1500    -8    -8    -8
## 140    2450    -8   980     0   820   250     0     0     0
## 141      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 142      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 143    2700    -8    -8    -8    -8    -8    -8    -8    -8
## 144    1500    -8  1300    -8    -9   200    -8    -8    -8
## 145    1240     0    -8    -8   840     0     0     0     0
## 146   16454    -9   640    -9    -9 15574   240    -9    -9
## 147    1800    -8   880    -8    -8   920    -8    -8    -8
## 148     800    -8   800    -8    -8    -8    -8    -8    -8
## 149      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 150    3300    -8   700    -8    -8   900  1700    -8    -8
## 151    3650    -9   950    -8    -8   500  2100   100    -8
## 152    9620  5667    -8    -8    -8  1000    -8    -8  2953
## 153    3500  2000  1100    -9    -9    -9    -8   400    -8
## 154    6200    -9   200    -9    -9    -9    -9    -9    -9
## 155      -9    -9    -8    -8    -8    -8    -9    -8    -8
## 156    4840    -9   540    -9    -9  1000    -9    -9    -9
## 157      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 158    7500    -8  1500     0     0  5000     0  1000     0
## 159    3000    -8   800    -8    -9  1000    -8    -8    -8
## 160      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 161      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 162      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 163      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 164      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 165    1070     0   320    -9    -9    -9   750    -9    -9
## 166    8849    -8   390    -8    -8    -8  7959   500    -8
## 167    3700    -8  1000    -8    -8  1500    -8   850    -8
## 168   11400    -8   400    -8    -8  8000  3000    -8    -8
## 169      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 170      -9    -9    -9    -9    -9    -9    -9    -9    -9
## 171     710    -8    -8    -8   710    -8    -8    -8    -8
## 172      -9    -8    -8    -8    -8    -8    -8    -8    -8
## 173    2120     0  1400    -8   720    -8    -8    -8    -8
```

```
## 174   1700   -8  1200   -8   -8   500   -8   -8   -8
## 175  17304   -8    -9  130  250 15574   -8  700   -8
## 176  21274   -8  3000   -8   -8 15574 2500   -8   -8
## 177     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 178   3190   -8    -8   -8  490    -8 2700   -8   -8
## 179   1670   -8   770    0    0   300    0    0    0
## 180     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 181     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 182     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 183     -9   -8    -9   -8   -9    -9   -9   -9   -8
## 184     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 185     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 186   3780 3500   280   -8   -8     0   -8    0   -8
## 187     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 188   1900   -8    -8   -8 1900    -8   -8   -8   -8
## 189   1400   -8    -8   -8 1400    -8   -8   -8   -8
## 190     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 191   7400    0     0    0    0     0    0    0    0
## 192     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 193     -9    0    -9   -9   -9    -9   -9   -9   -9
## 194  11400   -8  1400   -8   -8  9000 1000   -8   -8
## 195     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 196   8900   -8    -8   -8   -9  3800   -8   -8   -8
## 197   1600   -8    -8   -8   -8    -8 1600   -8   -8
## 198     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 199      0    0     0    0    0     0    0    0    0
## 200     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 201     -9   -8    -8   -8   -9    -8   -8   -8   -8
## 202     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 203     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 204     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 205     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 206    500   -8   500   -9   -9    -8   -8   -8   -8
## 207   3730   -8   500   -8   -8   530   -8   -8   -8
## 208     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 209    190   -8   190   -8   -8    -8   -8   -8   -8
## 210     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 211   7600   -8  4000   -8   -8  3200   -8   -8   -8
## 212   6900    0  1400    0    0  3000    0    0    0
## 213     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 214     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 215  12417 5667  2200    0  100   850 2900  150    0
## 216     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 217     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 218    350   -9   350   -9   -9    -9   -9   -9   -9
## 219   7200   -8  1000   -9   -9  1000 5200   -8   -8
## 220   3500   -8  1500   -8  500  1000   -8   -8   -8
## 221   4470   -8   170   -8   -8    -8 4200   -8   -8
## 222     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 223   4200   -8   500   -9   -9  1500 2000  200   -8
## 224     -9   -8    -8   -8   -9    -8   -8   -8   -8
## 225   6420   -9   320   -8   -9    -9 2300   -8   -8
## 226     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 227   6724    0  5524   -8   -8    -8   -8   -8   -8
```

```
## 228   3200   -9  3200   -9   -9   -9   -9   -9   -9
## 229   1980   -8   980   -8   -8 1000   -8   -8   -8
## 230   1120   -8     0   -8  720  400   -8    0   -8
## 231     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 232   3600   -8   400   -8   -9  200   -8   -9   -8
## 233     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 234   1100   -8   500   -9   -9  600   -8   -8   -8
## 235   2250   -8   350   -8   -9    0 1600  300   -8
## 236   8910   -8  3100 1800 1100  830 2000   -8   -8
## 237     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 238   6120    0  1700 3200  720  500    0    0    0
## 239   4550   -9   250   -9  300 1000   -9   -9   -9
## 240  11100 1800  3000   -9   -9 1500   -9   -9   -9
## 241   3070   -8   570   -8   -8 2000   -8   -8   -8
## 242   2490   -8   570   -8 1200   -8  720   -8   -8
## 243     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 244     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 245   6120   -8   820  100   -8 1500 2100   -8   -8
## 246     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 247     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 248     -9   -8    -8   -8   -9   -8   -8   -8   -8
## 249   5470   -8  1000   -9   -9  620 3700  150   -9
## 250   3030   -8   130   -8   -8  500   -8   -8   -8
## 251     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 252   2300    0  1100   -8   -8   -8   -8   -8   -8
## 253  27439    0  1000    0 2000 15574    0  500    0
## 254     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 255   2200   -8  1200   -8   -8 1000   -8   -8   -8
## 256   7000   -8  3900   -8   -8   -8 3100   -8   -8
## 257   4900   -8  1700   -8   -8 1000 2200   -8   -8
## 258     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 259   6500   -8  1500   -8   -9 5000   -8   -8   -8
## 260     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 261   2550   -8   150   -8   -9   -8   -9   -9   -9
## 262     -9   -8    -8   -8   -8   -8   -9   -8   -8
## 263     -9   -8    -8   -9   -9   -8   -8   -8   -8
## 264   1200   -8    -8   -8 1200   -8   -8   -8   -8
## 265   1700   -9   700   -9   -9   -9 1000   -9   -9
## 266     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 267    500  500    -9    0    0   -9   -8   -9   -8
## 268     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 269     -9   -8    -8   -9   -9   -8   -8   -8   -9
## 270     -9   -8    -8   -8   -8   -8   -8   -8   -8
## 271     -9    0    -9   -9   -9   -9   -9   -9   -9
## 272     -9    0    -9   -8   -8   -8   -8   -8   -8
## 273   3800   -8    -8   -8   -8 2000 1800   -8   -8
## 274     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 275     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 276     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 277     -9   -9    -9   -9   -9   -9   -9   -9   -9
## 278  13559    0   600    0    0 5000 7959    0    0
## 279     -9    0    -9   -9   -9   -9   -9   -9   -9
## 280   2850   -8   850   -8   -8 2000   -8   -8   -8
## 281     -9   -9    -9   -9   -9   -9   -9   -9   -9
```

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## 282     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 283   5000     -8    900     -8     -9   1500   2600     -8     -8
## 284     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 285     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 286   3500     -8   1500     -8     -8   2000     -8     -8     -8
## 287   7000     -8   1000     -9     -9     -9     -9     -9     -8
## 288     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 289   1400      0   1400     -8     -8     -8     -8     -8     -8
## 290   7550      0    450     -9     -9   1200     -9    900     -9
## 291     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 292   3000     -8   2500     -8     -8    500     -8     -8     -8
## 293     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 294    620     -8    170     -9     -9    450     -9     -9     -9
## 295   2550      0    870     -9     -9     50   1300     -9     -9
## 296   3200     -8    200   1500     -9    500    900    100     -8
## 297  17700   1500   1000   3200   2000  10000     -8     -8     -8
## 298   7300     -8   2300     -8   1800     -8     -8     -8     -8
## 299     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 300   4030     -8    170     -9     -9    990   2800     -8     -8
## 301     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 302     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 303     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 304     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 305     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 306     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 307     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 308   6370   2400   1400     -8     -9   2000    320    250     -8
## 309     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 310     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 311     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 312   4900     -8   1200     -8     -8   3700     -8     -8     -8
## 313     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 314     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 315     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 316     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 317   6230     -8    530     -8   1400   3200     -8     -8     -8
## 318  11000     -8   3500     -9     -8   5000   2500     -8     -8
## 319     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 320   2600     -8   1100      0   1200    300     -8     -8     -8
## 321     -9     -8     -9     -9     -9     -8     -8     -9     -8
## 322   1820     -8    560     -8     -8    130   1000    130     -8
## 323   2450     -8    450     -8     -8   1000     -8   1000     -8
## 324   1600     -8     -8     -8   1600     -8     -8     -8     -8
## 325   2300      0   2000      0    300      0      0      0      0
## 326   3620      0   3200      0      0    420      0      0      0
## 327  13000     -8     -9     -9     -9  10000   3000     -9     -9
## 328     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 329   4700     -9     -8     -8     -8   1600   2000     -8     -8
## 330     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 331   3100      0    600      0   1000      0   1500      0      0
## 332  11879     -8    920     -8     -8   2000   7959   1000     -8
## 333   8750     -8   3000     -8   5200     -8     -8     -8     -8
## 334     -9     -8     -9     -8     -8     -8     -8     -9     -8
## 335  12300     -8   1500     -9   1800   1500     -8     -8     -8
```

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## 336      -9     -8     -8     -9     -9     -8     -8     -8     -8
## 337      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 338      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 339      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 340    9100     -9     -9   1300     -9   7800     -9     -9     -9
## 341    9260     -8    680     -8     -9    380   1300     -8     -8
## 342    2000     -9   1200     -8     -8     -8     -8    200     -8
## 343   23024     -8   1600     -8     -8  15574   2500     -8     -8
## 344      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 345    2530   2000    250     -8     -9    280     -8     -8     -8
## 346      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 347   20165     -8    500     -8     -9   7500   3800     -8     -8
## 348      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 349    6000     -9     -9     -8     -8   5000     -8     -8     -8
## 350    6960     -8   2500     -8    960   3500     -8     -8     -8
## 351    2580     -9   1200     -9    660     -9     -9     -9     -9
## 352      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 353    5524      0   5524     -8     -8     -8     -8     -8     -8
## 354      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 355    7800     -9   1400     -9     -9   4000   2400     -9     -9
## 356    2500      0   2000      0      0      0     -8     -8     -8
## 357    8720     -8    510     -8     -9    860   5200    350     -8
## 358      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 359      -9     -9     -9     -8     -8     -8     -9     -9     -8
## 360    3800     -9     -9   1800     -9     -9     -9     -9     -9
## 361    6760     -8    940     40     -9     80   2100     -8     -8
## 362   26074     -8   1500   3200   1800  15574   4000     -8     -8
## 363    7575     -8   3000     -8     -9     -8     -8     -8     -8
## 364    7150     -8   1200     -8     -9     50   5900     -8     -8
## 365      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 366      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 367   10700     -8   2000   1500   5200   2000     -8     -8     -8
## 368    1000     -8     -8     -8     -9     -8     -8     -8     -8
## 369    1500     -8   1000     -8     -8     -8     -8    500     -8
## 370      -9     -8     -9     -9     -9     -8     -9     -8     -8
## 371    8140      0    790      0      0    500    400    750     -8
## 372    3200     -8    600     -8     -9   2000     -8     -8     -8
## 373    8100     -9   2400   1400   1200   2500     -8     -8     -8
## 374    6800   4000    800   1000     -9   1000     -9     -8     -8
## 375    4800     -8   2000    800     -9   1500     -8    500     -8
## 376   10580     -8   2200     -8     -8   2200   5400     -8     -8
## 377      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 378      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 379      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 380      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 381      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 382      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 383      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 384      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 385      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 386      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 387      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 388    5000      0     -9      0   3000      0   1500      0      0
## 389      -9     -8     -8     -8     -8     -8     -8     -8     -8
```

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## 390     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 391     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 392     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 393     -9      0     -9     -9     -9     -9     -9     -9     -9
## 394  23400     -8   5000   4000   1200  12000     -8     -8     -8
## 395     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 396  14700      0   3000   1700   2400   5000      0   1800      0
## 397     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 398     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 399   6500     -8   1400     -8    600   2000     -8     -8     -8
## 400   6600     -8   2600   2000   2000     -8     -8     -8     -8
## 401     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 402     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 403   4600      0     -9      0      0   3200   1400      0      0
## 404     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 405   5350     -9   3000     -8    250     -9     -9    300     -9
## 406     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 407     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 408     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 409   5400     -8   2000     -8    900   2000     -8    500     -8
## 410  10700     -8   2200   1700   2300     -9   3100     -9     -8
## 411   7990   1400   2100     -8    540   2400    740      0      0
## 412  15700     -8   1300     -8   1400   8000     -8   1000     -8
## 413      0      0      0      0      0      0      0      0      0
## 414   3600     -8   3000     -8     -8     -8     -8     -8     -8
## 415     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 416  17314      0     -9     -9     -9     -9     -8     -9     -9
## 417   4800     -8   1300     -8   2500     -8     -8   1000     -8
## 418   3000     -9   3000     -9     -9     -9     -9     -9     -9
## 419     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 420   1800     -8   1800     -9     -9     -9     -8     -8     -8
## 421     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 422  12500     -8   5000     -9   2500   5000     -8     -9     -8
## 423     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 424     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 425     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 426     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 427  30500  23550    650     -8   3300     -8     -8   2000     -8
## 428     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 429   2000     -8     -9     -9   2000     -9     -9     -9     -9
## 430     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 431   3300     -8   2000     -8   1000    300     -8     -8     -8
## 432   8100     -9   3600     -9   4500     -9     -9     -9     -9
## 433  10900     -8   2000   3700   5200     -9     -8     -8     -8
## 434     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 435  11200     -8   2000      0   1200   6000     -8    800     -8
## 436     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 437   5900     -8    400     -8     -8   3500   1600    100     -8
## 438     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 439   3500     -8   1500     -9   2000     -9     -8     -9     -9
## 440     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 441  13400     -8   1300     -8   6100   6000     -8     -8     -8
## 442   3300     -8    900     -8   1200     -9     -8   1200     -8
## 443     -9     -8     -8     -8     -8     -8     -8     -8     -8
```

```
## 444   6000   -8  1000   -8   -9  5000   -8   -8   -8
## 445   5320  450   800  100 1700  1800    0   50    0
## 446     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 447   2200   -8  1200   -8  400    -8   -8   -8   -8
## 448   2400   -9   900   -9   -9  1500   -9   -9   -9
## 449  13600    0  5000    0 5400  2000    0    0    0
## 450     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 451   4000   -8   500   -8  500  2500   -8   -8   -8
## 452     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 453     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 454     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 455     -9   -9    -9   -9   -9    -9   -9   -9   -9
## 456  10150   -8  1300   -8 1100  6500   -8   -8   -8
## 457     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 458     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 459  20040   -9  2200  240 1600 15000   -9 1000    0
## 460     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 461  36867   -8  3700 1500  840 28697   -8  130   -8
## 462     -9   -8    -9   -9   -9    -9   -9   -9   -9
## 463   2800   -8  1800   -8   -8  1000   -8   -8   -8
## 464   8300   -8  2000   -8   -9  5000   -8   -8   -8
## 465     -9   -8    -8   -8   -9    -8   -8   -8   -8
## 466   2600  500   750   -8  600   500   -8  250   -8
## 467   6960   -8  1000   -8  960  5000   -8   -8   -8
## 468     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 469     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 470     -9   -8    -8   -8   -9    -8   -8   -8   -9
## 471     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 472  44263 23550  8713   -8 2000 10000   -8   -8   -8
## 473   4480   -8  1300    0 1200  1100   -8  880   -8
## 474  10500   -8  5500   -8 4000    -8   -8   -8   -8
## 475     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 476   3400   -8    -8   -8 2400    -8   -8   -8   -8
## 477  27000   -8  4000 2000 3000 12000   -8 1200   -8
## 478     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 479  10700    0  2400   -8 1800  6200   -9   -9   -9
## 480  14483   -8  1600   -8 9683  2000   -9   -8   -8
## 481   2200   -8  1200   -8   -9  1000   -9   -8   -8
## 482     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 483  10400   -8  1500 4600 1200  2500   -8  600   -8
## 484     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 485     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 486   4120   -8   920  100   -8  2000   -8  650   -8
## 487   2200   -8  2200   -9   -9    -9   -9   -9   -9
## 488   8000    0  3300 1100 1400  1000    0 1200    0
## 489   7500   -8  2500   -8 5000    -8   -8   -8   -8
## 490  15500   -8  3500 4500 7500    -8   -8   -8   -8
## 491     -9   -8    -9   -9   -9    -9   -9   -9   -9
## 492     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 493     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 494     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 495   8000   -8  5000   -8 3000    -8   -8   -8   -8
## 496     -9   -8    -8   -8   -8    -8   -8   -8   -8
## 497  11900   -8    -9    0 2700  3300 2700   -9   -9
```

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## 498      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 499    4500     -8     -9     -8   2000     -9     -8   1300     -8
## 500    9940     -8    570   4200    260   3400     -8    920     -8
## 501      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 502   12200     -8     -8     -8   4000   3000   4000     -9     -9
## 503    7000     -8   2000   1500   3500     -8     -8     -8     -8
## 504    3100     -8    800     -8   2000     -8     -8     -8     -8
## 505      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 506    5500     -8   3000     -8   2500     -8     -8     -8     -8
## 507    9000     -9   4200     -8   4800     -8     -8     -8     -8
## 508      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 509   13000      0   4000      0   2700   4300      0   2000      0
## 510   23350     -8   4500      0   1700  12000     -8     -8     -8
## 511   18400     -8   2600   5000   1400   7200     -8     -8     -8
## 512      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 513   23900     -8   1500    300   1900   6500   8000   2200     -8
## 514   10500     -8   3000   1500   2500   2500     -8     -8     -8
## 515    7600     -8   2000     -8   1300     -9     -8    800     -8
## 516   11700     -8   3000     -8   1200   6000     -8     -8     -8
## 517      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 518      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 519      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 520   11350   5000   2000     -8    960   2200     -8     90     -8
## 521   17800   1700   2300    590   4300    470   4200   1000   1440
## 522      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 523      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 524      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 525      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 526    3500      0    500      0   2000   1000      0      0      0
## 527      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 528    4710     -8   1100     -8   2400    760     -8     -8     -8
## 529    7710     -8   1400     -8   2100    350     -8     60     -8
## 530   29100     -8   4000     -8   2600  15000   6300     -8     -8
## 531   17313      0   8713   3600    600   2400      0   1500      0
## 532   14900      0   3500      0      0   3000     -8   3250     -8
## 533    8200      0   1200   4000   2000   1000      0      0      0
## 534      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 535   16000     -8   5000   7000   4000     -8     -8     -8     -8
## 536   10500     -8   5000     -8     -9   5500     -8     -9     -8
## 537   65211   5000   4800   5000   2000  28697     -9      0      0
## 538   36627      0   3300   1200   9683   7500  14054    140     70
## 539   11400   5000   2200     -9   1200   3000     -8     -8     -8
## 540      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 541      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 542    5400     -8   4200     -8     -8    600     -8     -8     -8
## 543   41597   5000   4000   1500   2400  28697     -8     -8     -8
## 544   26000   1500    700   2000    800   1000   8000   2000     -8
## 545   38504     -8   4000    250     -8  18500  14054   1000     -8
## 546      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 547    1280     -8    760     -8     -8    520     -8     -8     -8
## 548   16820     -8   1000     -8    420  14800     -8    400     -8
## 549      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 550   14400     -8   2800     -8   1000  10000     -8     -8     -8
## 551   12700     -8   3600   3500   1400   3000     -8     -8     -8
```

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## 552   3720   580  1100    0   240  1300    0    0    0
## 553   1500    -8  1200   -8   300    -8   -8   -8   -8
## 554  12000    -8  5500 1500  5000    -8   -8   -8   -8
## 555  17400     0  4600    0  1800  6500 1200    0    0
## 556     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 557     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 558     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 559     -9    -8    -9   -9    -9    -9   -9   -9   -9
## 560  12500  4500  1200 1500   630   970    0 1000   -9
## 561  18200    -8  1200 9000    -9  8000   -8   -8   -8
## 562   1790    -8   750   -8   450   590   -8   -8   -8
## 563  17250    -8   650 2500  4000  9500   -8  600   -8
## 564     -9    -8    -8   -8    -8    -8   -8   -8   -8
## 565   4700    -8   500 1200    -9  3000   -8   -8   -8
## 566     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 567   2200    -8   700   -8    -8  1200   -8   -8   -8
## 568   1400    -8    -9   -8    -9   700   -8  500   -8
## 569   3150    -8  1800   -8    -9   350   -8   -8   -8
## 570   4700    -8  1700   -8    -9  3000   -8   -8   -8
## 571   4200    -8   640   -8   260  2300   -8 1000   -8
## 572     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 573   3900    -8  1900   -8    -9  2000   -8   -8   -8
## 574  10510     0   510 3000  2000  2000    0  600   -9
## 575   2100    -9  1000    0   600   500   -9   -9   -9
## 576     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 577   5500    -8  1000   -8  2500  2000   -8   -8   -8
## 578  10000  2000  1000 1000    -9  6000   -8   -8   -8
## 579     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 580     -9    -8    -8   -8    -8    -8   -8   -8   -8
## 581     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 582  14113    -8  8713   -8  2400  3000   -8   -8   -8
## 583    800    -9   800   -9    -9    -9   -9   -9   -9
## 584    800    -9    -9   -9    -9    -9   -9   -9   -9
## 585   7500    -8  2900   -8  1600  3000   -8   -8   -8
## 586   5000    -8    -9   -8    -8  5000   -8   -8   -8
## 587     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 588   2960    -8  1200  660    -9  1000   -8   -8   -8
## 589   3000    -8  3000   -8    -8    -8   -8   -8   -8
## 590     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 591     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 592     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 593     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 594     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 595     -9    -8    -8   -9    -9    -8   -8   -8   -8
## 596   6450    -8    -8   -8  1200  2500 2500   -8   -8
## 597  13700    -8   900 3500  3000  4000   -8  300   -8
## 598   6450    -8  1200   -8  3000  1600   -8   -8  400
## 599     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 600     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 601   2000     0     0    0   800     0    0    0    0
## 602   2990    -9  1700  250     0   590    0  450    0
## 603   2950    -8   450   -8   500  2000   -8   -8   -8
## 604     -9    -9    -9   -9    -9    -9   -9   -9   -9
## 605   1680   580  1100   -8    -8    -8   -8   -8   -8
```

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## 606   4450    0 4000    0    0  450    0    0    0
## 607   4230  600 1200   -9   -8 1500   -8   80   -8
## 608     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 609  15230   -8 4400  290 4300 5000   -8  180   -8
## 610     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 611   7170    0  690    0  830 5000   -8  350   -8
## 612   6480   -8    0 1200  380 2300   -8  800    0
## 613     -9   -8   -9   -9   -9   -9   -9   -9   -9
## 614   8200   -8 2900 1100 2700 1500   -9   -9   -9
## 615     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 616  15200   -8 1500   -8   -8 7900   -8 1900   -8
## 617   4400    0 1200 2200    0  500    0    0    0
## 618     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 619  20100   -8 5400 6800 1500 2600   -8 2800   -8
## 620   3740   -8 1800  170  570    0   -8    0    0
## 621   4200   -9   -9   -9 1200 1000   -8   -8   -8
## 622     -9   -8   -8   -8   -8   -8   -8   -8   -8
## 623     -9   -8   -8   -8   -8   -8   -8   -8   -8
## 624   3000   -8 1500   -9   -8  500   -8   -8   -8
## 625   7570   -8  460   -8 2600 2800 1200   -8   -8
## 626   4720 1400 2000   -8  320 1000   -8   -8   -8
## 627     -9   -8   -8   -8   -9   -8   -8   -8   -8
## 628  11220   -8  920 2400 2400 5500   -8   -8   -8
## 629     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 630   3400   -8 1200   -8   -8 2200   -8   -8   -8
## 631  13000   -8   -8 8900 2000 2100   -8   -8   -8
## 632   7200    0 1200    0    0 5000    0  500    0
## 633     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 634  11000   -9 1300   -8   -8 2600 3800 2300   -8
## 635     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 636   1200   -8 1200    0    0    0    0    0    0
## 637     -9   -8   -8   -8   -8   -8   -8   -8   -8
## 638  23130   -8 3900 1700 2900 5100   -8  730   -8
## 639     -9   -8   -8   -8   -8   -8   -8   -8   -8
## 640   7000   -9 7000   -9   -9   -9   -9   -9   -9
## 641     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 642  13500   -8 1800    0 1200 10000   -8   -8   -8
## 643     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 644   2660    0 1500   -8  660  500   -8   -8   -8
## 645   4000   -8 1100  300  200 2000   -8   -8   -8
## 646     -9   -8   -8   -8   -8   -8   -8   -8   -8
## 647    800    0   -9   -9   -9   -9  800   -9   -9
## 648     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 649  14000   -8 2400    0 1200 8000   -8    0 1200
## 650   6600   -8 1000   -8   -8 5500   -8  100   -8
## 651     -9   -9   -9   -9   -9   -9   -9   -9   -9
## 652  36197   -8 1400  480 2400 28697   -9  940   -9
## 653  14800   -8 1200   -8  450 12000   -8  150   -8
## 654  10100   -8 4000  630  770 2000 2300   -8   -8
## 655   2500   -8 1000   -8  500 1000   -8   -8   -8
## 656   3400   -9   -9   -9 1200   -9 2200   -9   -9
## 657   3000   -8   -8   -8   -9   -8   -8   -8   -8
## 658   2600   -9   -9   -9   -9 1200 1400   -9   -9
## 659     -9   -8   -8   -8   -8   -8   -8   -8   -8
```

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## 660   14500    -8  2000    -8    -8  10000    -8    -8    -8
## 661      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 662   13993    -8  8713    -8  3000   2000    -8    -8    -8
## 663    5800    -8  2700    -8  1900     -8    -8    -8    -8
## 664   10770    -9  6700    -9   500   3200    -9   370    -9
## 665      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 666   13500    -8  3000   500    -8  10000    -8    -8    -8
## 667    5900    -8  2200    -8    -8   2500     0     0     0
## 668   11850    -8    -9   900  3200   4000    -8  3250    -8
## 669   18350  2600  1800  4800  2400   5000    -9   250    -9
## 670   10500    -8  5000     0  1500   4000    -8    -8    -8
## 671    4900    -8  1500    -8  2900    500    -8    -8    -8
## 672    4800    -8  1800    -8    -8   2000    -8    -8    -8
## 673      -9    -8    -9    -9    -9     -9    -8    -9    -8
## 674      -9    -8    -8    -8    -9     -8    -8    -8    -9
## 675   18300    -8  2200   600  2400  12000    -8    -8    -8
## 676   27900    -8    -9    -8  3400   7500  4100    -9    -8
## 677      -9    -8    -8    -8    -9     -8    -8    -8    -8
## 678   13000    -8    -9    -9    -9     -9    -9    -9    -8
## 679    5400    -8  2000   300   600   2500    -8    -8    -8
## 680    9440    -8    -9  4300  2000   1500    -9   240    -8
## 681   18050    -8  3500  1000  1400   8000    -8   650    -8
## 682    6500    -8  2500   500  3500     -8    -8    -8    -8
## 683    7590   510  2000   240  1800   2600    -8    -8    -8
## 684    4060    -9    -9   200   900   1200    -9   360    -9
## 685    7500    -8  1800    -8  2400   3000    -8    -8    -8
## 686      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 687   15300    -8  1500  6500  2300   2500     0   500     0
## 688    6250     0  2400    -9   350   2000     0   500    -9
## 689    2500    -8  1100   400     0   1000     0     0     0
## 690    5730    -8  4100    -8    -9   1000    -8    -8    -8
## 691      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 692      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 693    6000    -9    -9    -9  3000     -9    -9    -9    -9
## 694   13113    -8  8713    -9  2400   2000    -8    -8    -8
## 695      -9     0    -8    -8    -8     -8    -8    -8    -8
## 696   15200    -8  6000  3000  1200   5000    -8    -9    -8
## 697   19610    -8  6500    -8    -8   8700  1200  1800    -8
## 698    7000    -8  3800    -8   600   1200    -8   200    -8
## 699    6940    -8  4000  1900   240    500    -8    -8    -8
## 700    6240    -8  1500    -8   940   2500    -8    -8    -8
## 701    8500    -8  3000  1500  4000     -8    -8    -8    -8
## 702      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 703    9280    -8  3000    -8  2800   3000    -8    -8    -8
## 704      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 705      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 706      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 707   27300    -8  1900  3300  2000  18300    -8  1000    -8
## 708      -9    -8    -8    -8    -8     -8    -8    -8    -8
## 709      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 710    3900    -8  2000    -8  1500    200    -8    -8    -8
## 711      -9    -8    -9    -9    -9     -9    -9    -9    -9
## 712      -9    -9    -9    -9    -9     -9    -9    -9    -9
## 713    4240    -8  2800    -8   890     -8    -8    -8    -8
```

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## 714     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 715  14100     -9   1500      0      0   8800   3800      0      0
## 716  14200     -8   4000   2400   1600   5000     -8     -8   1200
## 717   7150     -8   3000     -8   2500    150     -8     -8     -8
## 718  11800      0   1200   5800   2400   1800      0      0      0
## 719  21100   5500   2300   2300   2400   1800     -8   1600     -8
## 720   2100     -8     -8     -8     -9     -8     -8     -8     -8
## 721  12000      0   3500   2000   1500   2500      0      0      0
## 722   4000     -8     -9     -9   2000   2000     -8     -8     -8
## 723   9500     -8   4500    500   2500   2000     -8     -8     -8
## 724  35316     -8   5000  11933   9683   8000     -8     -8     -8
## 725  24750     -8   3500    500   4500   8000     -9   3250     -9
## 726     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 727     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 728  31864      0   1800      0   4900   3100   2300    750      0
## 729     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 730   6000     -8     -8     -8   2400     -8   2400     -8     -8
## 731     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 732   5000     -8   2500     -8   2500     -8     -8     -8     -8
## 733     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 734  12020      0   1700    320   2200   2400   4600     -9     -9
## 735     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 736     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 737   3000     -8     -8     -8     -9     -8     -8     -8     -8
## 738  18783      0   8713   2000   1900   5200      0      0    100
## 739   3400     -8   3000     -9    200    200     -8     -8     -8
## 740   8160     -8   1500    950   1700   3000     -8    110     -8
## 741     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 742  12420     -8   3900    770   2200   4400     -8    150     -8
## 743  11209     -8   1400    330   3700   2000     -8    330   1089
## 744  10500     -8   3000   2500   5000     -8     -8     -8     -8
## 745     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 746     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 747     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 748     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 749     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 750  13400     -8   5000     -8   1600   5000     -8   1200     -8
## 751  10200      0   4000   1000    600    600      0   1500      0
## 752  14300     -8   1900   1100   2700   3700   4700     -8     -8
## 753  25300     -8   4000   1100   4000  15000     -8   1200     -8
## 754  32353   2000   3200   1000   1000  15000     -8    550   5003
## 755  17200     -8   1200   3400   3400   5000   3200   1000     -8
## 756   3200     -9   1100     -8   1600    500     -8     -8     -8
## 757   6000     -8   2000   1000   2000     -8     -8     -8     -8
## 758   8000     -8     -9     -8   3000   5000     -8     -8     -8
## 759  15850    650   2500   7600   2500   2000     -8    300     -8
## 760  23100     -8   5800   7200   2900   3600     -8     -8     -8
## 761     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 762     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 763  16500     -8   2600   1200    700  12000     -8     -8     -8
## 764     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 765   8800     -8   1500   4000   3300     -8     -8     -8     -8
## 766     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 767  34897     -8   2600     -8   2200  28697     -8     -8     -8
```

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## 768   3500   -8  1500   -8  2000   -8   -8   -8   -8
## 769   6500   -8  2700 2000  1800   -8   -8   -8   -8
## 770     -9   -8    -9   -9    -9   -9   -8   -9   -8
## 771   3800   -8    -9   -9    -9   -9 3800   -9   -9
## 772   8800   -8  2500   -8  2000 3300   -8   -8   -8
## 773   8000   -8  5000   -8  2500  500   -8   -8   -8
## 774     -9   -8    -8   -8    -9   -8   -8   -8   -9
## 775     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 776  10000   -9  4000 3500    -8 2500   -8   -8   -8
## 777  10600   -8   900 5400  1700 2600   -8   -8   -9
## 778   2750   -8  1500   -8  1000  250   -8   -8   -8
## 779  10700   -8  3000 1500  5000   -8   -8   -8   -8
## 780   6800   -8  3000 1000  1800   -8   -8   -8   -8
## 781  11950 3800  2200    0  1800 2500   -8  150   -8
## 782   8480   -8  4800  480    -9 2000   -8   -8   -8
## 783  21683   -8  3500 8500  9683   -8   -8   -8   -8
## 784   2500   -8    -8   -8  2500   -8   -8   -8   -8
## 785  10150   -8  3000  550  3000 2000   -8   -8   -8
## 786     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 787     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 788  13600   -8  1500   -9  3200 5100   -8 2700   -8
## 789   1000   -8    -9   -8    -8 1000   -8   -8   -8
## 790  30154   -8  4100 8500    -8 2000 14054   -8   -8
## 791  42297   -8  1100   -8    -8 28697   -8 1500   -8
## 792  22383   -8  5200   -8  9683 5500   -8   -8   -8
## 793  12850   -8   650   -8  1500 8000   -8  300   -8
## 794     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 795   5300   -8  3500   -8  1000   -8   -8   -8   -8
## 796     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 797  11000   -8    -9 4000    -9 7000   -8   -8   -8
## 798     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 799   6500   -8  3000   -8  1500 2000   -8   -8   -8
## 800   5800   -9  3300   -9    -9 2500   -9   -9   -9
## 801     -9   -8    -8   -8    -8   -8   -8   -8   -8
## 802   8600   -8  4000 1200  2400 1000   -8   -8   -8
## 803  20300   -8  3000 5200  3000 7100   -8   -8   -8
## 804     -9   -8    -9   -8    -8   -9   -9   -9   -8
## 805     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 806   6000   -8  2300  200   200 1500   -8   -8   -8
## 807  21483   -8  2000 11933 1600   -9   -8   -8   -8
## 808   5600   -8  1000   -8  3600 1000   -8   -8   -8
## 809   1250    0    -8   -8    -9 1000    0  250    0
## 810  12820   -8  2400 3500    -9 4000   -9 2200   -8
## 811  17410   -8  3500   -8   960 6000   -8 3250   -8
## 812  23783   -8  2600 11933 1200 6000  900  500  150
## 813  13500   -8  2000  500  3000 7000   -8   -8   -8
## 814     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 815  11300   -8  3500 2000  4200 1600   -8   -8   -8
## 816   9700    0  2400   -8  2800 2500   -8   -8   -8
## 817     -9   -9    -9   -9    -9   -9   -9   -9   -9
## 818  67850 23550  5300 8000  6900 13200 6200   -8   -8
## 819  16000    0     0 2000  2000 2000 8000    0    0
## 820  23713 2800  6700   -8    -9 2600 5500  680 5003
## 821   9200   -8  2200 1800  1200 3000   -8 1000   -8
```

```
## 822    12900    -8  4000   300  3600  3000    -8    -8    -8
## 823    22200    -8  2000  2000  2000  8200  6800    -9    -9
## 824    17900    -8  4100  2600  4100  5000    -8    -8    -8
## 825    38336  6000  7000  5500  9683    -8    -8    -8  5003
## 826       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 827       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 828       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 829       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 830       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 831       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 832       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 833       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 834       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 835       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 836       -9    -9    -9    -9    -9    -9    -9    -9     0
## 837       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 838       -9    -8    -8    -8    -9    -8    -8    -8    -8
## 839       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 840       -9    -8    -9    -9    -9    -9    -9    -9    -9
## 841       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 842       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 843       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 844       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 845    22600    -9  2300  9700  2000  6800  1800    -9    -9
## 846       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 847       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 848       -9    -8    -8    -8    -9    -8    -8    -8    -8
## 849       -9     0     0     0     0     0     0     0    -8
## 850    19730  3600   850   280  4500 10500    -8    -9    -8
## 851       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 852    55550    -9    -9 15000  6000 14100 13000   450    -8
## 853       -9    -8    -8    -8    -8    -8    -8    -8    -8
## 854    74000    -8  8000  6400  6100 18400    -8  1000    -8
## 855       -9    -8    -8    -8    -9    -8    -8    -8    -8
## 856       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 857    76900    -8 13300  2300  9400  1000 10700   200    -8
## 858    68300    -9 17800 17300 26500    -9    -9    -9    -9
## 859       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 860       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 861       -9     0     0     0     0     0    -9    -9    -9
## 862       -9     0     0     0     0     0     0    -8    -8
## 863       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 864       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 865       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 866       -9    -8    -8    -8    -8    -8    -8    -8    -8
## 867       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 868       -9    -8    -8    -8    -9    -8    -8    -8    -8
## 869       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 870       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 871       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 872       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 873       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 874       -9    -9    -9    -9    -9    -9    -9    -9    -9
## 875       -9    -9    -9    -9    -9    -9    -9    -9    -9
```

```
## 876       -9      -9      -9      -9       -9      -9      -9      -9       0
## 877       -9      -9      -9      -9       -9      -8      -8      -9      -9
## 878       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 879       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 880       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 881       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 882       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 883       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 884   114880    9200    4100     190    15500   60000   14100      -8      -8
## 885       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 886       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 887       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 888   101000       0   16000      -9    10000   50000   20000    5000      -8
## 889       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 890       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 891       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 892       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 893       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 894       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 895       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 896       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 897    55600   31000    7500    2200     9400      -9      -9      -9       0
## 898       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 899       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 900       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 901       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 902       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 903       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 904       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 905       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 906    89000   30000    7200    2400    10600   14700   13600    6300      -8
## 907       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 908    15620      -8    2400      -8     3200    8200      -8     320      -8
## 909       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 910       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 911       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 912       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 913     7240      -8    5800     720       -9      -8      -8      -8      -8
## 914       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 915    22300      -8    6500      -8       -8   10000    5800      -8      -8
## 916    19000   11000    8000      -8       -8      -8      -8      -8      -8
## 917       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 918    65500   30000    9000    5000     4000   15000      -8      -8      -8
## 919       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 920       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 921       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 922    26270       0    4000    8800     1200    7800       0     220       0
## 923       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 924       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 925       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 926       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 927   120000      -9      -9      -9       -9      -9      -9      -9      -9
## 928       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 929    69746      -8   18200    5500    38546    7500      -8      -8      -8
```

```
## 930      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 931      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 932  106300   6600   7800  24500  23900  33000     -8   1700     -8
## 933      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 934  193271  44000     -9  23000  27000  27000     -8   3800  11880
## 935   51880     -8   3000   1100   6500  18800  18000     -8     -8
## 936      -9     -8     -8     -9     -8     -8     -8     -8     -8
## 937  134900      0  11000   2300   6200 100000  15000      0      0
## 938      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 939   85500     -9  22000     -9  24000  12000  24000     -9     -9
## 940      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 941  151098 139098     -9  12000     -9     -9     -9     -9     -9
## 942  110600  46000  12000      0  12000  30000  10000      0      0
## 943      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 944      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 945    2700     -9     -9     -9     -9     -9     -9     -9     -9
## 946      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 947      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 948      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 949      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 950      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 951      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 952      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 953      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 954    3000      0     -9   3000     -9     -9      0      0      0
## 955      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 956      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 957      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 958   12000     -9     -9     -9     -9     -9     -9     -9     -8
## 959    1200     -8   1200     -9     -9      0     -8     -8     -8
## 960      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 961      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 962  105095  32000  33295     -8  12900   6000  12500     -8     -8
## 963      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 964      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 965      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 966      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 967      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 968      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 969      -9     -8     -8     -8     -9     -8     -8     -8     -8
## 970  208600  45000  18000  15600  25000  44000  36000   6000     -8
## 971      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 972      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 973   60460   9100   4200  12400   6200   9800   3800   3200     -9
## 974      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 975      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 976      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 977      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 978      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 979      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 980      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 981      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 982      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 983      -9     -8     -8     -8     -8     -8     -8     -8     -8
```

```
## 984       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 985       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 986       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 987    57250  35000   5000    250   5000  12000     -8     -8     -8
## 988    17000   5000   4800     -8   1000   5000     -8     -8     -9
## 989     7900     -8     -8     -9     -9     -8     -8     -8     -8
## 990    21700     -9   2900   2300   5000   4500     -9     -9     -9
## 991       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 992   215916  18800  23100   6700   7800 147516     -8     -8     -8
## 993       -9     -9     -9     -9     -9     -9     -9     -9     -9
## 994     2540     -8   2100    240      0     -9     -8    200     -8
## 995   102400  12800   6400  15000  10300   9000  20000   6900   2000
## 996   103100  21600  14900   4900  14900  18700     -8   2100     -8
## 997       -9     -8     -8     -8     -8     -8     -8     -8     -8
## 998    10640     -8   2000    240   3000   5000     -8    400     -8
## 999    89600   9000   6900   9700   6700  13000  20600   5100     -8
## 1000      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1001  274546  75000   4000  25000  38546 100000     -8  12000   8000
## 1002      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1003      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1004      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1005   99400  19700  14000  19800  15700   2000  10700  10900     -8
## 1006   59000     -8     -8     -8     -8  41800  13000     -8     -8
## 1007      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1008  125940     -8  10700   8000  10300  35400  13100   4400      0
## 1009   46420  22900   4400    300   3900   4500   8700      0      0
## 1010      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1011      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1012      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1013    5540   2500     -8     -8     -8   2500     -8     -8     -8
## 1014  101000  41000   8400  10300   8900  10900   9100   2700      0
## 1015      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1016   66500  10500   8100   1800     -9   4900  16400   6600     -8
## 1017      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1018      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1019      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1020   10500      0   1200   2000     -9   6500      0     -8     -8
## 1021   77800     -8   5900  12400   5000  24200  14800   5600     -8
## 1022      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1023    4600     -8   4600     -9     -9     -8     -8     -8     -8
## 1024   45700     -9   3100   8000   5000  21600   5000   3000     -8
## 1025      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1026      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1027      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1028      -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1029      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1030      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1031   20550     -9   2500     -9   6000   5000   4300    750     -9
## 1032    2400     -9     -9     -9     -8     -9     -8     -9     -8
## 1033   79461     -8  18000  43461  18000     -9     -9     -9     -8
## 1034   41100   9700   7800   2700   4300   2400      0    500      0
## 1035      -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1036   92800  45300   6600   6900  10100  14900      0   2700      0
## 1037      -9     -8     -8     -8     -8     -8     -8     -8     -8
```

```
## 1038    34566       0      -9    6000    11500    3700    6100      -9     966
## 1039    86869      -8   14900   12100    11300    5100      -8   31369       0
## 1040    24570       0    4200     400     2100   13100      -8     770      -8
## 1041   191949      -8      -8      -9       -9      -8      -8      -8      -8
## 1042       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1043    32400       0    5400    1100     2700    4900    4200    6600      -9
## 1044    28600      -8    3500    5700       -9   14000      -8    1000     300
## 1045    15000      -8      -9      -8     3000    5000      -8    1000      -8
## 1046       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1047       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1048   153332      -8   23300   32100     5600   44800   16600   13600     132
## 1049    73200      -8    9600   20400    13500   21500    6700    1500      -8
## 1050    40800      -8    9300    5200     5000   12700      -8    1200      -8
## 1051    28680      -8    7400     680     1500    4200   10600     400      -8
## 1052       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 1053       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1054   179591   59000   15100    6800     3800   48200   14000   18500      -8
## 1055   196516      -8    7100    7300     9600  147516   22000    3000       0
## 1056       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1057       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1058    63790       0   25000   15900     9500    6500    5000     490       0
## 1059       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1060       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1061   102330   12400   11200   11900    12000   20500   11800     530      -8
## 1062    27620   12000    6200      -9     1800     700      -8     620       0
## 1063    58000   15900   10900    2500     4800    4400    5200    2700       0
## 1064    32600      -8    4000    7500     4500   11200      -8    1800      -8
## 1065   309240   23400   12100   11600    15100   19600   18800    2500       0
## 1066    35400    5000    8000    9600     8000    3000      -8      -8      -8
## 1067   123500   31700   11200   21500     6200   18500   11800    7100       0
## 1068    65500      -8    3200    4700     4600   26000   15000    8400      -8
## 1069    89900      -8    8300   29700     7800   33000    4100    5000      -8
## 1070       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1071       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1072    51800      -8    7000    1500    15000   25000      -8      -8    1500
## 1073    92661    9200   20000   43461    10000    8000      -8    2000      -8
## 1074    17990      -8      -8     600     6300    3300    5800     650       0
## 1075    18300      -8    6100    1500     2600    7500      -8      -8      -8
## 1076       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1077       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1078    30200      -9    9300    3300     3900    7700      -8    1000      -9
## 1079    32200   12000    2500      -8     6000   10000      -8     600      -8
## 1080       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1081    55000      -8    3000    5000     5000   20000   18000    1000      -8
## 1082       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1083       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1084       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1085       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1086       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1087       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1088       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1089   183180   13100   16000    8700    12800   41200   17600    8800    2880
## 1090       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1091       -9      -8      -8      -8       -8      -8      -8      -8      -8
```

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## 1092      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1093  148012   16600   33295    2000   17500   10900   25400    3800    7317
## 1094   26000       0    5000    3000    6000    3000    8000    1000       0
## 1095   34000       0    8000    1400    2600   10000      -8      -8      -8
## 1096  210295   73000   33295    3800    9500    9800   22400    7800      -8
## 1097      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1098  118800    9600   13000    3000    7600   63000   18700      -8      -8
## 1099   89800   17000   12800   12100   13000   13500      -8    1800    3600
## 1100  231100  107000   23800   31800    6700   51000      -8   10800      -8
## 1101  300999      -8   13900    3200   11900   10600   50750   11200      -8
## 1102   81400      -8   10900    3100   12300   51000      -9    1300      -9
## 1103      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1104  136800      -8   12300   28800      -9   67000   27800     900      -8
## 1105   58730   20000    9400     480    2900   13200      -8     150      -8
## 1106      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1107  130000   12000   12000   23000   23000   30000   25000    5000      -8
## 1108   73490       0    9100    3700   20700   10500   20200     890      -8
## 1109      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1110      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1111      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1112    2600       0    1800       0       0       0       0       0      -8
## 1113      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1114   72000      -9   10000   34000    8000   20000      -9      -9      -9
## 1115      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1116  216065    5400   16400   13600    9300   87000    1200    3800    3065
## 1117   23300      -8      -8    4000      -9    5000   10500    1800      -8
## 1118      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1119      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1120      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1121  196769   34100    6300   27200    9100   65000   21900   31369       0
## 1122   41100      -8    7900    4900    9700   14500      -8      -9      -8
## 1123   48200      -8   13200    2400    7600   18000    6000    1000       0
## 1124      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1125      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1126   26000    5000    3500    5000   10000    2500      -8      -8      -8
## 1127      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1128  256890   53000   33295   43461   11700   57000   34300   11000    4334
## 1129  257016   75000   12000   10000    2500  147516      -8   10000      -8
## 1130      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1131   62000      -8    5000   12000   24000   10000      -8      -8      -8
## 1132      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1133      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1134      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1135      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1136  189369   15000   15000   20000   10000   30000   30000   31369    8000
## 1137      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1138      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1139      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1140  120546      -8      -8   34800   38546   30000    7200   10000      -8
## 1141   33000    3600    5000       0   15000    7000       0       0       0
## 1142      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1143  106300   18900    7400    1400   24300   17700   27600      -9      -9
## 1144  232500   67000   16600   10000   24000   80000   24000    7000      -8
## 1145   86840   30000    9800    2600   12600   14100   12200     540      -9
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## 1146   119195   34000   33295    5600     1500    9100   31000      -9      -9
## 1147       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1148       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1149       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1150       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1151       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1152       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1153    49140      -8    2000     140     7900   33100      -8      -8      -8
## 1154       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1155    40000      -8   10800   18000     8200    3000      -8      -8      -8
## 1156   230616   23000   13100    2000    13400  147516   20100      -8      -8
## 1157     8500      -8    3500    2000     3000      -8      -8      -8      -8
## 1158    75700      -8    3300    1100     7400   47000    9600     800      -8
## 1159     2800      -8    1200      -9       -9     400      -8      -8      -8
## 1160       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 1161    43300      -8    5000    7000     1300   30000      -8      -8      -8
## 1162    23000      -8    5500    2400     4100    2000      -8      -8      -8
## 1163       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1164    50030      -8    3900     230     7900    3100   15900      -9      -8
## 1165    37600    6000    5000    1600    11000   10000      -8    1500      -8
## 1166       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1167       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1168    47500      -8    6500      -8     4000   20000   12000    2000    3000
## 1169    98290   56000   10700   13600     9800    5200      -8      -8      -8
## 1170    12000      -9   12000      -8       -8      -8      -8      -8      -8
## 1171     7200      -8      -8      -8     5400      -8      -8      -8      -8
## 1172       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1173    25000      -8    4000    5000     3000    8000    5000      -8      -8
## 1174       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 1175    20980      -8      -9   11000     5600    4300      -8      80      -8
## 1176    32350    5300     950    1900     8500   10500      -8      -8      -8
## 1177    11400      -9    3400      -9       -9      -9      -9      -9      -9
## 1178    61100   29000    3300      -9     2500   25000      -9    1300      -9
## 1179    63620   25000    7000     300     2400   24000      -8      -8     720
## 1180       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1181    31800      -8    7000    4000     4300   15000      -8      -8      -8
## 1182    84700   65000      -8    3900    15800      -8      -8      -8      -8
## 1183    18000      -8    2500    7500     5500    2500      -8      -8      -8
## 1184     6070      -8      -9     670       -8    5400      -8      -8      -8
## 1185    76000   19200    6300   10000     2000   34000       0     500      -9
## 1186       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1187       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1188    19000      -9    2900    1400     2200    2400    7700    1200      -9
## 1189     1300      -8      -9     300       -9    1000      -8      -8      -8
## 1190        0       0       0       0        0       0       0       0       0
## 1191   177469      -8      -9   10600     8000   84000   38000   31369      -9
## 1192       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1193     1740      -9      -9      -9       -9     540    1200      -9      -9
## 1194       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1195    34940      -8      -8     640       -8   31000      -8      -8      -8
## 1196    19600    6000    7000       0     3600    1500       0       0       0
## 1197    69600       0    8100   10500     4700   35000       0    6600       0
## 1198    96928   27300    6000   22100     4300    9900   15900    2800     128
## 1199       -9      -9      -9      -9       -9      -9      -9      -9      -9
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## 1200     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1201  12500     -8  10000     -8     -8     -9     -8   2000     -8
## 1202  32310     -8   2300    610   4500  21900     -9    600      0
## 1203  37200     -8   4000   5600   1400  15500     -8     -9     -8
## 1204     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1205     -9     -8     -9     -9     -9     -9     -8     -9     -8
## 1206  15000     -8   6900   1000   2800   1400     -8    910     -8
## 1207  46600      0   5000    600   7000  30000      0      0      0
## 1208 126000      0  19200  14000   8800  74000   7600   2400      0
## 1209     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1210     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 1211  37000  10000   3500  10000   2900   5000     -8   2000     -8
## 1212     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1213  51600     -8   8300   4600   6000  19800   5300   1600     -8
## 1214  63100     -8   2900   3100   3100  47700     -8    100     -8
## 1215     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1216     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1217  24130     -8   1500    630   6200  11500     -8     -8     -8
## 1218  41600      0   5200  11400      0  12000   3300   3800      0
## 1219  29000     -8   3800     -8   8200   3500   8500     -8     -8
## 1220     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1221     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1222  38500     -9   3100  11300     -9  16000   8100     -9     -9
## 1223  55400     -8  13000  11100   1900  10000  11700   1700     -8
## 1224  45500     -8   7000   3000   7700  25000     -8     -8     -8
## 1225  24100     -8   5600   2000   1500  10000     -8   5000     -8
## 1226  19150     -8   1500    450   4500   7000   5700     -8     -8
## 1227  10300   7200     -8     -8   3100     -8     -8     -8     -8
## 1228  28500     -8   7200   3800   1000   7700   2000   1800     -8
## 1229     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1230     -9     -9     -9     -9     -9     -9     -9     -9      0
## 1231  52250     -8   7700   3500   7300  16000  17000     -8     -8
## 1232  25900      0   4100   5300   2900  12000      0      0      0
## 1233     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1234 142200  50000  19000   5000   9000  25000     -8  14000     -8
## 1235  10000     -8   2400   1100   3300     -9     -8     -9     -9
## 1236 125550  38500   5000  20700  11600  30600   6400   1800   8550
## 1237  15880     -8    500    480   2300  11300     -9     -9     -9
## 1238     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1239  27900     -8   4800     -8   9400   6500     -8   3100     -8
## 1240   3750     -8     -8   3000     -9    750     -8     -9     -9
## 1241     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1242  37800  15000   3000   1400   7200  10000     -9     -9     -9
## 1243  71890     -8   3500   3700   3500  56000   2200    590     -8
## 1244 181800  43700   5000  34200   8100  43000  22400   8000     -8
## 1245  51000      0  10000   6000   5000  10000      0   5000      0
## 1246     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1247     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1248  15900     -8   6500     -9   3500   5000     -8     -8     -8
## 1249     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1250   6600      0   1800   2000   1800    500     -8     -8     -8
## 1251     -9     -8     -9     -9     -9     -9     -8     -9     -8
## 1252  29300     -8     -9  18500     -9     -9   9000     -9     -8
## 1253     -9     -9     -9     -9     -9     -9     -9     -9     -9
```

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## 1254     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1255  53400     -8  14200   2100   3600  31700     -8     -8     -8
## 1256     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1257  51300      0   4500   4000   1200  40000      0      0    400
## 1258  84300     -8   5900  17100  19800  37000     -8     -8     -8
## 1259  14800      0   1800   7000   2000     -8   4000      0      0
## 1260  25700     -8  11000   1200   7500   2000   3500    500     -8
## 1261  46000     -8  15000     -8  12000   6000     -8   4000     -8
## 1262     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1263  46070     -8  19900   1500  17200   7300     -8      0      0
## 1264  13300  10800   2500     -9     -9     -9     -9     -9     -9
## 1265  70000     -9     -9     -9     -9     -9     -9     -9     -9
## 1266  24300   1200   7000    600   1400  12000     -8    900     -8
## 1267 158600     -8   7100  11500  24700  95000  13600   2100     -8
## 1268     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1269  12000      0   2000   1000   2000   3000     -8      0     -8
## 1270     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1271  14500   5000   2700   2100   2700   1000     -8     -8     -8
## 1272     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1273     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1274     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1275  36200  10000   4000   1200  10000  10000     -9   1000     -9
## 1276  46500  15000   5000   3500  10000   3000  10000     -8     -8
## 1277  21000     -8   3000   5000   3000   6000     -8   1000     -8
## 1278     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1279 113200  80000   8000   7200   6000  12000     -8     -8     -8
## 1280  19600     -8   7000    600  12000     -8     -8     -8     -8
## 1281     -9     -8     -8     -9     -9     -8     -8     -8     -8
## 1282  43000     -8   6000     -8   6000  25000     -8   1000     -8
## 1283     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1284   7500     -9   7500     -9     -9     -9     -9     -9     -9
## 1285  22760      0   3000   2100   2600   2800   5700    260      0
## 1286     -9     -8     -8     -8     -9     -8     -8     -8     -8
## 1287  27600     -8   7600     -8   2900  10500   4400    900     -8
## 1288  13820     -8   2500    600   4000   6000     -9     -9    360
## 1289     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 1290  49000     -8   5500   1500   5000  30000   4000   3000     -8
## 1291 185746      0  24500  36900  38546  48200  27600   4200      0
## 1292 117200  98000     -8     -8  19200     -8     -8     -8     -8
## 1293     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1294  22800      0   9500   3000   4200   4000      0      0      0
## 1295     -9     -8     -8     -9     -9     -9     -9     -8     -8
## 1296  48500  14600   3500   5200   7200  13000      0   4000     -9
## 1297     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1298 116700     -8  21500  25800   8500  22300  27700   2500     -8
## 1299  18700     -8   8500   6000   4200     -9     -8     -9     -8
## 1300  25000  25000     -9     -9     -9     -9     -9      0      0
## 1301     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1302  82400      0   7500   9600  24600  19300   6000   3000     -8
## 1303  85300  11000   3300  10000   9000  15000  10000  12000      0
## 1304  17450     -8   6000     -8   2800   5000   2400    750     -8
## 1305  46300     -8   5600   8000   4600   5700  14800   1800     -8
## 1306     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1307     -9     -8      0      0      0      0     -8      0      0
```

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## 1308     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1309     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1310  15480      0    580   3800     -9   8200   2900     -9     -9
## 1311  32100      0   6600   7000   4000   2000   7800   1200   2500
## 1312  32500  15000  10000   2500   5000     -8     -8     -8     -8
## 1313  53400      0   5000  21200   9000   6000     -8   3000     -8
## 1314  24000     -8   4000     -8     -9  20000     -8      0      0
## 1315     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1316     -9     -8     -9     -9     -9     -9     -9     -9     -9
## 1317  37750   5000   2300     -9  10500   8000     -8    750     -8
## 1318  23380     -8   5200    580   1300   8500     -8   1300     -8
## 1319  15800     -8   7700      0    500     -9      0      0     -9
## 1320  27300     -8   4300   3900   4900  10800     -8   1000     -8
## 1321  95500     -8  23000  32000     -9  30000     -8   4000     -8
## 1322     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1323  66800     -8   5500   8800   3500  16800  15000   1500     -9
## 1324  50900  45000     -8     -8   3800     -8     -8     -8     -8
## 1325     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1326  21000   8000   1500   4000   3500   4000     -9     -9     -9
## 1327  28100      0   9000   7500   5000   3000      0      0      0
## 1328   6000   6000     -8     -8     -9     -8     -8     -8     -8
## 1329  15300     -8     -8   3800  11500     -8     -8     -8     -8
## 1330 105247     -8   7000  43461  38546  15000     -8   1000     -8
## 1331  51250  20000   5500   2000   1500  15000     -8   3000     -8
## 1332  12900      0   3400      0   3000   1600      0      0      0
## 1333  22710      0   7600    700    610   8800     -9     -9     -9
## 1334  24400     -8  12000   2400     -8  10000     -8     -8     -8
## 1335 185546   9000  15000  24000  38546  50000     -8     -8     -8
## 1336     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1337     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1338  42300      0   2100   1500  10400  11500  12800   2000      0
## 1339     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1340     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1341     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1342  63701     -8   4000  43461     -9  12200     -8    540     -8
## 1343 115540     -8  14700   1700  16100  49000  25500   2500     -8
## 1344     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1345  56800     -8  12900  21600   8300  14000     -8     -8     -8
## 1346 237861 100000   6400  43461   3600  14400  26600   2800      0
## 1347     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1348  25550     -9   3500     -8     -8  13800   4300    350     -8
## 1349  94495     -9  33295   4000  24000  30000     -9    700     -9
## 1350  18300     -8     -8   3700   2700   2300     -8     -8     -8
## 1351 139200  21900  21300     -9  11200  16200   6100     -9     -9
## 1352  56500     -8   4400   2500   8000  18000  15100   3500     -8
## 1353 244616  32500   7500  21500  18600 147516  15000   2000      0
## 1354  54500   3500  13000   3000  12000   5000  13000   2000     -8
## 1355     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1356 172200  80000   5200   5000  10000  30000  20000   1000     -9
## 1357  45710     -8   3500      0   8300  13600  16600    410     -9
## 1358 175441      0  25000   3500  22700  56000  50750   3300      0
## 1359     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1360     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1361  19300   3000   4500   1000    800   9000      0      0      0
```

```
## 1362      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1363      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1364    8500      -8    3500    1500    3500      -8      -8      -8      -8
## 1365  128000   17900   23700   14500   16200   38600      -8      -8    7600
## 1366      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1367   45800      -8    7200      -9    4100    3500   25000      -8      -8
## 1368      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1369   75900   20000    5000    5400   12000    5000   25000      -8      -8
## 1370      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1371      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1372   20560      -8    1500     960   13200    3500      -8      -8      -8
## 1373   29400      -8    3400    6000    6000   11000      -8     600      -8
## 1374  107600    7700   13200    1700   13800   50000   16200    1800      -9
## 1375   71500   11900    6900    3400    4200    8500    9100   15900      -8
## 1376   39500      -8    9800      -8   11000    2500   15000      -8      -8
## 1377  191949      -9      -9      -9      -9      -9      -9      -9      -9
## 1378      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1379      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1380   17170      -8    1600     420   10300    1700      -8    1000      -8
## 1381   34600      -8   18000    5800    4800    6000       0       0       0
## 1382   37840      -8    4700     610    4600   16200      -8    7100      -8
## 1383   63200    3000      -9    4600    7200   20800   21600    6000      -9
## 1384   22160      -8    3100    3700     560   12200      -8      -9      -8
## 1385   24700      -8   10500     200    8000    5000      -8     500       0
## 1386      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1387   37300    7000   11500     500    5400    8800      -8    1900      -9
## 1388   84500   12000   16000   28000    8000   15000      -8    2500      -8
## 1389   86500   22000    9000    6000    5500    7000   10000   12000       0
## 1390  158746       0   13900   18200   38546   70000   11800     900      -9
## 1391   37600      -9    6500    1400     800   28900      -9      -9      -9
## 1392   38200      -8    4200    7400    5700   10800    7900      -8      -8
## 1393      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1394   84000   10000    8000   13000    7000   40000      -8    2000       0
## 1395      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1396      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1397      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1398   27000      -9    5000    4000    5000   10000      -9      -9      -9
## 1399   44500    9300    2400   12800    1900   15000      -8      -8      -8
## 1400   22700    8000    6400      -8    2400    3000      -8    1500      -8
## 1401   64110      -8    4900     810   13700   18500    7800    3400       0
## 1402  125691      -8   20200    2400   12200   55000      -9    9200      -9
## 1403      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1404      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1405      -9      -8      -9      -8      -9      -9      -8      -8      -8
## 1406   35350      -9    6200    3500   13800    8600      -9     750      -9
## 1407   75800   25000    3500     300    3000    6000   15000   10000       0
## 1408   36000      -9      -9      -9      -9   36000      -9      -9      -9
## 1409      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1410      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1411   34100      -8    3100    1200    3200    8200   11000    3000      -8
## 1412   34900   17000    4900    3000    3000    3000    2000       0       0
## 1413      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1414      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1415  125500   20000   17500   10000   10000   20000   20000   15000       0
```

```
## 1416    29220    2500    3600    4400    7500   11000       0     220       0
## 1417       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1418    83746      -8   12000   17000   38546   15000      -8    1200      -8
## 1419    78800      -8   23200      -8    7500   29000   19100      -8      -8
## 1420    42000       0    4000   11000      -9   15000      -9    6000      -9
## 1421       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1422       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1423       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1424    50200      -8    9100    8600    3500   25000      -8    1200      -8
## 1425       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1426       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1427    30000      -8    6100    3300    5600    3500    4500    4000      -8
## 1428       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1429   141568   16100   14500   16400    8400   38600   20300    2100    1868
## 1430    38500      -9   12000      -9    5000    7400   10000     500      -9
## 1431       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1432    95000   50000   12000    5000    5000    4000    4000    5000       0
## 1433    25000      -8    5000      -8    2200    6000    9800      -9      -8
## 1434    30390       0    3000    1100     720   10900   10900     770      -8
## 1435   257593  139098   33295   26900   15600   12400      -8    5100      -8
## 1436    47700      -8    1800     800   15400    5100   15600      -8      -8
## 1437    48800      -9   13000      -8      -9    7200   25000      -9      -9
## 1438   244107      -8   14500    5700   21500  147516   35400    4100      -8
## 1439   253578  139098    1400   30400   10400   16000   35100    4500   11880
## 1440    45900      -8      -8    1000    7900   19400   15700      -8      -8
## 1441    18710   14400      -9     530      -9    2100      -9     480      -9
## 1442    45500      -8    7000   14200    9800    6000    6000      -8      -8
## 1443    89000   18000   25000    6500   12000   15000      -9    5000      -9
## 1444       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 1445     8000      -9      -9      -9      -9      -9      -9      -9      -9
## 1446    62800    9800    5000   12700    4800   16400   10500    1200       0
## 1447   112300   15000    7500    5600   13000   48000   20500    2700       0
## 1448    32700      -9    6500      -9    8200   10000      -8      -8      -9
## 1449   148900      -8    8500    4300    9900   69000   23600    1600      -8
## 1450   122600      -8    3700      -8   20000   91000      -8    3000      -8
## 1451   104800   64000    4000    2600    4000   10200    8700    7800       0
## 1452    81000   70000    4000       0       0    1000    5000       0       0
## 1453     6400      -8      -8      -8    6400      -8      -8      -8      -8
## 1454       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1455       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1456       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1457       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1458    29690      -8   11900      -8   13000    3600      -8      -8      -8
## 1459       -9       0      -9      -9      -9      -9      -9      -9      -9
## 1460   107800   35700    6600    8200   11000   25600   13000    1300      -8
## 1461       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1462   107700   37500    7500   10400    5600   19500    7500      -9      -9
## 1463    67400       0    6000    7000   26400   10000   12000    6000       0
## 1464       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1465   151400   65000   10800    5100   21500   19000      -9    5000      -9
## 1466    51500       0   14900    4000    4800    6000   17000    4800      -9
## 1467       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1468       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1469    97400   19900    9900    2700   16200   19800   11300    1600      -8
```

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## 1470    17000   17000      -9      -9       -9      -9      -9      -9      -9
## 1471   129750      -8   15000    7000     7000   50000   50750      -8      -8
## 1472   197061   47000   23000   43461    21000   40000   14000    6600    2000
## 1473    30400      -9      -9      -9       -9      -9      -9      -9      -9
## 1474   227800   12000    5000    5000    17400   25000   31000    4000      -8
## 1475       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 1476       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1477   264330   26000   19500   43461    23000   68000   28000   31369      -8
## 1478   167480      -8   21000    7200    23000   75000   12600   15000   11880
## 1479       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1480    94100   44200    9400    2400    13900   15900      -8      -8      -8
## 1481       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1482       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1483       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1484   138700      -8   16700   25400     5600   36900   19900   11100      -8
## 1485   160750       0      -8   10000       -9  100000   50750      -8      -8
## 1486       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1487       -9      -8      -9      -8       -8      -9      -9      -8      -8
## 1488       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1489       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1490       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1491       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1492       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1493       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1494       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1495       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1496       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1497       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1498       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1499       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1500       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1501       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1502       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1503       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1504       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1505       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1506       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1507       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1508       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1509       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1510       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1511       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1512       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1513       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1514       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1515       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1516       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1517       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1518       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 1519       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1520       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1521       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1522       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 1523       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 1524      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1525      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1526      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 1527      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1528      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1529      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1530      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1531      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1532      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1533      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1534      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1535      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1536      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1537      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1538      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1539      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1540      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1541      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1542      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1543      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 1544      -9      -9       0       0       0       0       0       0       0
## 1545  212000   29200   30100   47000   10700   41600   25100    6300       0
## 1546      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1547  192500   30200   20900   21900   10500   74000   17000    7400      -8
## 1548      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1549      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1550  379648  164000   21400   12900   15800   67000   31700    8700   22848
## 1551  202200      -9   45100   20100    8700   12000   35700    5400       0
## 1552  281096   30000   10000   18000   12000  206096       0       0      -9
## 1553      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1554  595000  162000   47000   92000   65000  153000   65000   10000      -8
## 1555  209800   20000   22700    6300   24400   86000   14200    3200      -8
## 1556      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1557  197300      -8   26000    8900   18300   98000   24500    4000      -8
## 1558      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1559      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1560  189400   50000   33000    5400   16000   30000      -8    3000    4000
## 1561      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1562      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1563      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1564      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1565      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1566      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1567      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1568      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1569      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1570      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1571      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1572      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1573      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1574      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1575      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1576  402571      -8      -8      -9      -9      -8      -8      -8      -8
## 1577      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

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## 1578      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1579      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1580      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1581      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1582  212900   44600   40400    9200    6100   68000   30000    4800       0
## 1583      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1584  216206      -9   23800    8700   77516   10200   71000    8200    1990
## 1585      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1586  289470   23700   24000    7100   48400  181000      -9     470      -8
## 1587      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1588      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1589      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1590      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1591      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1592      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1593      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1594      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1595  116500   48000      -9   26900   25600   16000      -9      -9       0
## 1596      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1597      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1598      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1599      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1600      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1601  377653  239253   23800   18900   28800   37200      -8   10400      -8
## 1602      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1603      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1604  206400   28600   32000   49100   25300   23600   21800    6600      -9
## 1605      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1606      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1607  688653  112000   61432   91000   77516  206096   53000   52761   22848
## 1608      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1609      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1610      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1611  192000      -9   44000   55000   20000   65000      -8    8000      -8
## 1612  156800   44800   14100   16300   16700   46900   11500    6500      -8
## 1613      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1614      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1615      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1616      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1617      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1618      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1619   91100       0   21500    7300   23500      -9   33800     500      -9
## 1620      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1621      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1622  282696      -8   20800    5000   12000  206096   25000    5000       0
## 1623      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1624  167111   45900    5000   20300    4000   27700   14400   14100    1711
## 1625      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1626  488294      -8   41800   65000   22100   70000   47000   23000    4057
## 1627      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1628      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1629      -9       0       0      -8      -9      -8      -8      -8      -8
## 1630      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1631      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

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## 1632   152761      -8   24400    3000    5900   40700   14600   52761       0
## 1633       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1634       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 1635       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1636       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1637       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1638   204300   63000   11100    5200   14700   48900   36100    4600      -8
## 1639       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1640       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1641   145400   31900   13200    5300    6000   41100   17200    7200      -8
## 1642       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1643   179500   35800   12500   13700   11500   69000   25900    2200       0
## 1644       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1645       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1646       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1647   101500       0   24000    2500    9200   27500   14800    5800       0
## 1648       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1649       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1650   158700   25600   12600    6700   21400   17500   24600    8500      -8
## 1651   201634   62000   17000    3900   12000   32600   20300   20000    2034
## 1652   203100   16200   31500    2200    2600   38400   28700    1100      -8
## 1653   242356   72000   23600    8300   28900   23800   22700    8700     656
## 1654       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1655       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1656   250100   60000   19700   14800    8100   83000   11000   20000      -8
## 1657   208200   20000   27700   42000   33000   32500   34000   19000      -8
## 1658       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1659   165200      -8   13100   28500   14900   48900   30300    9500      -8
## 1660       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1661       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1662   120700      -8    6200   34400    5100   50000   18000    7000      -8
## 1663       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1664   145200   25800   31900   10000    7600   51000      -8    5000      -8
## 1665       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1666   282000  150000      -9      -9   32000   75000      -9    5000      -9
## 1667       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1668   200261   37700      -8   50000    6700   30600   22500   52761      -9
## 1669       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1670       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1671       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1672    77600      -8   15300      -8   27500   15000   11400    1800      -8
## 1673       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1674   380878  107000   61432   15400    9100   73000   38500    9800    3346
## 1675       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1676   308632       0   22400   24000   10600  104000   18000   11300     932
## 1677   128700    8000   47000   18500   17500    7300    5000    5000       0
## 1678   125000    5100   21600    8100    6300   36800   20000    5900       0
## 1679       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1680   387707   73000   27900   69000   14800  106000   29700    8200    1407
## 1681   198241   34300    5900  139741    3800    4300      -8    1800    1200
## 1682   270300      -8   10300   71000      -9      -9      -9      -9      -9
## 1683   277300   61000   10200   46400   25900   45200   20200    7200      -8
## 1684   209100   47500   21800   14500   25100   25400   40100    5600      -8
## 1685       -9      -8      -8      -8      -8      -8      -8      -8      -8
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## 1686   331200   82000    7600   11800    14800   78000      -9      -9      -9
## 1687   314700  102000   27900   30800    20100   40900      -8      -8       0
## 1688   224300   14500    9800      -9       -9      -9   22000      -9      -9
## 1689   284656      -8    9100   33300    15000   58000   35200   14800    2256
## 1690   217900   53000   14200    4300    41800   36200   17700   21800      -9
## 1691   200600  109000   28100   31900       -9      -8      -8      -8      -8
## 1692       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1693   108148      -8      -8   22600       -9   45100   14000      -8   13548
## 1694   205895   58000   35800    7300     9800   15700   18500   16700    1495
## 1695      300      -8      -9      -9       -9      -9      -9      -8      -8
## 1696       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1697       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1698   211118   47300   21200   13100     4500   52000   59000    8700    5318
## 1699       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1700       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1701       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1702       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 1703   165600   18600    9100   28800     8200   45900   16100    7000       0
## 1704       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1705       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1706   425582      -8   16500   89000     6400  109000   97742     940      -8
## 1707   235601   22500   16600  139741    10200   19700   13100     560      -9
## 1708   219000  125000   15000    5000    40000   25000      -8    2000      -8
## 1709   196900  112000   10700   19100    15800   18900      -8    7500      -8
## 1710   158000      -8   17700   20600    17700   47100   14100    3900       0
## 1711   143900   27900   14800   21200     9400   30600   14900    3400       0
## 1712   124830   26000   20100    6300    24200   10800   14300    2600      -8
## 1713       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1714   277500   42200   12700   16000    12000  148000   34700    4500    3400
## 1715   386896   57000   26700    2900    21800  206096   18000   18300     200
## 1716   261541      -8   34000  139741     4000   23500   15600    2300       0
## 1717   272960      -8   27100   66000    38000   39500   16400   12900     660
## 1718       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1719       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1720   260500  108000   31600      -8    25200   63000      -8    7200      -8
## 1721   245800   37600   18100    8200    24600   88000   18100   11300       0
## 1722   217080   75000   28900   21200     4400   66000    5800     680      -8
## 1723       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1724   126000   47000   18000   30000    10000   15000       0    2500       0
## 1725   448571       0       0    8700    12200    8600       0      -8      -8
## 1726       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1727   439610   51000   44900  139741    63000   30100       0   12600    3269
## 1728   235600   69000    8600    4700    18000   81000    8000    2500      -8
## 1729   363800  135000   40200   25500       -9   35500   19100   22500      -8
## 1730       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1731    17500      -8   12000    3500     2000      -8      -8      -9      -8
## 1732       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1733   281000  175000   50000   10000     5000    5000    5000   10000    3000
## 1734    18100   18100      -9      -9       -9      -9      -9      -9      -9
## 1735       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1736   141500      -8   26800    3500     3000  104000      -8      -8      -8
## 1737    33500      -8   13500      -9       -9   10000      -8      -8      -8
## 1738   334703   61000   16000   10100    44400  103000   56000    3900    2203
## 1739       -9      -8      -8      -8       -8      -8      -8      -8      -8
```

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## 1740   480193   99000    6500   16900    20600  151000   23800   16100    8493
## 1741       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1742   115300   19000   17000    4000    15000   13900   16000    9400      -8
## 1743       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1744       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1745   252742   25000   50000   25000    55000      -8   97742      -8      -8
## 1746   162000      -8    8700    5100    15000  104000   26000    1200      -8
## 1747   297000   94000   27500    3600    17900   40600   76000      -8      -8
## 1748       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1749       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1750   345900  133000   34100   51000    25800   51000   27200    2800      -9
## 1751   176912   30000   19600    7700    17800   35100   23700   11600    3612
## 1752       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1753       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1754   178342      -8   10000   11200    13300   37300   97742    4200      -9
## 1755       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1756    75000   50000   10000    5000        0    5000    5000       0       0
## 1757       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1758   187420   20100   19300    7200    13900   53000   37800     820      -8
## 1759       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1760   276100       0   15800    9800    18200  116000   25300    8200      -9
## 1761       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1762   426900  130000   36700   91000    34000   68000   38000   27000      -8
## 1763       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1764   133000      -8   15000      -8    18000   60000   33000      -8      -8
## 1765   228000   31600   38000   25000    20000   66000   43800      -8      -8
## 1766   337900   92000   26400    9900    47900   79000   52000    3200      -8
## 1767   299000  185000   21000   63000       -9   10000      -8    8000      -8
## 1768       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1769       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1770       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1771   408600  114000   14800   42500    21100  114000   27300   31200       0
## 1772   430533  239253   61432   48000    27000   12000      -8      -8   22848
## 1773   153500   35500   23100   16300    43800   32600      -9    2200      -9
## 1774   258639   23000   28000    9300    27700  106000   44300     790    5349
## 1775       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1776   135000      -8   10000   25000    50000      -8   50000      -8      -8
## 1777       -9      -8       0       0        0       0       0       0      -8
## 1778   561257  140000    9900  139741    77516  170000      -9   20200      -8
## 1779    38600      -9   20000    5000     3000   10000      -8     600      -8
## 1780       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1781       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1782   205500   50000    7500   52000    48000   25000   18000    5000       0
## 1783   237500   13000   50000   25800    17600   38800   58000   24700      -8
## 1784       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 1785   675958   49700   37000  100000    77516   79000   97742   23000      -8
## 1786     3000      -9      -9      -9       -9    3000      -9      -9      -9
## 1787   219500   36000    8400   50000    11100   45000   30500    2500       0
## 1788   186900      -8      -8   13700    23000  116000   34200      -8      -8
## 1789       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1790       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1791   138000      -8      -9    5000    50000   50000   33000      -8      -8
## 1792       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 1793       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

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## 1794      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1795      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 1796      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1797   32000   10000   10000    4500    7500      -8      -8      -8      -8
## 1798  239100   92000   15700    9400   17800   51000   21900    3800      -8
## 1799  277380  133000   28200   25400   18100   13400   19900   12800       0
## 1800      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1801  162500      -8   15700    6200   66000   52000   15000    1500      -8
## 1802  142200      -8      -8   44300   15700   13300   38800    7300      -8
## 1803  112600   47000    7600   17500    9500   22200      -8       0       0
## 1804  467561  159000   43400   38200   32200  138000      -8   52761      -8
## 1805   39000       0      -9   12000    8000   12000      -8    2000      -8
## 1806      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1807  350500      -8   18300   38500   35700  123000   69000      -9      -8
## 1808      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1809  170686   35900   13400    4200       0   68000   31500    7400     486
## 1810  172500   39400   16200   12700   20000   56000   22400    2600      -9
## 1811      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1812      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1813   53300      -8   11000    2900    6900    2800   25300    1200      -9
## 1814      -9      -8      -8      -8      -9      -8      -8      -8      -9
## 1815  109866   23900   41900      -8      -9   12000   25400    4400    2266
## 1816      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1817      -9      -8      -8      -8      -9      -8      -9      -8      -8
## 1818  244000      -8   15000    5000   45000  150000       0   25000     500
## 1819  374037   92000   24500    6000   51000   49100   24600    2500       0
## 1820      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 1821   13500      -8   12000      -9      -9    1500      -9      -8      -8
## 1822    7000      -8      -8      -8    7000      -8      -8      -8      -8
## 1823      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1824  228700   46400   16900    9800   20500   33700   33000   17700       0
## 1825      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1826  106000      -9      -9      -9      -9      -9      -9      -9      -9
## 1827  191860   45800   17900     860   16000   67000   14400   11000      -8
## 1828  100500   50000      -9   10500      -9   30000      -8      -8      -8
## 1829      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1830      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1831      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1832      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1833  274400   15000   22000   48000   36000  120000   26400    3000      -8
## 1834      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1835      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1836  469171      -8   20000      -8      -9   10000   36000     600      -8
## 1837      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1838  117400      -8   20000    5100   36000   37000      -8    3000    6300
## 1839  179400   24000   43800   13200    8400   41600   42400    1600      -8
## 1840  156200   10800   10600   11300   37000   59000   21400    3500      -8
## 1841      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1842      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1843      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1844      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1845  207340   44300    9500    3800   28100   61000    6000   17200    3540
## 1846  134500   14000   12100    3200   30000   31000   27000    2500      -9
## 1847      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 1848   178500  112000      -8    3500   63000      -8      -8      -8      -8
## 1849       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1850       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1851   214000   48000   12000   18000   25000   62000   24000    5000      -9
## 1852    92970      -8      -9   21400   12300   37500   14800     870      -9
## 1853       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1854    24000   24000      -9      -9      -9      -9      -9      -9      -9
## 1855    60000      -8      -8      -8      -9      -8      -8      -8      -8
## 1856       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1857       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 1858       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1859       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1860   203900      -8   15500   37500   21500   35000   69000   13900      -8
## 1861   114100   12100    3900    4000   19100   44300      -8    4000      -8
## 1862       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1863   196716   31200   18900    7200   77516   14000      -9      -9      -9
## 1864   281500      -8   16000    6100   43600  112000   63000    4800      -8
## 1865       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1866    93000      -8   10000    6800   16100   27800   16800    1500      -8
## 1867   122685   11600   21400   18800   21700   13400      -9   10400     285
## 1868   213600      -9   12400   36800   28300   54000   37900    8500      -9
## 1869   107700      -9   10800   33500   21400   31000      -9      -9      -9
## 1870   237500  110000   13600   20100   21000   53000       0   17800       0
## 1871   343300   79000   32100   51000   58000  107000       0    6700       0
## 1872       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1873    94246       0    8400    3900   29000   27600      -9       0   10546
## 1874   136700       0   13500   14500   22400   61000   19400    1800       0
## 1875   169000   85000   43000    5000    5000    8000   10000    5000       0
## 1876       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1877   184900      -8    8700   34000   26500   77000      -8    2800      -8
## 1878   160200   81000   30000   20000   16000    4200      -8    9000      -8
## 1879   332742  100000      -8   75000   35000   15000   97742      -8      -8
## 1880       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1881       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1882       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1883       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1884   183700    9000   23000   11300   22600   52000   52000    3800      -8
## 1885     2400      -8      -8      -8      -9      -8      -8      -8      -8
## 1886   230300   45900   29800   25900   14100   58000      -8   19800      -8
## 1887    80000    5000   13000   51000      -9   11000      -8      -8      -8
## 1888    89900      -8   15000    2600    7300   35000   30000      -9      -8
## 1889   207400   16400   21400   20300   28800   36500   19600    1600      -8
## 1890       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1891   399789   58000   13700   60000    6600  111000   27900    6600    1689
## 1892       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1893       -9      -8      -8      -9      -9      -8      -8      -8      -8
## 1894   171700   38600   18500   11700   17000   15700   40000    3200   10000
## 1895   266500   38500   30200   23600   22000   45400   35200   16600      -8
## 1896   152000      -8   13000   20000   14000   50000   40000    5000      -8
## 1897       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1898    42300   34000      -8      -9    4500      -8      -8      -8      -8
## 1899       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 1900       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1901   161400      -9   13000    4000   35000  100000      -9      -9      -9
```

```
## 1902     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1903 331397     -8  23900  30100  29800  82000  97742  10500   4355
## 1904     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1905     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1906     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1907 417253 239253  40000  15000  10000  15000  25000  35000  10000
## 1908     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1909 123200  10200  14400  39600  17000  36000     -8     -8     -8
## 1910 294932  19800  61432  39500  30700  65000  36000  12500     -9
## 1911 303200  15800  10600   5500  30300 180000  35100   2400     -8
## 1912 281900  10500  45000  47900  52000  67000  40000  19500      0
## 1913     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1914     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1915  27700     -8     -8   4900  22800     -8     -8     -8     -8
## 1916  68500     -8   7300   6600   4300  28300  15300     -8     -8
## 1917  53300     -9  14000     -8  16800   2500  20000     -9     -8
## 1918 163800  18500  46000   6000  47000   4800  33000   6000     -8
## 1919 164500     -9  20000  26000  15500  61000  39000   3000     -9
## 1920 166600     -8  13000  35600  17500  41100  18300   4100      0
## 1921 218700  42000  19700  12600  25100  64000  38800   9300     -8
## 1922     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1923 169800  13000  32900  42400  35400  25800     -8   8400     -8
## 1924 331569  37200  61432  23900  63000  17800  42900   8000   2000
## 1925     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1926 104600  14600  26800  25200  12300  19100     -8     -8     -8
## 1927     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1928 224700  12900  20700   5000  19900 105000  34000   3400     -8
## 1929 337533     -8  11100  30000  57000 206096     -9     -9     -8
## 1930 211680  19300  11700  16900  11600  84000  56000    300    380
## 1931     -9     -8     -8     -8     -8     -8     -8     -8     -8
## 1932  81800  12000   4800  25000  10500  26500     -8     -8     -8
## 1933     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1934     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1935     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1936     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1937     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1938     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1939     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1940     -9     -9     -9     -9     -9     -9     -9     -9      0
## 1941     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1942     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1943     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1944     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1945     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1946     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1947     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1948     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1949     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1950     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1951     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1952     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1953     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1954     -9     -9     -9     -9     -9     -9     -9     -9     -9
## 1955     -9     -9     -9     -9     -9     -9     -9     -9     -9
```

```
## 1956      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1957      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1958      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1959      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1960      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1961      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1962      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1963      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1964      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1965      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1966      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1967      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1968      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1969      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1970      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1971      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1972      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1973      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1974      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1975      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1976      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1977      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1978      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1979      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1980      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1981      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1982      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1983      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1984      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1985      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1986      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1987      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1988      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1989      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1990      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1991      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1992      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1993  655000      -9      -9  436000  219000      -8      -8      -9      -9
## 1994      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1995      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1996      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1997      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1998      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 1999      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2000      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2001      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2002      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2003      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2004      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2005      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2006      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2007      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2008      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2009      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2010      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2011      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2012      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2013      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2014      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2015      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2016      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2017      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2018      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2019      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2020      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2021      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2022      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2023      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2024      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2025  684800  160000   37600   71000   71000  225000   93000       0       0
## 2026      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2027      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2028  548300      -8   35000  185000   47800  200000   45000   15500      -8
## 2029      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2030      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2031      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2032      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2033      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2034      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2035      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2036      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2037  255500   79000   38400   62000   49000      -9   24000      -9      -9
## 2038      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2039  384300   78000  120000   14500  123000   32000      -8   12300      -8
## 2040      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2041      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2042      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2043      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2044      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2045      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2046      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2047      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2048      -9      -9      -9      -9      -9      -9      -8      -9      -8
## 2049   88200   23900      -9    2400   22200   12200   21000      -9      -8
## 2050      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2051      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 2052      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2053  269000  181000      -9      -9   88000      -9      -9      -9       0
## 2054      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2055      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2056      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2057      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 2058      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2059      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2060      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2061      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2062      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2063      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2064      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2065  285000      -9  225000      -9      -9      -9      -9      -9      -8
## 2066  440500      -8   47000   23000   78000  110000      -8  149000    7500
## 2067      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2068      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2069      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2070      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2071 1120300      -8  299000   90000  352000   52000  225000   21700      -8
## 2072      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2073      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2074    2100    2100      -9      -9      -9      -9      -9      -9      -9
## 2075      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2076      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2077      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2078      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2079      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2080      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2081      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2082 3554740      -8  182000   35400  759371 1180000 1148153  166000   48216
## 2083      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2084      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2085      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2086      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2087      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2088      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2089      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2090      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2091      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2092      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2093      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2094      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2095      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2096      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2097      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2098      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2099      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2100      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2101      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2102      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2103      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2104      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2105      -9      -8      -8      -8      -9      -9      -9      -9      -9
## 2106      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2107      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2108 1337247  358000  104000  302000  189000  161000   87000   49600   53747
## 2109  672000      -8      -9  240000      -9  120000      -9      -9      -9
## 2110      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2111      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2112      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2113      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2114      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2115      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2116      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2117      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2118  1472100      -9  297000  225000   240000  574000      -9   42100      -9
## 2119   401000  168000   45000   19000    59000   36000   42000      -8   20000
## 2120       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2121       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2122       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2123       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2124       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2125       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2126       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2127       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2128       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2129       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2130       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2131       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2132  2820000  542000  405000   45000   306000  528000  329000  148000       0
## 2133       -9      -8       0       0        0       0       0       0       0
## 2134   355600   92000   27400   16100    32800  126000   39600    1300      -8
## 2135       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2136       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2137       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2138       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2139   218800   50000   43700   19700    26300   18900   39100    6000       0
## 2140       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2141       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2142       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2143       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2144       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2145       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2146       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2147       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2148       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2149       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2150       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2151       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2152       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2153       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2154       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2155       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2156       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2157       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2158       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2159       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2160       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2161   184500      -9      -9   19200   133000      -9      -9      -9      -9
## 2162  1666900  195000  110000   52000   103000 1100000  102000    4900      -8
## 2163  2611371 1020000  277000  118000   759371  287000      -8   52000      -8
## 2164       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2165  4066017      -8  183000  157000   395000 1602217   67000  101000   47000
## 2166       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2167       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2168       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2169       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2170       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2171       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2172   317900   77000   24800   38300    36400   84000      -8   35200      -8
## 2173       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2174       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2175       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2176       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2177       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2178   231046      -9   81000   24600    96000      -9      -9      -9   11146
## 2179       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2180       -9       0       0       0        0       0       0       0      -8
## 2181       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2182       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2183  1515000  467000  285000   25000   121000  213000      -8  153000    6000
## 2184       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2185       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2186       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2187       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2188       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2189    40000      -9      -9      -9       -9   40000      -9      -9      -9
## 2190  1376400  457000  174000   40300   153000  292000  159000      -8   38100
## 2191       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2192       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2193       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 2194       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2195       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2196       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2197  1583214  474000  153000   59000   208000  265000  195000  114000   16214
## 2198       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2199       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2200       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2201       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2202       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2203       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2204       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2205       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2206  2447000 1020000      -8  247000   393000  560000      -8   65000   60000
## 2207  1976350  468000  130000  486000   205000  266000  175000   17200   12150
## 2208       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2209       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2210       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2211   876000      -9      -9  426000   231000  107000      -9    4300      -9
## 2212  2053000  753000  369000   92000   131000  312000  240000   48000      -9
## 2213       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2214   417000      -8   73000      -9       -9  118000  149000    4000      -8
## 2215   198500      -8      -9    9100    53000   79000      -9      -9      -8
## 2216       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2217       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2218       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2219       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2220       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2221       -9      -9      -9      -9       -9      -9       0      -9      -9
## 2222       -9      -8      -8      -9       -8      -8      -8      -8      -8
## 2223       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2224       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2225       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2226      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2227      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2228      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2229      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2230      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2231      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2232      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2233 1514000  451000  318000   46000  305000  141000  191000   14000      -9
## 2234      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2235      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2236      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2237      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2238      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2239      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2240      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2241      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2242      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2243       0       0       0       0       0       0       0       0       0
## 2244      -9       0       0      -8      -8       0       0       0       0
## 2245      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2246      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2247      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2248      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2249      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2250      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 2251      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2252      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2253      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2254      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2255      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2256      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2257      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2258      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2259      -9      -9      -9      -9      -9      -9      -9      -9       0
## 2260      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2261      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2262      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2263      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2264      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2265  424800      -9      -9   26100  148000      -9      -9    2700      -9
## 2266      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2267  395000  100000   91000   18100   68000   39700   50000    3500      -8
## 2268      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2269      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2270      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2271      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2272      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2273      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2274      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2275      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2276      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2277   40000      -9      -9      -9      -9      -9      -9      -9      -9
## 2278      -9      -8      -9      -8      -9      -8      -8      -8      -8
## 2279      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

```
## 2280       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2281       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 2282       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2283       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2284       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2285       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2286       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2287  1979153      -9  240000   70000   120000  256000 1148153      -8   42000
## 2288       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2289    36500      -8      -8      -8       -9      -8      -8      -8      -8
## 2290       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2291       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2292  2180500  920000  151000   73000   328000  329000  331000    8300    9000
## 2293       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2294       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2295       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2296       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2297       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2298       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2299       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2300   383800      -8      -8   38000    73000  218000      -8    1800      -8
## 2301       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2302       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2303       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2304       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2305       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2306       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2307       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2308       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2309       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2310       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2311       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2312       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2313       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2314       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2315       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2316       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 2317       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2318       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2319       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2320       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2321  2687217      -8  266000   75000   380000 1602217  264000      -8      -8
## 2322       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2323       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2324       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2325       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2326  1808950  837000  127000  102000   257000  290000      -8   10000   40950
## 2327       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2328       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2329       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2330       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2331       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2332       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2333       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2334       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2335       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2336   205100      -9      -9   23800   140000      -9      -9   24500      -9
## 2337  1067000  271000  299000   86000       -9  411000      -8      -8      -8
## 2338       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2339       -9      -8      -8      -9       -8      -8      -8      -8      -8
## 2340  2669046  800000  149000   87000   438000  320000      -8      -8   16000
## 2341       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2342        0       0       0       0        0       0       0       0       0
## 2343    32000      -8      -8      -9       -9      -8      -8      -8      -8
## 2344  1282201  516000  132000   46300    60000  123000  141000    7100    7801
## 2345       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2346       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2347       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2348   547900  300000      -8   15900    33000   52000   90000      -8    5000
## 2349       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2350       -9      -9      -9      -9       -9      -9      -9      -9      -8
## 2351       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2352       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2353       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2354  8292116 4105867  655222  847124   759371  723000      -8  102000  293486
## 2355  3539054  369000      -9   61000    73000  237000  138000   56000      -9
## 2356       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2357       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2358       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2359       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2360       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2361  1313200      -9  133000  134000   402000   24400  400000   87000      -9
## 2362       -9      -9      -9      -9       -9      -9      -9      -9       0
## 2363       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2364       -9      -8      -8      -8       -8      -8      -9      -8      -8
## 2365       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2366  1344000  593000  128000   85000   180000   90000  193000      -9    6000
## 2367       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2368       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2369       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2370       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2371       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2372   684605      -8   42200   19500   110000   63000  149000    9000    1605
## 2373       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2374  1110000  296000  188000   60000    85000   88000  227000   27000      -8
## 2375       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2376       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2377       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2378       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2379    77000   40600      -8      -8       -8      -8      -8   25000      -8
## 2380       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2381       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2382  1570012  526000  156000   63000   209000  210000      -8   35100   39212
## 2383       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2384       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2385   105920   59000    8100   16300    11400    8500      -8     120      -8
## 2386       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2387       -9      -8      -9      -8       -8      -8      -8      -8      -8
```

```
## 2388       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2389       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2390  1590313  633000  141000   24400   150000  277000  216000   36300   45613
## 2391       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2392       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 2393       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2394       -9      -9      -9      -9       -9      -9      -9      -9      -8
## 2395       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2396        0       0       0       0        0       0       0       0       0
## 2397   805600   64000   69000   55000   276000  150000      -8  150000      -8
## 2398       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2399       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2400   626100   63000   44500   61000   126000  198000  103000    3000      -8
## 2401       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2402       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2403       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2404       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2405       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2406       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2407       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2408       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2409       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2410       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2411   774900  100000   28000   75000    38000  289000  140000   42600      -8
## 2412       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2413       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2414       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2415       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2416       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2417  6740630 4105867   44500  212000   112000 1602217      -8   79000   57000
## 2418  2153622  400000  655222  129000   215000  384000      -9    4400      -9
## 2419       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2420       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2421       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2422       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2423  1380400  412000  130000   37400   158000  385000      -9    2000      -8
## 2424       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2425       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2426       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2427       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2428       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2429       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2430       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2431       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2432       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2433       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2434       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2435  1642100  132000  203000  144000   130000  625000  100000   51000   52100
## 2436       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2437   627900  307000      -8      -8       -8  133000  161000   13200      -8
## 2438       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2439  1241500  485000  170000   99000   147000   99000  146000   46000      -9
## 2440       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2441       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2442  1673600  521000  147000   45500    53000  136000  205000   48700    3400
## 2443       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2444       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2445       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2446       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2447       -9      -9      -9      -9       -9      -9      -9      -8      -9
## 2448  2060800  779000  308000      -9       -9  100000  211000      -9   36000
## 2449       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2450       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2451       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2452       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2453       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2454       -9      -9      -8      -8       -8      -9      -8      -9      -8
## 2455       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2456       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2457       -9      -9      -9      -8       -8      -8      -8      -8      -8
## 2458       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2459       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2460       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2461   692500  215000   60000   67000    82000  129000  104000   10000      -8
## 2462       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2463       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2464       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2465   554050  248000      -9   11200    98000  133000      -8   14700   14650
## 2466       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2467  1766900  961000  150000  162000   379000      -8      -8      -9   72900
## 2468       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2469       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2470       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2471       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2472       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2473       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2474  6142313 4105867  432000  414000   248000  555000      -8   28400      -8
## 2475  1877883  319000   83000  130000   126000  300000  111000  771944    6039
## 2476       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2477       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2478       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2479       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2480       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2481       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2482       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2483       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2484       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2485  1374400 1140000       0   98000    59000   47400       0       0       0
## 2486   163500      -9      -9      -9       -9   50000      -9      -9   73000
## 2487       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2488       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2489   582420   52000   37900   39800    43100  271000  100000      -8    6020
## 2490       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 2491       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2492       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2493   655000      -9  100000   85000   100000  340000      -9      -9      -9
## 2494  1287000  416000   52000   97000   218000  180000  132000   22000      -8
## 2495   336934  116000   66000   40900    53000   27000      -8    6600    3134
```

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## 2496       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2497       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2498  1245780  458000   30800   66000   152000   83000  249000   83000   14880
## 2499       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2500       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2501       -9      -9      -9      -8       -8      -9      -8      -9      -9
## 2502       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2503    18200      -9      -9      -9       -9      -9      -9      -9      -9
## 2504       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2505       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2506    64000      -9    7000      -9       -9      -8   57000      -8      -8
## 2507   403400   33000      -9      -8   360000      -9      -9      -9      -8
## 2508    22000      -9      -9      -9       -9   10000      -9      -9      -8
## 2509  2887796  755000  371000   62000   192000  862000  248000   70000  214796
## 2510       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2511   517500  181000   20200   32900   126000   52000   42800    5300   13600
## 2512   982814  295000  152000   84000   121000  147000  130000   13800    3514
## 2513       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2514       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2515       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2516       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2517       -9      -9      -9      -9       -9      -9      -8      -9      -9
## 2518       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2519       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2520       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2521       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2522       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2523       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2524   465600  158000   33900   14300    72000   47700   66000   13500   26400
## 2525       -9      -9      -9      -9       -9      -9      -9      -9      -8
## 2526       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2527       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2528       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2529   831773      -8   71000   97000    45700  298000  164000   46700   34373
## 2530       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2531       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2532       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2533       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2534       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2535       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2536       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2537       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2538    12000      -9      -9      -9       -9      -9      -9      -9      -9
## 2539   286000      -9  100000  104000       -9   20000   41000   12000      -9
## 2540       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2541       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2542  1357682  375000  129000  211000    43100  194000  168000   40800   22782
## 2543  1097812  227000  167000  235000       -9  202000      -8   65000   91812
## 2544       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2545       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2546       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2547   855088  283000  109000   17100   273000   77000      -8   12100   13888
## 2548       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2549       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2550   374000  182000      -9      -9       -9      -9      -9      -9  192000
## 2551       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2552   553436  188000   57000   14200    66000   99000   31700   90000    7536
## 2553       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2554       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2555       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2556       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2557       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2558       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2559       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2560       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2561       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2562       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2563       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2564       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2565       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2566       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2567       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2568       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2569   254212      -8   15600   12000    56000   68000   72000    6700    1912
## 2570       -9      -9      -9      -9       -9      -9      -9      -9      -8
## 2571       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2572       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2573       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2574   496067      -8   34300   57000    41500  134000   91000   93000    8267
## 2575       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2576       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2577       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2578       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2579       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2580       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2581       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2582       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2583       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2584       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2585       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2586       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2587       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2588       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2589       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 2590  1206000  363000  179000   21000   100000  323000  127000      -8   13000
## 2591  1629100  402000  298000  370000   128000  200000  170000    5100   16000
## 2592       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2593   915000  391000  262000   75000   187000      -9      -9      -9      -9
## 2594       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2595  1350629  309000  302000  237000   115000  144000  105000      -9  123529
## 2596       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2597       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2598       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2599       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2600       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2601       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2602   505000      -9   99000   80000   150000   87000      -9      -9   49000
## 2603       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2604       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2605  1315640  446000  298000   74000       -8  333000  113000      -8   14640
## 2606  1678210  425000  157000   64000   214000  510000  199000   24700   17510
## 2607       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2608  1136965  232000   28700  283000   169000  370000      -9   17400    4065
## 2609       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2610       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2611       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2612       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2613       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2614       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2615    40000      -9      -9      -9       -9      -9      -9      -9      -9
## 2616   687300  377000  260000      -9       -9      -9      -8    6300      -8
## 2617  1066000  200000  242000  330000       -9   83000  112000      -8   20000
## 2618  2213000  358000  250000   75000       -9  511000  150000    5000      -8
## 2619   811160  380000  177000   20700    47300   18600  122000     460      -8
## 2620       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2621       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2622       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2623  1267900  476000  160000  135000       -8  305000  106000      -8      -8
## 2624       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2625       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2626       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2627       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2628       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2629       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2630       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2631       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2632       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 2633       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2634       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2635       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2636  1205500  526000   54000   54000    41600  196000  138000    5200      -8
## 2637       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2638       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2639       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2640       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2641       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2642  1805000  373000  249000   27000   218000  652000  266000   20000      -8
## 2643       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2644       -9      -9      -8      -8       -8      -8      -9      -8      -8
## 2645       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2646       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2647       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2648   934963      -9      -9  328000   451000      -9      -8   44700   86363
## 2649  7060020 4105867  178000  412000   117000  899000 1148153   58000      -8
## 2650  1170000      -9   86000  317000    79000  559000      -8  129000      -8
## 2651  1056200   77000  205000  100000    26700  480000   80000   39700      -9
## 2652       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2653       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2654       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2655   804900      -8  332000   73000   122000  231000      -8      -8      -8
## 2656       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2657  1106900  490000  140000   47900       -9  187000  175000      -9      -9
```

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## 2658  1902587  506000  655222   42200   187000  123000  123000   15600    9565
## 2659   546000      -9  125000      -9       -9  342000      -9      -9      -9
## 2660   413000      -9  413000      -8       -9      -9      -9      -9      -9
## 2661  4099240 1290000  397000  288000   759371  192000  290000  771944   57925
## 2662       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2663  7453365 4105867  655222  287000   759371  969000  513000    5400   49505
## 2664       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2665       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2666       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2667       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2668       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2669       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2670       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2671       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2672       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2673       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2674       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2675       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2676  1234000  438000  313000   18000   180000   88000  137000   15000      -8
## 2677       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2678       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2679       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2680   588000  160000      -8   18000   320000      -8      -8      -8      -8
## 2681       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2682       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2683       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2684       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2685       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2686       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2687       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2688       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2689       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2690       -9      -8      -8      -9       -8      -8      -8      -8      -8
## 2691       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2692       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2693       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2694       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2695       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2696       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2697       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2698       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2699       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2700       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2701       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2702       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2703       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2704       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2705       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2706       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2707       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2708       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2709       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2710       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2711       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2712       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2713       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2714       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2715       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2716       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2717       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2718       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2719       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2720       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2721       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2722       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2723       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2724   224000      -9      -9   84000       -9  110000      -9      -9    6000
## 2725       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2726       -9      -8      -9      -8       -8      -8      -8      -8      -8
## 2727  2665000  600000  400000  325000   200000  315000      -8  160000  140000
## 2728       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2729       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2730  2548054      -9      -9      -9       -9      -9      -9      -9      -9
## 2731       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 2732       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2733       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2734       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2735   321466   37700   28300   17600    40900  121000   49500    8000   18466
## 2736   337728      -8      -9   31500       -9   66000  176000      -9   13228
## 2737       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2738       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2739       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2740       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2741       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2742       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2743       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2744    15000      -9      -9    9000     6000      -9      -9      -9      -9
## 2745       -9       0       0      -9       -9      -9      -9      -9      -9
## 2746       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2747       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2748       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2749       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2750       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2751       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2752       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2753       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 2754       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2755       -9      -9      -9      -9       -9      -9      -8      -9      -9
## 2756       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2757       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2758       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2759       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2760       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2761       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2762       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2763       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2764   219000      -9      -9    7000    75000  110000      -9      -9      -9
## 2765       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2766       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2767       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2768       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2769       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2770  3503684  433000  160000   43300   246000 1602217  179000   80000  596367
## 2771       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2772       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2773   251796      -8      -9      -9   148000   55000      -8      -9   14796
## 2774       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2775       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2776       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2777       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2778       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2779       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2780       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2781       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2782       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2783   548300      -9   61000   63000    62000  109000      -9   13000      -9
## 2784       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2785       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2786       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2787       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2788       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2789       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2790       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2791       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2792       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2793       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2794       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2795       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2796       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2797   638400  210000   68000   24400    38000   71000   97000   73000    2500
## 2798       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2799       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 2800       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2801   628000  146000   79000  147000   112000   42500   79000   17100      -8
## 2802       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2803       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2804   469243  143000   52000      -8    50000   69000   48500   17300    2743
## 2805       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2806       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2807       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2808       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2809       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2810       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2811       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2812       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2813       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2814       -9      -8      -8      -9       -9      -8      -9      -9      -9
## 2815       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2816       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2817       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2818       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2819       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 2820       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2821       -9       0       0      -8        0       0       0       0      -8
## 2822  2880040  335000  655222  324000   255000  414000  277000  113000  211818
## 2823       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2824  1634000  500000  128000   70000   214000  372000  160000      -9  109000
## 2825       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2826       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2827   908000  250000   44000  330000    70000   92000   70000   12000      -8
## 2828    60000      -9      -9      -9    60000      -9      -9      -9      -9
## 2829       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2830       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2831       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2832       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2833       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2834       -9      -9      -9      -9       -9      -9      -9      -9      -9
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## 2836       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2837       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2838       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2839       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2840       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2841       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2842   304500      -8      -9   15500   128000  161000      -8      -9      -8
## 2843       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2844       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2845   558700  181000  143000   16500    44000   96000   28800   28400      -8
## 2846       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2847       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2848       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2849       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2850   935900  168000   59000  115000   162000  299000   76000   24000   32900
## 2851       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2852       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2853       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2854       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2855       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2856       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2857       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2858       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2859       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2860       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2861       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2862       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2863       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2864       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2865       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2866       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 2867       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2868   787799      -9      -9   29600    18000  733000      -9      -9    7199
## 2869       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2870       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2871       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2872       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 2873       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

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## 2874      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2875      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2876      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2877    6000      -9      -9      -9      -9      -9      -9      -9      -9
## 2878      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2879      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2880  831224  354000   16300  100000  101000   61000      -8  105000    1924
## 2881      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2882      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2883      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2884      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2885      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2886      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2887      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2888      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2889      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2890      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2891      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2892      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2893      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2894      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2895      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2896      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2897      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2898      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2899      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2900      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2901  130000  120000      -9      -9      -9   10000      -9      -9      -9
## 2902  886832  403000  104000   31700  128000  193000      -8      -8   15132
## 2903      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2904      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2905      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2906      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2907      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2908      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2909      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2910      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2911      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2912      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2913      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2914      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2915      -9      -9      -9      -9      -9      -9      -9      -9      -9
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## 2917      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2918      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2919      -9      -9      -9      -9      -9      -9      -9      -9      -9
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## 2922      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2923      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2924      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2925      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2926      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2927      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

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## 2929      -9      -9      -9      -9      -9      -9      -9      -9      -9
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## 2931      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2932      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2933      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2934      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2935      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2936      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2937      -9      -8      -8      -8      -8      -8      -8      -8      -8
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## 2939      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2940      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2941      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2942      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2943      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2944      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2945      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2946      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2947      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2948      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2949      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2950      -9      -9      -9      -9      -9      -9      -9      -9      -9
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## 2952      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2953      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2954      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2955      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2956      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2957      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2958      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2959      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2960      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2961      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2962      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2963      -9      -9      -9      -9      -9      -9      -9      -9      -9
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## 2966      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2967      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2968      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2969      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2970      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2971      -9      -8      -8      -8      -8      -8      -8      -8      -8
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## 2973      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2974      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2975      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2976      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2977      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2978      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2979      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2980      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2981      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

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## 2982      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2983      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2984      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2985      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2986      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2987      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2988      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2989      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2990      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2991      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2992      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2993      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2994      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2995      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 2996      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2997      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2998      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 2999 5883349 1280000  259000  847124  463000 1602217 1148153   20200  131655
## 3000      -9      -8      -9      -9      -9      -8      -8      -8      -8
## 3001      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3002      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3003      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3004      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3005      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3006      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3007 1680653       0   62000  363000   45800   13600 1148153    2100       0
## 3008      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3009      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3010      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3011      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3012      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3013      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3014 2087097      -8      -8   60000   40000   67000 1148153  771944      -8
## 3015   18900       0      -9      -9      -9      -9   18900      -9      -9
## 3016      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3017      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3018      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3019      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3020      -9      -8      -8      -8      -8      -8      -9      -8      -8
## 3021      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3022      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3023  425000      -9   23000   19000   55000  170000   35000   96000      -8
## 3024      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3025      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3026      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3027      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3028      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3029      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3030      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3031      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3032      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3033      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3034      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3035      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

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## 3036      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3037 1950000  822000  144000  272000  100000  518000       0   15000       0
## 3038      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3039      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3040      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3041      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3042      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3043   98000      -9      -9      -9      -9   98000      -9      -9      -9
## 3044  179200   52000   21500      -8   38400   43000      -8   10000    6600
## 3045      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3046      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3047      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3048      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3049      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3050      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3051      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3052 2076768  686000  215000  108000      -9  624000  235000   28300   52568
## 3053      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3054      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3055      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3056       0       0       0       0       0       0       0       0       0
## 3057      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3058      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3059  915700  369000      -8   38000  249000  208000      -9   15500      -9
## 3060      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3061      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3062      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3063      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3064      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3065      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3066      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3067      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3068      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3069      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3070      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3071      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3072  846000      -9      -9   96000      -9      -9      -9      -9      -9
## 3073      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3074      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3075      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3076      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3077      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3078      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3079      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3080      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3081      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3082 2094900  161000  108000   54000  426000 1150000  148000   47900      -8
## 3083      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3084      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3085 2548054      -9      -9      -9      -9      -9      -9      -9      -9
## 3086      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3087      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3088      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3089      -9      -8      -8      -8      -8      -8      -8      -8      -8
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## 3090      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3091      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3092  178000      -8  120000      -9      -9      -9      -9      -9      -8
## 3093      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3094      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3095      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3096      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3097      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3098      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3099      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3100      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3101      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3102      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3103      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 3104      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3105   95430      -9      -9      -8      -9      -9      -9      -9   13430
## 3106      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3107      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3108      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3109 3253000  838000  181000  248000  207000 1140000  522000   64000      -8
## 3110      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3111      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3112      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3113      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3114      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3115 1918853  238000  163000  202000  127000  778000  224000    8200   59153
## 3116      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3117      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3118      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3119      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3120      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3121      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3122      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3123      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3124      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3125      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3126      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3127      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3128      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3129      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3130      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3131      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3132      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3133      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3134      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3135      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3136      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3137      -9      -9      -9      -8      -9      -9      -9      -9      -9
## 3138      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3139      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3140      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3141  509900   70000  172000   25000   26300  166000   32400    8400      -8
## 3142      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3143      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

```
## 3144      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3145  292500  100000   10000   10000   75000      -8   75000      -8    7500
## 3146      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 3147      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3148      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3149  666164  211000   89000   42000   42000   67000   52000    1500    5064
## 3150      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3151  174371   41400   21500   11100   10600   22400   17800   21800    2271
## 3152  452600   61000   22600    6000    6000   65000  258000   10000      -9
## 3153      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3154      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3155      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3156      -9      -9      -9      -9      -9      -9      -9      -9      -8
## 3157      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3158  405000  185000   20000   18500   14000   75000   45000   10000   17500
## 3159  714997  164000   52000    9100   38000  144000   47600   15600    6197
## 3160      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3161      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3162  177600   27300   25400    3600   25300   11000   36100    6600       0
## 3163      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3164      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3165      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3166      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3167      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3168      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3169  125000   50000   10000   35000   20000      -8      -8      -8   10000
## 3170      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3171      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3172      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3173      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 3174      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3175   68760   19000    8900    8300      -9   16200   11900     960      -8
## 3176  115000   25000   15000   15000   25000      -8   15000   10000      -8
## 3177      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3178      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3179      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3180      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3181      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3182      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3183  142500   30000      -8    2500   50000   40000      -8   10000      -8
## 3184      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3185  492823  105000   73000    7400   72000   99000   89000    4100    1023
## 3186       0       0       0       0       0       0       0       0       0
## 3187      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3188  270700   96000      -8    8200   14500      -9   53000      -9      -8
## 3189      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3190      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3191  360900  199000    5900      -8    9700  120000   10100      -8      -8
## 3192  491117  166000   27800   13100   58000  119000   43800      -8   25217
## 3193      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3194  507665  138000   79000   13500   35500   67000   66000   16300    2365
## 3195  569000  175000   57000   19000   32000  142000   53000   18000   44000
## 3196  613929  180000   38500   10000   35800  150000   46300   16700  136629
## 3197      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

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## 3198      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3199      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3200      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3201      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3202      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3203      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3204  892190  437000   12700   18100   93000  288000      -9     490      -9
## 3205      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3206  808400   85000   90000  148000   49500  164000  181000   30900      -8
## 3207  313800      -8   22600   10700   55000   84000  121000      -8      -8
## 3208  320700  135000   53000   11400   20500   23800   21500   30000      -8
## 3209      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3210      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3211      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3212      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3213      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3214      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3215      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3216  609000  107000  100000  143000      -9  154000   29000   26000      -8
## 3217      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3218      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3219      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3220  477100       0   42600   16800   29800  128000  145000   72000      -8
## 3221      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3222  356000   80000    7000   74000   52000   31400   46500    4200      -8
## 3223      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3224  497200  148000  138000   14200      -8  149000   24000      -8      -8
## 3225 1040000  250000  190000   80000   90000   70000   70000   80000   50000
## 3226  545116  129000   56000   23500   45100  123000   56000   11200    1816
## 3227      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3228  189250   72000   20000    6300   35600   26500    9700     550      -8
## 3229      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3230  312458   56000   18200   15900    8300   35000   29000   33000     458
## 3231      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3232  336655  105000   29200   46800   22200   75000   42200    7100    5655
## 3233      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3234  137500   30800   11300    5400    9800   17800   10400      -9       0
## 3235      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3236  518000   70000   26300   52000   39000  145000   82000    1600       0
## 3237      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3238      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3239  399577      -8   39500   28300   10700   43000   30100   12700   12077
## 3240  407400      -9   10000   14400   15000  290000   24000   18000      -9
## 3241      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3242      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3243      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3244      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3245  591300   88000   56000   17300  150000  150000  110000      -8      -8
## 3246      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3247      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3248 3143624  302000  193000  847124  218000  633000  155000   18500      -8
## 3249      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3250      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3251 1735095  347000  101000  168000  235000  435000  148000   18600   31695
```

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## 3252      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3253      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3254      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3255      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3256  193200   62000   12200   21500   20600   37800   22500    7400      -8
## 3257      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3258      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3259      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3260      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3261  382900   76000   30400   11000   57000  121000   40000    7500    9000
## 3262      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3263      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3264  782200      -9  187000   27500  167000  304000   43000    1700   16000
## 3265  464119  148000   81000   25000   47600   62000   40000   11100     419
## 3266  296100  115000   23800   51000    9300   97000      -8      -8      -8
## 3267  403953  121000   39800   29500   51000   47500       0   28100   24353
## 3268  853100  291000   61000   77000   68000  190000   80000   30100      -9
## 3269  464700  143000    5900   83000   49600  137000   14600   16600       0
## 3270      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3271  777100  137000   71000   81000   77000  241000  130000   25100      -8
## 3272      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3273 1489000  923000  333000      -8      -8  136000      -8      -8      -8
## 3274  109000      -8      -8   50000   49000      -8      -8      -8      -8
## 3275   10000      -8      -8      -8      -8      -8      -8      -8      -8
## 3276      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3277      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3278      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3279      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3280  340100  114000   37000  145000   44100      -8      -8      -8      -8
## 3281  655500  235000   37500  104000   42700  133000   56000    2600       0
## 3282      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3283      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3284  596500  142000   54000   58000   59000  108000   98000    5500      -8
## 3285      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3286 4605867 4105867  130000      -8      -8  200000  170000      -9      -8
## 3287      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3288      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3289      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3290      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3291  281100  140000      -9   32400   78000      -9       0    3000    2500
## 3292      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3293 1261222  180000  655222   84000  130000  140000      -8   20000      -8
## 3294      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3295      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3296      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3297      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3298      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3299      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3300      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3301      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3302  628300  127000   75000   22100   77000  151000   49500   10000       0
## 3303      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3304  760000  137000   55000   81000   77000  240000  130000   25000       0
## 3305 1635300  278000  153000  241000  126000  308000  208000   31200       0
```

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## 3306   331000  210000      -8      -8       -9   25000      -8      -8      -8
## 3307       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3308   468000  300000   55000   20000    50000   10000   10000   10000    5000
## 3309       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3310       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3311   611000  200000   53000   51000    58000  200000   10000   19000       0
## 3312       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3313       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3314       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3315       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3316  7053921 4105867   46000  139000    22000   25000      -8   90000      -8
## 3317  1287300  328000   61000   74000    62000  381000      -8   97000  149000
## 3318       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3319   202600      -8      -8   27600    96000      -8      -8      -8   62400
## 3320   239200   57000   62000    2800     2200   33500   46500    7900      -8
## 3321   387000   22400   41100   60000    50000  133000   79000    1500      -8
## 3322   414404       0   51000   36700    33900   52000   54000   35000    3804
## 3323   407500  111000   14100   22700    36400  105000   59000   18600      -8
## 3324   174000   35000   50000   10000    10000   50000   10000      -8    5500
## 3325       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3326    92000      -8   20000   25000    15000   10000   15000    4000      -8
## 3327       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3328  1555900  446000  143000  314000    72000  247000  126000   32400      -8
## 3329       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3330       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3331   495267  177000   58000   94000     8100   34000   44900   33200    4567
## 3332       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3333       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3334       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3335   918200  235000  110000   84000   136000  178000  133000   40000      -8
## 3336   737730   96000       0   39000    33800   89000   43400   15300    3830
## 3337       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3338   474500      -8       0   97000        0   89000  112000   30000  115200
## 3339       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3340       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3341   720950  268000   93000   14300    61000  140000   70000    3500   10150
## 3342       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3343   558400      -8   29700   75000    36400   57000   69000    5900    1000
## 3344       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3345       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3346       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3347       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3348       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3349   735000  110000   62000  167000       -9  182000  182000   32000      -8
## 3350       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3351       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3352       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3353  1138375  350000   76000  128000    45900  251000   79000   20100   35275
## 3354  2262200  369000  174000  193000   122000 1000000      -9   72000  288000
## 3355   326300   49600   23700   35800    22900  112000   39200   33100      -8
## 3356   332500   88000   14000   38800    16700   97000   29400   16800      -8
## 3357       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3358       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3359   556142   99000   47600   25100    12300  174000   71000   15800    3742
```

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## 3360   568136  146000   31100   71000    4700   43100   36400   19300  123136
## 3361   819064   97000   50000  113000   87000  132000   69000   99000   37164
## 3362   574520   93000  111000   28100   44600  147000   86000   40100    1920
## 3363       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3364  1011000  228000   77000  102000   74000  322000  188000   10000       0
## 3365       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3366       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3367       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3368   200000      -9      -9      -9      -9      -9      -9      -9      -9
## 3369       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3370       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3371       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3372       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3373       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3374       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3375       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3376       -9      -8      -8      -9      -9      -8      -8      -8      -8
## 3377       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3378  1806000  388000  305000   79000  164000  404000  151000       0       0
## 3379       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3380   276000   65000   36000   25000   18000   78000   28000   13000       0
## 3381   566500  103000   28000   67000   33800  121000   73000   19700      -8
## 3382  1843100  278000  122000  337000  170000  717000  158000   27800       0
## 3383       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3384    67000   25000   10000   15000    7000    5000      -8      -8      -8
## 3385       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3386       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3387   252000  163000   48000      -8      -9   20000      -8      -8    6000
## 3388       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3389       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3390  1316100  227000   75000  112000   63000  496000  114000   29400      -8
## 3391       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3392       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3393       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3394       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3395       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3396       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3397   437000   98000   35000   32000   28000  126000   81000   17000      -8
## 3398       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3399       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3400  1104300      -8  310000   49300  155000  152000  186000      -9      -8
## 3401       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3402  1080070   71000   76000   70000  224000  422000  116000   31100   36670
## 3403       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3404  1255600  272000  106000  135000   49600  344000  148000   41000      -8
## 3405  2415046  514000      -8  504000  193000  727000  172000   46000      -8
## 3406  1069000  255000  177000  224000      -9  220000   92000   37000      -8
## 3407   323500  173000    7700   19000   39000   13200   59000    3200       0
## 3408       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3409       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3410       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3411       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3412       -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3413       -9      -8      -8      -8      -8      -8      -8      -8      -8
```

```
## 3414      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3415      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3416      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3417 1077000  295000   85000  230000   50000  175000  125000   25000      -8
## 3418   52200      -9      -9   33300      -9   18900      -9      -9      -9
## 3419      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3420      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3421      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3422 1643900  539000  113000   79000  138000  343000  210000  118000   25900
## 3423      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3424      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3425      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3426      -9      -8      -8      -9      -9      -8      -8      -9      -8
## 3427      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3428      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3429  500000  135000   40000   60000   58000   53000   53000   18900      -8
## 3430      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3431      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3432  589683  141000   43800   17200   48100  120000   44400   38500   37983
## 3433      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3434  606088  156000  103000   18600   61000   80000   69000    3900    9788
## 3435      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3436  639293  181000  139000   20600   30100   26400   70000   10000  118093
## 3437  585000  100000   72000   80000   20000  190000   50000   20000   12000
## 3438 2818000      -8  240000  530000   55000 1010000  332000   91000      -8
## 3439  541000  143000   47700   15000    8400  284000      -8    2900      -8
## 3440      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3441      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3442 1182400  226000  298000  104000   46000  164000  121000   96000      -8
## 3443   32200      -9      -9      -9      -9      -9      -9      -9      -9
## 3444      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3445 1218000  330000  161000   19200  136000  203000  148000      -8      -8
## 3446      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3447 2548054      -8      -8      -8      -8      -8      -8      -8      -8
## 3448      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3449      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3450  571700      -8   61000   28700   74000  324000   80000      -8      -8
## 3451  445600   84000      -9    3600   30000  288000      -9   10000      -9
## 3452      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3453      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3454      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3455      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3456 1144352  245000   62000   20000  115000  368000       0  136000   76752
## 3457  516100  157000   17800   48600   66000  100000   46500   21700      -8
## 3458  504500      -8   42600   39100   20200   47500   41800   22300      -8
## 3459  520000  205000   36000  249000      -9   17000      -8      -9      -8
## 3460      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3461      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3462  842100  230000   36000   13200   79000   43000   57000   26700      -8
## 3463  457700  156000   28000    2900   64000   34800      -8   28500       0
## 3464      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3465      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3466      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3467  387224      -9  120000   72000   75000   35700   62000      -9   12124
```

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## 3468      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3469      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3470      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3471      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3472  945388  185000  119000   28100  138000  197000   93000   20600    1688
## 3473 1280225  487000  139000  105000   61000  395000      -8   38400   48225
## 3474      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3475      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3476      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3477  714200  187000   87000   44400  124000  116000       0    8900       0
## 3478  107000  107000      -8      -9      -9      -8      -8      -8      -8
## 3479  522684  161000   80000   16800   60000   78000   42900   15800     584
## 3480      -9      -9      -9      -8      -9      -9      -9      -9      -9
## 3481      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3482      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3483      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3484      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3485 1091263  344000  132000   44700   35200  108000   93000   16200   12963
## 3486      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3487  891800       0   82000   19900   45400  454000  162000   44000       0
## 3488  369335  143000   38900   21700   57000   22600      -9    6900    4535
## 3489      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3490      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3491      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3492  319500   95000   30000   14700   24000  113000   38700    4100      -9
## 3493      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3494      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 3495  266600   80000   33000   30000   28000   31000      -8   48000    1600
## 3496      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3497      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3498 1108900  525000  108000   54000   30700  288000      -8   14400   52800
## 3499      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3500 1664000  500000  350000   35000  220000  250000  150000   74000      -8
## 3501  859600  236000  193000   22000  116000  108000   91000   20800      -8
## 3502  762819  209000  106000   36000   58000   74000   79000   16200   90719
## 3503      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3504  783608   88000   54000  132000   36400  330000       0  114000   21608
## 3505 1191000  475000  117000   36000  125000  400000      -9      -9      -9
## 3506      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3507      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3508      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3509 4293380  605000   79000  123000  183000  369000       0   96000   31280
## 3510      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3511      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3512      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3513  891500  157000   39900  100000   93000  114000  106000   47300      -8
## 3514      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3515      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3516      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3517      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3518  461000  118000   45000   20000   28000   87000  104000   19000      -8
## 3519 2716217  499000  250000   41000  148000 1602217  158000   18000      -8
## 3520      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3521 1677395  624000   68000  166000   65000  222000   90000   21500   16195
```

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## 3522      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3523  698300  170000   74000   12300   81000  222000      -9      -9      -9
## 3524      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3525      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3526      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3527      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3528 2517700  674000  159000  243000  170000  593000  390000  104000      -8
## 3529      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3530      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3531      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3532      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3533      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3534 1673857  403000  279000   37300  266000  247000  142000    7800  114757
## 3535      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3536      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3537      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3538  881599      -8   99000  255000   98000  288000   86000   12500   10899
## 3539  687907  144000   65000   20100   91000   82000   77000   81000   63907
## 3540 1413200  348000  142000   25200  105000  639000   62000   92000      -8
## 3541 3850305  414000  112000  349000  183000  167000       0   23200   50551
## 3542      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3543 1634000  500000   20000  300000  200000  200000  350000   35000    9000
## 3544  159000       0   40000   11000   34000   10000   40000    4000       0
## 3545 1670000      -8   90000  127000  138000  686000  249000  109000      -8
## 3546 1227000  379000   62000      -8   91000  337000  153000    1800      -8
## 3547      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3548  907000  360000   95000  136000   36000  165000   75000      -9      -8
## 3549  298900   85000   46000   52000   25000   68000      -8    2800       0
## 3550      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3551      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3552 1083523       0  250000  269000   86000  271000  156000   16600     923
## 3553  668000  460000      -8      -8      -8   80000   45000      -8   23000
## 3554      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3555 1989800  364000  161000  179000  301000  583000  158000   33200  169100
## 3556  817458  139000   56000   17900   94000  253000  152000   10200    1458
## 3557      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3558      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3559      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3560      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3561  607300  251000   75000   14500   74000  110000   63000    3600      -8
## 3562 2831268 1130000  147000  170000  186000  524000  223000      -9   38268
## 3563      -9      -8      -8      -9      -9      -8      -8      -8      -8
## 3564      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3565 2731687  809000  297000  466000  161000  394000  305000   31400   81287
## 3566  976993  193000  141000  262000  134000   34800      -9   20600    1093
## 3567  984800  192000   97000   10300  107000  286000  195000    7900      -8
## 3568 1745000  150000  200000  350000  165000  350000  240000  100000      -8
## 3569 3647730  312000   76000   78000   95000 1602217  244000   12300   35213
## 3570      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3571      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3572      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3573      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3574  255500   30000  100000   10500  100000   10000      -8      -8      -8
## 3575      -9      -8      -8      -8      -8      -8      -8      -8      -8
```

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## 3576   370000  110000   38100   23700    37800   67000   75000    8500       0
## 3577       -9       0      -9      -9       -9      -9      -9      -9      -9
## 3578   763217  115000   18000  137000    68000  274000      -9      -9   76717
## 3579       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3580   955413      -9      -9      -9       -9      -9      -9      -9  596367
## 3581   488700  145000   23700   32600    39500   24000   98000    5200      -9
## 3582   506010  153000   46000   36400    12200   65000   88000   18300   12410
## 3583   733000  733000      -9      -9       -9      -9      -9      -9      -9
## 3584  1397100  432000  356000   42700   124000   63000  105000   75000       0
## 3585   108000       0   10000    6000    40000   10000       0    3000       0
## 3586       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3587   527322   92000   23500   10200    61000  202000   67000   21600    1622
## 3588       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3589       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3590       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3591       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3592       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3593  1663200  216000   87000  118000   404000  613000  161000   25000      -8
## 3594   801700   66000   83000  137000    71000  203000   99000   84000      -9
## 3595  4566391  871000  161000  847124   381000 1080000  243000   20900  596367
## 3596  1923305  463000  222000   11900   257000  567000  180000    6700    6705
## 3597       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3598       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3599   759364      -8   25000   72000    30600  306000  180000    6900    2264
## 3600       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3601       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3602  4459457  683000  655222  847124   125000  424000  249000  771944  596367
## 3603       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3604  3048124  674000  339000  847124   215000  660000  128000   30000   60000
## 3605  1707000  486000  158000   72000   168000   76000  149000   10000      -9
## 3606       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3607       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3608       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3609       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3610  1306100  401000  171000  263000   100000  160000   93000    7100      -9
## 3611       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3612       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3613       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3614       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3615       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 3616       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3617  1766000  159000  350000  272000   106000  420000  249000   92000      -8
## 3618       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3619       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3620       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3621  1769671  284000  252000  221000   270000  235000   95000   23700   68971
## 3622       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3623  2834667  431000  241000  274000   176000  833000  106000   39300  596367
## 3624   809900  167000   23400  126000    56000  142000   70000    5500      -8
## 3625  1166888  406000  139000   36200   107000   96000   94000    7400   86688
## 3626   913600  370000   46000   45800    48100   60000  136000   17700      -9
## 3627   953600  144000   79000   93000    65000  380000   93000   19500      -9
## 3628       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3629       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

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## 3630       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3631       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3632       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3633  2099300  641000  254000      -9       -9  543000  143000    5400    2900
## 3634  3445317  603000  183000   80000   339000 1602217  317000   46600      -8
## 3635  1416559  180000   77000   11500   405000  383000  241000   15200   13959
## 3636       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3637  1551300  517000   92000  307000    78000  110000  176000   47200      -8
## 3638  2325096  547000  230000   37700   353000  431000  347000   13500    6850
## 3639       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3640  2548054      -9      -9      -9       -9      -9      -9      -9      -9
## 3641  3848848 1230000  116000  847124   351000  885000      -8   45700   87924
## 3642  2205353  309000  126000   31000   181000  347000 1148153    6200      -9
## 3643   900546      -8  147000   40000   109000   97000  152000   35600    4846
## 3644   639828  116000  108000   53000    84000  143000   85000   16500    5128
## 3645  4569710 1210000  263000  403000   326000  169000 1148153  771944   19567
## 3646       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3647  1267426  465000   67000   49300   189000  140000  152000   20100    9026
## 3648       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3649       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3650       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3651       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3652       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3653       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3654       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3655       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3656   656065  185000   39600   21400   141000   92000   61000   16100   10265
## 3657  1275051  402000  153000   33000   141000  214000  229000   20400   28651
## 3658  4363668  952000  276000  847124   489000  243000  559000  771944      -9
## 3659   937856      -9  192000  478000   194000      -9      -9      -9   17856
## 3660       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3661       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3662       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3663       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3664       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3665   915800      -8  116000   69000    38800  270000  250000   82000   40000
## 3666  2097046 1000000  220000   50000   100000  240000      -8      -8  228000
## 3667  1015688  262000   48700  149000    94000  272000  136000   11800   37388
## 3668       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3669       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3670  2826222  432000  655222   50000   169000 1000000  285000   30000   15000
## 3671       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3672       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3673   544861  115000      -8  152000       -9   51000   95000   22600   79761
## 3674       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3675  1835700  482000  115000  242000   174000   72000  156000   46700      -8
## 3676       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3677   290000      -8   38000  176000       -8   42100      -8    7000      -8
## 3678   910000      -9  140000   60000    45000  110000  470000      -9      -9
## 3679       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3680       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3681       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3682       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3683   410000  360000   50000      -8       -9      -8      -8      -8      -8
```

```
## 3684      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3685 1107508  312000   82000   36000  203000  182000      -8   84000    2508
## 3686      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3687  521300      -8  108000   40000   66000   86000  174000   12500      -8
## 3688  832215  254000   57000   36200   82000   58000  119000   16800   26215
## 3689      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3690      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3691      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3692 3682124 1330000  302000  847124  218000  655000  278000       0       0
## 3693      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3694      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3695      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3696 1164600  379000   90000   56000   55000   61000  130000   22000    9600
## 3697      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3698  682900       0   97000    8400   90000  148000  124000    8500       0
## 3699      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3700 3499217  532000  440000   58000  227000 1602217  371000   30000   19000
## 3701      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3702      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3703 1233200  421000   77000  115000       0  188000  201000   12200       0
## 3704      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3705      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3706      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3707      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3708      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3709      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3710      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3711      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3712 2659371  500000  250000  500000  759371  500000      -8   50000      -8
## 3713      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3714  996215  141000   48600   33400  213000  283000  130000   37600     915
## 3715 3327944  800000  400000  150000  116000  300000  200000  771944  100000
## 3716      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3717      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3718 1016800  334000   96000   34800   78000  211000  157000    5300      -8
## 3719      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3720 1031000  520000  342000   86000   83000      -8      -8      -8      -8
## 3721 2088933  422000  186000   28700  194000  370000  569000    1600    2133
## 3722      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3723 2616083  950000  162000   98000  105000  196000  405000   27800   87283
## 3724      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3725      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3726      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3727      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3728      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3729      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3730      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3731      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3732      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3733      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3734      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3735  889160  318000  123000   64000   64000   81000  161000   16200   11960
## 3736      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 3737 1421320  572000  171000   55000  198000   63000  194000  133000   10820
```

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## 3738  8232387 4105867  274000  545000   167000  129000 1148153  100000  596367
## 3739  1645000  306000  198000  117000   195000  232000  242000   68000      -8
## 3740       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3741       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3742       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3743       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3744       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3745       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3746       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3747       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3748  1609000  685000  110000  100000   256000  165000  205000   40000      -8
## 3749       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3750       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3751  1004799      -8  141000      -8   233000  221000  351000      -8   18799
## 3752       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3753       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3754  1087200  386000   44900  319000       -9  321000      -9   16300      -9
## 3755       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3756  1667500  439000   80000   26500   283000   89000  211000    4100   24900
## 3757  7371867 4105867   73000  308000   120000  295000  385000      -8      -8
## 3758       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3759       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3760  3556341      -8  410000  847124   360000 1602217   33000      -8  150000
## 3761       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3762  1527352  562000  145000   11800   193000  398000  171000    4000    3752
## 3763  3026059      -8  118000  205000   159000  738000  339000   13800   20259
## 3764  1466600  444000  436000      -8   283000   42000  216000      -8      -8
## 3765  7521032 4105867  332000  349000   759371  427000  505000  771944  102850
## 3766  1947746  454000  134000   76000    82000  586000  186000  125000   10700
## 3767       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3768       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3769       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3770  4771183 1130000  369000  255000   394000 1150000  508000  771944   50239
## 3771       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3772       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3773       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3774  2256423  535000  164000  148000   367000  373000  601000   47900   20523
## 3775       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3776  1638000  634000  194000   43000   205000      -9  291000      -9      -9
## 3777   781565  260000  103000   97000    27400  154000       0   55000   28265
## 3778       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3779   742000  390000   97000  108000   101000   28000      -8      -8      -8
## 3780       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3781       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3782       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3783  1737400  606000   81000   47500    38100   97000  156000    1800  108000
## 3784       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3785       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3786       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3787       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3788  2953367  200000  180000  500000   270000  750000  357000   50000  596367
## 3789  2198639  744000   96000   87000   254000  219000  234000   37200   76439
## 3790       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3791  1046000   85000  101000   86000    32000  437000  118000  136000   12000
```

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## 3792       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3793       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3794   450180  108000   23300   62000   102000   57000       0   44500    9480
## 3795        0       0       0       0        0       0       0       0       0
## 3796       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3797       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3798       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3799       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3800       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3801  3136804      -8  168000  241000   439000  261000   30300   20900  102604
## 3802       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3803   520300  270000      -8   14600   223000      -8      -8      -8      -8
## 3804       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3805       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3806  1209000  157000   78000   61000   441000  360000      -8      -8   12000
## 3807   948935      -8  190000   46900   345000  156000  144000   10200   15735
## 3808       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3809  6111387  573000  213000  847124   300000 1602217  637000   60000       0
## 3810       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3811  9984094 4105867  655222  847124   759371  841000 1148153  771944  596367
## 3812       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3813       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3814       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3815       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3816       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3817       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3818       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3819       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3820       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3821       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3822  5059488 1330000  290000  478000   247000 1602217  326000   12700   57571
## 3823  5858867 4105867  277000  352000   121000  456000  320000   52000      -8
## 3824       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3825       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3826       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3827       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3828   933492  446000   30000  106000    28400   83000       0   18300  168492
## 3829       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3830       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3831   312875   43100   51000   20000    23900   40100   37800    1900   48375
## 3832       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3833   200000      -9      -9      -9   200000      -9      -9      -9      -9
## 3834       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3835   337100      -9   37700   10000   122000  152000      -9      -9      -9
## 3836       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3837  7802237 4105867  248000  346000   225000 1602217 1148153      -9      -9
## 3838       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3839       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3840       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3841       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3842  5400887 1130000   92000  229000   759371  865000  379000   30800   96670
## 3843   363800  183000      -9   19600    87000   72000       0       0       0
## 3844   372500   82000   75000   20000    24000   54000   28000    6000      -8
## 3845       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 3846   352100  116000   49700    8400   21200   67000   26200    5400       0
## 3847       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3848       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3849   400000   80000  100000  100000   50000   50000       0    5000       0
## 3850    68000      -8      -8      -8   68000      -8      -8      -8      -8
## 3851   605000   50000  200000  150000   50000      -8  100000      -8   55000
## 3852       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3853       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3854       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3855       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3856       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3857   562200       0   30700   41200  129000  219000  106000    3500       0
## 3858       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3859       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3860       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3861       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3862       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3863       -9      -8      -8      -9      -9      -8      -8      -8      -8
## 3864       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3865       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3866       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3867       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3868       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3869   187160      -8   43600      -9   29800   59000   23100     760      -8
## 3870       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3871   352000   55000   47000   40000   84000  120000      -8    2000      -8
## 3872   315500   60000      -9   20500   92000   78000   49000      -9      -9
## 3873       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3874       -9      -9      -9      -9      -9      -9      -9      -9       0
## 3875     1800      -8      -8      -8      -9      -8      -8      -8      -8
## 3876   365000  100000   15000  100000   50000      -8  100000      -8      -8
## 3877   303200  150000  108000      -8      -8   18000      -8      -8      -8
## 3878  2232600  264000   70000   54000   31600   86000  119000      -9      -9
## 3879   117900   45000   35000    9600    7200   15600      -8    3000      -8
## 3880       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3881   231000  100000   15000    6000   50000   10000   20000   10000    5000
## 3882  2284921  546000  655222   85000  323000  530000       0    8700   41999
## 3883   476000   80000   35000  105000  109000   82000      -8    8000      -8
## 3884   209400       0      -8   44000   18000   40000  100000    1000    6400
## 3885   750000  300000   40000   30000   30000   60000   70000   80000   50000
## 3886  1318000  900000  100000  100000  100000   30000   20000   20000   15000
## 3887   494400      -9   20200   11000   78000   32000   64000    6100       0
## 3888       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3889       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3890       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3891   942200  330000  167000   40700  112000   89000   99000   79000      -8
## 3892       -9      -8      -8      -8      -9      -8      -8      -8      -8
## 3893  1173086   36400   49100   90000   54000  722000  131000   43900    1786
## 3894       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3895       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3896       -9      -9      -9      -9      -9      -9      -9      -9      -9
## 3897   405000  117000   61000   58000   19000   64000   42000   34000       0
## 3898   134300      -8   23000   16700   25500   24600    7200    3400      -8
## 3899       -9      -9      -9      -9      -9      -9      -9      -9      -9
```

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## 3900   423700  150000   20000    9000    36700   75000   40000   21000       0
## 3901       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3902       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3903   280000   37000   30000  102000       -9  102000      -8      -8      -8
## 3904   443000  176000   60000   60000    65000   18000   52000   12000      -8
## 3905   318200   81000   23400    9100    32100   99000   29000   10400       0
## 3906       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3907       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3908    19000      -8      -8    3800    12000      -8      -8      -8      -8
## 3909       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3910       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3911   419000  300000   20000   15000    15000   10000   20000   15000    8000
## 3912  1008520  523000      -8  123000   246000   40000      -8      -8   11520
## 3913   775060  236000   45700   77000    30200  161000   94000   52000    8160
## 3914   282000  201000   56000       0        0   10000    5000       0       0
## 3915   265200   80000   15000    4200   156000   10000      -8      -8      -8
## 3916       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3917   731600  182000   53000   17000    57000  215000  178000    3900       0
## 3918       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3919  1365000  286000   38000   89000    70000  116000   99000   13000    5000
## 3920   486900   92000   73000   36000    96000   94000   76000    1600      -8
## 3921       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3922       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3923       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3924       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3925  3078534  551000  655222  179000   255000  688000  551000   21200  178112
## 3926   602000  106000   38000   97000    29000  242000   46000    6000      -8
## 3927       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3928    32400      -8      -8      -8    32400      -8      -8      -8      -8
## 3929   125000   70000   12000    6000     6000   12000    7000    3000    2000
## 3930       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3931   298700  143000   16500   17100    26600   34400   42100    4900       0
## 3932   677000   58000   70000   87000    99000  288000   67000    8000      -8
## 3933   600000  500000   12000    5000    50000   10000   10000    5000       0
## 3934       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3935   718000  250000   40000   60000    60000   65000  200000       0    6000
## 3936   566000  240000      -8   16000   150000   20000  110000      -9   30000
## 3937   270000   60000   50000   35000    50000      -8   75000      -8      -8
## 3938       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3939       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3940       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3941   649650   76000   26500      -9       -9   77000  443000     750   26400
## 3942   307600   84000   56000    9500    40800   63000   29200    4100      -8
## 3943   335000      -9   48000   52000    41000  145000   49000      -9      -9
## 3944   350380   44200   14800   48500    43200  102000      -8    3600   29080
## 3945   207300      -8    8500    8100    33700  140000      -8      -8      -8
## 3946   819000  141000  113000  109000    66000  136000   93000    9300       0
## 3947       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3948       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3949   337500      -8   28700   13400    37800  113000  105000   36100      -8
## 3950   808434  292000  190000   24700    11700   61000   67000    9300    2134
## 3951   177100   48000      -8   60000    24000      -8   33600      -8      -8
## 3952   532500  100000  200000  100000   100000      -8   25000      -8      -8
## 3953  1549899      -8   88000   60000   229000  201000  204000  147000   12399
```

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## 3954   393300   82000   62000   75000    86000   46200   30000   12100      -9
## 3955       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3956  1333500  144000   58000  255000   146000  483000  138000   14600      -9
## 3957  1217000  598000   39000  130000    50000  184000   89000      -9       0
## 3958   374500  143000   55000   49600    48000   65000       0    8500       0
## 3959   596977  136000   43000   62000    40900  105000  109000   11800   13377
## 3960   704834  247000   45800   26200    75000   89000   69000    2600     534
## 3961   241100   88000      -9   18600    52000   48400      -8      -8      -8
## 3962       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3963   652400  292000   10800   36500   118000  138000      -8   25800      -8
## 3964   377000  160000      -8  180000    37000      -8      -8      -8      -8
## 3965       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3966       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 3967       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3968  1067696  334000   39500   58000   162000  222000   89000   94000    8196
## 3969   416600  105000   25900    8900    74000   27700   44500   35800       0
## 3970       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3971   491500  156000  106000   37900    35900  119000      -8   29700      -8
## 3972       -9       0       0       0        0       0      -9       0       0
## 3973       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 3974   198200   50000   17000   24500    22500   49000      -8    2200      -8
## 3975       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3976       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3977       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3978       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 3979   415000  250000   50000   20000    10000   50000       0    5000      -9
## 3980   787500  208000   41400  123000    58000  106000   67000   33000      -8
## 3981  1814000  580000   81000  517000       -9  112000  187000      -8      -8
## 3982  1326003  353000  101000   87000   105000  400000  220000    9600    9303
## 3983       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 3984       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3985       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3986       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3987       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3988  1069521  353000   98000   97000   108000  138000   23300   25500    2721
## 3989       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 3990  1924496  460000   91000  130000   144000  204000      -9  771944     552
## 3991       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3992       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3993   966698  194000  193000   91000   107000  227000   63000   60000    3198
## 3994       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 3995   825500       0   48800   45700    36300   56000   80000    3500       0
## 3996   254000   84000      -8      -9    40200   20000   55000      -8   31200
## 3997       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 3998  1624000  415000  302000  296000   270000  200000      -9      -9      -9
## 3999       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4000       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 4001       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4002   351100   39500   59000   52000    38600   74000   24900   14800   12000
## 4003   154500      -8   90000    6000    54000      -8      -8    2500      -8
## 4004   518900  176000   54000   19700    41800   51000   65000      -9      -9
## 4005     7000      -9      -9      -9       -9      -9      -9      -9    7000
## 4006   351200   55000   14400   18000    26800  136000   41700    6200       0
## 4007   852000      -9   88000  248000       -9   56000  425000      -9   15000
```

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## 4008      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4009  493800  148000  111000   32900   28300  139000      -8    1500      -8
## 4010  585000  120000   50000   10000   80000  100000  150000   50000    5000
## 4011      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4012 1833589  420000  149000   19500  233000  658000  174000   54000    2089
## 4013  776371      -8      -8   17000  759371      -8      -8      -8      -8
## 4014  365000  266000      -8   22000   77000      -8      -8      -8      -8
## 4015  407550  149000   27700    8900   58000   90000   45000     750       0
## 4016 1148216  283000   46300  206000   64000   28800  283000   20400  142016
## 4017      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4018  769737   51000  128000   11800   54000  212000  146000   10500    1337
## 4019      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4020      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4021      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4022  237100   37500      -8    8000   70000  100000      -8      -8      -8
## 4023  164800   95000      -9    3600   48000   11000      -8      -8      -8
## 4024  617724  144000   83000    2800   72000  136000      -8   64000   25924
## 4025      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4026 3085137  675000  416000  118000  464000  208000  225000  771944   39193
## 4027 2382000  581000  123000  246000   74000  760000  423000  158000    5000
## 4028  167400   57000   86000    5800   18600      -9      -9      -9      -9
## 4029  789000  183000   47600   64000  266000   26500   59000    3400      -8
## 4030  745940      -8   62000  155000   88000  225000  187000      -9    1040
## 4031 2362879      -8  114000   79000  285000  425000 1148153   44300   19426
## 4032      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4033 2418315      -8  171000  291000  132000  547000  202000  177000   25315
## 4034      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4035      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4036  602000  360000   30000   12000   60000   20000  100000      -8      -8
## 4037      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4038  672500  125000  104000   18000  102000  119000   48800    8600      -9
## 4039  446200  154000   46200   35900   33200  169000      -8    2200      -8
## 4040      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4041      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4042      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4043      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 4044      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4045      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4046  621600  227000  116000      -8  182000   69000      -8      -8      -8
## 4047      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4048      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4049      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4050      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4051  813000  242000   51000   14000   54000  190000   52000   13000    8000
## 4052      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4053  148000   15000   10000    6000   70000   12000   15000   10000       0
## 4054      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4055      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4056      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4057  615900  208000   25900   17200   59000   92000   79000   14900       0
## 4058  410000   71000   28700    9200   68000   23800   69000    2500      -8
## 4059      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4060 2536761      -8   88000   80000  286000  594000 1148153   41400   41208
## 4061      -9      -9      -9      -9      -9      -9      -9      -9      -9
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## 4062      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4063 2485500  522000  175000  376000  226000  543000  233000   20500       0
## 4064      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4065 1145050  282000  135000   45500  125000  143000      -8    6500    3050
## 4066      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4067      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4068  399000  225000   20000   14000  110000   15000       0      -8      -8
## 4069      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4070      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4071      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4072      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4073 1208500      -9      -9   60000  413000  284000  141000   16500   22000
## 4074      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4075      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4076  392000  189000      -8   10000   80000   20000   60000      -8   15000
## 4077      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4078      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4079 3130124  810000  153000  847124   85000  862000  130000   61000  170000
## 4080  332000  170000   23000   12000  100000   15000       0       0       0
## 4081      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4082      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4083  878033  193000   48000   40000   89000  137000   78000   26500   27533
## 4084      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4085 1477100  373000  185000  166000  130000  180000  180000   10800      -9
## 4086      -9       0      -9      -9      -9      -9      -9      -9      -9
## 4087    3300      -9      -8      -8      -8      -8      -8      -8      -8
## 4088      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4089 1265700  384000   66000  300000  144000  250000   52000   50000      -8
## 4090  986700  240000  113000   24000   89000  341000  112000    2700       0
## 4091   51000   33000      -9      -9      -9      -9      -9      -9      -9
## 4092  853000  400000  130000   30000  110000  110000      -8   50000      -8
## 4093      -9      -8      -8      -8      -9      -8      -8      -8      -8
## 4094      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4095      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4096      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4097  970000  160000  124000  162000      -9  288000  121000   35000       0
## 4098  242000       0      -9    7000   97000  120000      -9       0       0
## 4099      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4100      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4101      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4102  324300      -8      -8  214000   50000      -8      -8   31300      -8
## 4103  989300  274000  187000   11800  240000   70000  114000   20000      -8
## 4104  781200   84000   96000      -9  201000  218000  125000   15200      -9
## 4105      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4106 1601244  420000  218000   32000  131000  296000  151000   71000   17444
## 4107   80500   15000   10000    5500   25000      -8   15000      -8   10000
## 4108  614400      -8  164000   59000  144000   70000   92000   50000     400
## 4109  169000      -8   17000   13000  100000   20800      -9   11000      -8
## 4110      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4111      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4112      -9      -8      -8      -8      -8      -8      -8      -8      -8
## 4113      -9      -9      -9      -9      -9      -9      -9      -9      -9
## 4114  438600  125000   60000  150000   15000   10000   60000      -8     600
## 4115      -9      -9      -9      -9      -9      -9      -9      -9      -9
```

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## 4116   756900  184000   67000   20700   161000  295000      -8      -8      -8
## 4117       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4118   161000      -8   14000      -8    36000   70000   15000   20000      -8
## 4119  1476027  452000   41900   51000    83000  351000  161000   16000   20127
## 4120       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4121   838600      -8  459000   49700    26900  113000  190000      -8      -8
## 4122   790200  144000   54000    5600    26100  112000   86000    5500      -8
## 4123   360200      -8   95000   17400    99000      -8   67000    6000      -9
## 4124       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4125       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4126  1272193  371000   77000  108000   125000  210000   96000  153000   78193
## 4127       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4128   154700      -9      -9   15900    15800   55000      -9      -9      -9
## 4129   347200   66000   47200   42900    79000   55000      -8   16200      -8
## 4130  1863331  361000  227000   18000   134000  127000  137000   32300    1031
## 4131   602500  207000   82000  179000    43100   58000   20400      -8      -9
## 4132       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4133       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4134       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4135   398200   22800   60000  129000    86000   90000      -8      -8      -8
## 4136       -9      -9      -9      -8       -8      -9      -9      -9      -8
## 4137       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4138       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4139   795500  177000   67000   85000    88000  192000  131000   12300   11400
## 4140       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4141       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4142  1611756  519000   91000   75000   205000  129000  181000   98000   46056
## 4143   597840  214000   92000   26000    72000  155000   16000     640      -8
## 4144   428600  156000   79000    3900    17100   37600   54000    1200      -8
## 4145       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 4146       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4147       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4148  1299000  250000   79000  100000   120000  100000  300000   50000  100000
## 4149       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4150       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4151       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4152   549500      -8   43000  130000    31500   35000  285000      -8   10000
## 4153       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4154       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4155       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4156       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4157  3621800  294000  419000  222000    35800  734000  406000   51000      -8
## 4158  1263119  216000  114000  114000    41200  383000   94000  116000   13319
## 4159       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4160       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4161   369000      -9   30000   84000    31000  220000      -9    4000      -8
## 4162       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4163       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4164       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4165       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4166  1362000      -8  124000  326000   101000  151000  155000   54000      -8
## 4167  1038000       0  107000   75000    95000  166000   82000   13100      -8
## 4168       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4169  1002043      -8  134000  268000    80000  208000  130000      -9   16043
```

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## 4170       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 4171  1074600       0   91000  161000    57000  194000   32400   27200      -9
## 4172       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4173  4290352  787000  655222  337000   319000  886000  424000   43900  193630
## 4174  3247190  949000  232000  330000   152000  816000  243000    5700  105490
## 4175  2958091  753000  147000   19300   759371  564000  250000   26200   20220
## 4176       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4177       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4178       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4179       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4180       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 4181   850358  234000   99000   55000    61000  162000  102000     550   20808
## 4182  1230000      -9  300000   29000   477000  190000  150000      -8      -8
## 4183   358650   40000   67000   60000    51000   22800   63000   37000     950
## 4184   609400  260000   46000   64000    68000   15000  130000      -8      -8
## 4185   790700  107000   45200  159000   142000  167000  148000      -8      -8
## 4186   200000      -9      -9      -9       -9      -9      -9      -9      -9
## 4187  2600600  765000      -9  186000   304000  653000  318000  179000       0
## 4188       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4189       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4190       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4191  1526909  494000  139000   63000   176000  501000   81000   15100    5809
## 4192       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4193       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4194   124000      -9      -9   20000    22000   30000      -9   30000      -9
## 4195       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4196       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4197       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4198       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4199       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4200  1601700  558000  178000   45800   230000  240000  143000      -9      -9
## 4201  3629254  375000  655222  491000   381000  349000  243000   72000   24032
## 4202       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4203  1561000  289000  180000  115000   128000  500000  100000    5000  144000
## 4204       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4205       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4206   569600  159000   59000   15100    68000   74000   89000    6400      -8
## 4207       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4208       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4209       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4210  1480000  270000  210000   50000   115000  150000  235000   95000   10000
## 4211  4733906  969000      -9  283000   111000  331000  339000   40900   54952
## 4212       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4213       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4214  1965000 1200000  100000  100000    75000  200000  250000      -8      -8
## 4215  1054993  285000   97000   69000   109000  291000  140000    8100   11493
## 4216       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4217   830000  700000   43000      -8       -8   87000      -8      -8      -8
## 4218   325000      -9   25000   25000   150000   20000   60000      -8      -8
## 4219  1388900      -8  173000  374000   251000  148000  255000       0  162600
## 4220       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4221  1078000  301000  251000   87000   168000  106000   95000      -8   10000
## 4222       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4223       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

```
## 4224       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4225       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4226       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4227       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4228       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4229       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 4230       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4231       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4232   560000  160000   30000   65000   180000   10000   85000      -8      -9
## 4233       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4234       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4235       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4236       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4237  3256524  732000  146000  171000   186000  582000  395000  771944   13534
## 4238       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4239  1404753  409000   43400   52000   234000  328000  225000   15600   17753
## 4240       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4241  2893875  497000  415000   47600   348000  245000  263000   41600   56875
## 4242       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4243       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4244    66000      -9      -9      -9       -9      -9      -9      -9      -9
## 4245  1900400 1110000      -9  290000   160000  287000      -8    5000      -8
## 4246       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4247       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4248       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4249  1150400  336000  113000   60000   175000  180000  203000   30400       0
## 4250       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4251   823000  166000  101000   52000    95000  299000      -8      -8      -8
## 4252       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4253       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4254       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4255       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4256   749000  214000  330000   15000   100000   80000      -8      -8      -8
## 4257       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4258  1618603  255000  180000   48700   348000  435000  165000   22700   10203
## 4259   867895       0  167000   62000   239000   95000  188000    4700    6195
## 4260       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4261       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4262  1463286  430000   54000   54000   144000  219000  246000   27300   27986
## 4263       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4264       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4265   959623  118000  103000  113000   142000  328000   55000   64000    2923
## 4266  1472000  341000  162000   90000   125000  379000  169000   60000       0
## 4267       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4268       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4269       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4270   871000  270000   59000  113000    48000  142000   74000  165000      -9
## 4271       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4272  1863000  441000  210000  118000   460000  462000      -8      -8      -8
## 4273       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4274       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4275  1105000  169000  147000  249000    84000  347000      -8      -8      -8
## 4276       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4277  1252000  458000  131000  165000   255000  100000      -8      -8      -9
```

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## 4278       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4279       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4280   791000  345000  133000      -8   180000   50000      -8   24000      -8
## 4281       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4282       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4283       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4284       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4285       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4286       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4287  1561000  500000   95000   50000   130000  410000  200000   15000    1000
## 4288       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4289       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4290       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4291  3479600  608000  157000  286000   115000  545000   86000   22600      -8
## 4292       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4293  1346000  173000      -8  351000       -8  748000      -8      -8      -8
## 4294       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4295  2724074  593000  172000  173000   475000  337000  327000   70000   65274
## 4296  4841590  796000  655222   42400   759371   48800  390000   94000   40797
## 4297       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4298  1379523  528000  116000  163000   241000   75000  189000   18200    1023
## 4299   649740  331000   96000   15900    98000   36400      -8    4000    5940
## 4300       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4301   826000  300000   30000   58000   274000   15000   96000      -8      -8
## 4302  1090750  350000  136000   60000   110000  220000      -9     750   14000
## 4303       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4304       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4305       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4306       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4307  1475500  449000  300000  149000   144000  186000  152000       0       0
## 4308       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4309  1343000  600000   40000   55000   338000  200000  100000      -8      -8
## 4310       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4311       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4312  2303000  580000  170000  105000   330000  560000  300000   16000   13000
## 4313  1040800  455000   67000   39600    88000  176000  151000    3200       0
## 4314       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4315       -9      -8      -8      -9       -9      -8      -8      -8      -8
## 4316  2098838  549000  288000  290000   210000  445000  222000    5200   22638
## 4317       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4318       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4319       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4320       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4321       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4322       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4323       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4324  1322850  482000   62000   76000   167000  202000  173000    8600   19250
## 4325  1611100  504000  151000   75000   187000   98000  197000    3400      -8
## 4326       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4327       -9      -8      -8      -8       -9      -8      -8      -8      -8
## 4328       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4329       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4330  1017263  312000  162000   17300    52000  128000  148000   70000    5363
## 4331       -9      -9      -9      -9       -9      -9      -9      -9      -9
```

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## 4332       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4333       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4334  1801000  500000  111000    5000   250000  500000  120000   15000   50000
## 4335       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4336       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4337       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4338       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4339  1034022  280000  161000   40300   126000  165000  149000   19900    8222
## 4340  2394231  602000  655222  127000   457000   89000  104000    1400   69009
## 4341  1923000  420000  365000   88000   123000  810000      -8      -8   72000
## 4342   963828      -8  187000      -8   249000   45400  159000      -8   16428
## 4343   673200      -8  315000   51000   125000   79000   78000      -8      -8
## 4344       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4345  1019040  148000      -9   78000   349000  126000  163000      -8   13140
## 4346    64500   12600    5700    5500     4800   21000    5800    1600      -8
## 4347       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4348       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4349  8744554 4105867  655222  847124   759371 1602217  388000   31200   29553
## 4350  3564487 1400000  220000  122000   759371  277000  320000   39200  188916
## 4351  1606000  484000  120000   75000   190000  250000  168000  109000      -8
## 4352       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4353       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4354       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4355       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4356  2342000  600000  288000   47000   270000  337000  260000   27000   36000
## 4357       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4358       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4359       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4360  1534232  393000  172000   56000   240000  139000  129000  187000   10232
## 4361   182300   80000   44500      -8       -8   37500      -8    5300      -8
## 4362       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4363       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4364   450400  110000  127000   75000    33100   47700   36500    2700    7000
## 4365       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4366  1818900  578000   90000  140000   223000  468000  151000   73000      -8
## 4367  6575867 4105867  200000  100000   500000 1000000  500000   50000       0
## 4368  2045103  621000  243000   13700    96000  503000  209000   25100    9303
## 4369       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4370       -9      -8      -9      -9       -9      -9      -9      -9      -9
## 4371       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4372       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4373       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4374       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4375       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4376       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4377   482000  200000      -8   12000   260000   10000      -8      -8       0
## 4378       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4379       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4380       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4381  3267124  698000  197000  847124   420000  800000  203000   58000   17100
## 4382  5543017      -8  655222  847124   759371 1602217 1148153  125000  146884
## 4383       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4384       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4385       -9      -8      -8      -8       -8      -8      -8      -8      -8
```

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## 4386       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4387       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4388   800971      -8      -8   21000   759371      -8      -8      -8      -8
## 4389  2548054      -9      -9      -9       -9      -9      -9      -9      -9
## 4390       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4391       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4392       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4393       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4394       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4395       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4396  2111700  507000  295000   43000   166000  590000  169000   24700      -8
## 4397       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4398       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4399  3349341  500000      -8  847124   100000 1602217      -8      -8  200000
## 4400       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4401  3773571  846000  219000  472000   759371 1140000  276000   26700      -8
## 4402   147201      -9      -9      -9    76000      -9      -9      -9   28501
## 4403  2263944  753000  261000  113000   143000  222000      -8  771944      -8
## 4404       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4405       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4406       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4407       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4408  2620160  741000  209000  128000   306000  518000      -8  134000   20160
## 4409       -9      -8      -8      -8       -8      -8      -8      -8      -8
## 4410       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4411       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4412  2689939  695000  145000  847124   759371  167000      -8   56000    8244
## 4413       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4414  1927300      -8  327000  486000   344000   96000  346000   36300   74000
## 4415       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4416  1770666  555000  133000  102000   132000  436000  281000   34300   43366
## 4417       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4418       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4419  5032307  958000  351000  847124   284000  463000  392000   30000  207983
## 4420 10858717 4105867  655222  220000   100000 1602217      -8  771944  596367
## 4421       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4422       -9      -9      -9      -9       -9      -9      -9      -9      -9
## 4423  2367000  844000  355000  421000   391000  281000      -8      -8      -8
## 4424  2836217  377000  266000  123000   261000 1602217      -8      -8      -8
## 4425       -9      -8      -8      -8       -8      -8      -8      -8      -8
##      OPLAND OPOTHEX
## 1        -9      -9
## 2        -9      -9
## 3        -9      -9
## 4        -9      -9
## 5        -9      -9
## 6        -9      -9
## 7        -9      -9
## 8        -9      -9
## 9        -9      -9
## 10       -9      -9
## 11       -9      -9
## 12       -8      -8
## 13       -8      -9
```

```
## 14       -9      -9
## 15       -9      -9
## 16       -8      -8
## 17       -9      -9
## 18       -8      -8
## 19       -8      -8
## 20       -8      -9
## 21       -8      -8
## 22       -9      -9
## 23       -8      -8
## 24       -9      -9
## 25       -9      -9
## 26        0      -8
## 27       -9      -9
## 28       -8      -8
## 29       -8      -8
## 30     4575      -9
## 31       -9      -9
## 32       -8      -8
## 33       -9      -9
## 34       -8      -8
## 35       -9      -9
## 36       -8      -8
## 37       -9      -9
## 38       -9      -9
## 39       -9      -9
## 40       -8      -8
## 41       -8      -8
## 42      200      -9
## 43       -8      -8
## 44       -9      -9
## 45       -9      -9
## 46       -8      -8
## 47        0       0
## 48       -8      -8
## 49       -8     500
## 50       -8    4100
## 51       -8      -8
## 52       -9      -9
## 53      500     500
## 54       -8      -8
## 55       -9      -9
## 56       -9      -9
## 57       -9      -9
## 58       -8      -8
## 59       -9      -9
## 60      900      -8
## 61       -8      -8
## 62       -8      -8
## 63      590      -8
## 64       -9      -9
## 65       -8      -8
## 66       -8    4500
## 67      600     250
```

```
## 68       -9      -9
## 69       -9      -9
## 70      700       0
## 71      110     190
## 72       -8      -8
## 73       -8      -8
## 74       -9     500
## 75       -9    8365
## 76       -8      -8
## 77       -8      -8
## 78       -8      -8
## 79       -8      -8
## 80      400     360
## 81     1200      -8
## 82     1000      -8
## 83        0    1900
## 84       -9      -9
## 85     1000      -8
## 86       -9      -9
## 87      770      -9
## 88       -9      -9
## 89       -9      -9
## 90        0       0
## 91       -8      -8
## 92       -8      -8
## 93       -8      -8
## 94       -9      -9
## 95      500      -8
## 96       -8      -8
## 97     4575    6400
## 98       -8      -9
## 99     2400      -8
## 100       0       0
## 101    4575      -9
## 102     720       0
## 103       0      -9
## 104      -8      -8
## 105      -8      -8
## 106      -8      -8
## 107      -8      -8
## 108      -8      -8
## 109     600      -9
## 110       0       0
## 111       0       0
## 112      -9      -9
## 113      -9      -9
## 114    2700       0
## 115     380      -8
## 116      -9      -9
## 117     670       0
## 118      -9      -9
## 119      -8      -8
## 120     500      -8
## 121      -8      -8
```

```
## 122       0       0
## 123      -9      -9
## 124      -8    6600
## 125      -8      -8
## 126      -9      -9
## 127      -8      -8
## 128      -8      -8
## 129      -9      -9
## 130      -9      -9
## 131       0    4800
## 132      -8      -8
## 133    1200       0
## 134       0       0
## 135       0       0
## 136      -8    6600
## 137      -9      -9
## 138      -9      -9
## 139    1200      -9
## 140     400      -8
## 141      -8      -8
## 142      -9      -9
## 143      -8    2700
## 144      -8      -8
## 145     400      -8
## 146      -9      -9
## 147      -8      -8
## 148      -8      -8
## 149      -8      -8
## 150      -8      -8
## 151      -8      -9
## 152      -8      -8
## 153      -8      -8
## 154      -9    6000
## 155      -8      -8
## 156      -9    3300
## 157      -9      -9
## 158       0       0
## 159      -8    1200
## 160      -9      -9
## 161      -8      -8
## 162      -9      -9
## 163      -8      -8
## 164      -9      -9
## 165      -9      -9
## 166      -8      -8
## 167     350      -8
## 168      -8      -8
## 169      -9      -9
## 170      -9      -9
## 171      -8      -8
## 172      -8      -8
## 173      -8      -8
## 174      -8      -8
## 175      -8     650
```

```
## 176      -8     200
## 177      -9      -9
## 178      -8      -8
## 179     600      -8
## 180      -9      -9
## 181      -9      -9
## 182      -9      -9
## 183      -8      -8
## 184      -9      -9
## 185      -9      -9
## 186      -8      -8
## 187      -9      -9
## 188      -8      -8
## 189      -8      -9
## 190      -9      -9
## 191       0    7400
## 192      -9      -9
## 193      -9      -9
## 194      -8      -8
## 195      -9      -9
## 196      -8    5100
## 197      -8      -8
## 198      -9      -9
## 199       0       0
## 200      -9      -9
## 201      -8      -8
## 202      -8      -8
## 203      -9      -9
## 204      -8      -8
## 205      -9      -9
## 206      -8      -8
## 207      -8    2700
## 208      -9      -9
## 209      -8      -8
## 210      -9      -9
## 211     400      -8
## 212    2500       0
## 213      -9      -9
## 214      -9      -9
## 215     250     300
## 216      -9      -9
## 217      -9      -9
## 218      -9      -9
## 219      -8      -8
## 220     500      -8
## 221      -8     100
## 222      -9      -9
## 223      -8      -9
## 224      -8      -9
## 225      -9    3800
## 226      -9      -9
## 227    1200      -8
## 228      -9      -9
## 229      -8      -8
```

```
## 230      -8       0
## 231      -8      -8
## 232      -8    3000
## 233      -9      -9
## 234      -8      -8
## 235      -8      -8
## 236      80      -9
## 237      -8      -8
## 238       0       0
## 239      -9    3000
## 240    1200    3600
## 241     500      -9
## 242      -8      -8
## 243      -8      -8
## 244      -8      -8
## 245    1600      -9
## 246      -9      -9
## 247      -9      -9
## 248      -8      -8
## 249      -8      -8
## 250    2400      -8
## 251      -9      -9
## 252    1200      -8
## 253       0    8365
## 254      -8      -8
## 255      -8      -8
## 256      -8      -8
## 257      -8      -8
## 258      -9      -9
## 259      -8      -8
## 260      -9      -9
## 261      -8    2400
## 262      -8      -8
## 263      -8      -8
## 264      -8      -8
## 265      -9      -9
## 266      -9      -9
## 267      -8      -9
## 268      -9      -9
## 269      -8      -8
## 270      -8      -8
## 271      -9      -9
## 272      -8      -8
## 273      -8      -8
## 274      -9      -9
## 275      -9      -9
## 276      -9      -9
## 277      -9      -9
## 278       0       0
## 279      -9      -9
## 280      -8      -8
## 281      -9      -9
## 282      -8      -8
## 283      -8      -8
```

```
## 284      -8      -8
## 285      -8      -8
## 286      -8      -8
## 287      -9    6000
## 288      -9      -9
## 289      -8      -8
## 290      -9    5000
## 291      -9      -9
## 292      -8      -8
## 293      -9      -9
## 294      -9      -9
## 295      -9     330
## 296      -8      -8
## 297      -8      -8
## 298    3200      -8
## 299      -9      -9
## 300      -8      70
## 301      -9      -9
## 302      -9      -9
## 303      -9      -9
## 304      -8      -8
## 305      -9      -9
## 306      -9      -9
## 307      -9      -9
## 308      -8      -9
## 309      -9      -9
## 310      -8      -8
## 311      -9      -9
## 312      -8      -8
## 313      -9      -9
## 314      -8      -8
## 315      -9      -9
## 316      -9      -9
## 317    1100      -8
## 318      -8      -8
## 319      -9      -9
## 320      -8      -8
## 321      -9      -8
## 322      -8      -8
## 323      -8      -8
## 324      -8      -8
## 325       0       0
## 326       0       0
## 327      -9      -9
## 328      -9      -9
## 329    1100      -9
## 330      -9      -9
## 331       0       0
## 332      -8      -8
## 333     550      -8
## 334      -8      -8
## 335    2500    5000
## 336      -8      -8
## 337      -9      -9
```

```
## 338      -9      -9
## 339      -9      -9
## 340      -9      -9
## 341      -8    6900
## 342      -8     600
## 343    2500     850
## 344      -9      -9
## 345      -8      -8
## 346      -9      -9
## 347      -8    8365
## 348      -9      -9
## 349    1000      -9
## 350      -8      -8
## 351     720      -9
## 352      -9      -9
## 353      -8      -8
## 354      -9      -9
## 355      -9      -9
## 356     500       0
## 357      -8    1800
## 358      -8      -8
## 359      -8      -9
## 360      -9    2000
## 361      -8    3600
## 362      -8      -8
## 363    4575      -8
## 364      -8      -9
## 365      -8      -8
## 366      -8      -8
## 367      -8      -9
## 368    1000      -8
## 369      -8      -8
## 370      -8      -8
## 371      -8    5700
## 372     600      -8
## 373     600      -9
## 374      -8      -9
## 375      -8      -8
## 376     780      -9
## 377      -9      -9
## 378      -9      -9
## 379      -9      -9
## 380      -9      -9
## 381      -9      -9
## 382      -9      -9
## 383      -9      -9
## 384      -8      -8
## 385      -9      -9
## 386      -8      -8
## 387      -9      -9
## 388     500      -9
## 389      -8      -8
## 390      -9      -9
## 391      -9      -9
```

```
## 392      -9      -9
## 393      -9      -9
## 394    1200      -8
## 395      -9      -9
## 396     800      -8
## 397      -8      -8
## 398      -9      -9
## 399    2500      -9
## 400      -8      -8
## 401      -9      -9
## 402      -9      -9
## 403       0       0
## 404      -9      -9
## 405    1800      -9
## 406      -9      -9
## 407      -8      -8
## 408      -8      -8
## 409      -8      -8
## 410    1400      -9
## 411     470     340
## 412    4000      -8
## 413       0       0
## 414     600      -8
## 415      -9      -9
## 416      -9   17314
## 417      -9      -8
## 418      -9      -9
## 419      -9      -9
## 420      -8      -8
## 421      -9      -9
## 422      -8      -9
## 423      -9      -9
## 424      -9      -9
## 425      -8      -8
## 426      -8      -8
## 427    1000      -9
## 428      -8      -8
## 429      -9      -9
## 430      -9      -9
## 431      -8      -8
## 432      -9      -9
## 433      -8      -8
## 434      -9      -9
## 435    1200      -9
## 436      -8      -8
## 437     300      -8
## 438      -8      -8
## 439      -8      -9
## 440      -8      -8
## 441      -8      -8
## 442      -9      -9
## 443      -8      -8
## 444      -8      -8
## 445     420      -8
```

```
## 446      -8      -8
## 447     600      -9
## 448      -9      -9
## 449    1200       0
## 450      -8      -8
## 451     500      -8
## 452      -8      -8
## 453      -9      -9
## 454      -8      -8
## 455      -9      -9
## 456     900     350
## 457      -8      -8
## 458      -8      -8
## 459       0       0
## 460      -8      -8
## 461    2000      -8
## 462      -9      -9
## 463      -8      -8
## 464    1300      -8
## 465      -8      -8
## 466      -8      -8
## 467      -8      -8
## 468      -8      -8
## 469      -8      -8
## 470      -8      -8
## 471      -8      -8
## 472      -8      -8
## 473      -8      -8
## 474    1000      -8
## 475      -8      -8
## 476      -8    1000
## 477    1800    3000
## 478      -8      -8
## 479     300      -9
## 480    1200      -8
## 481      -8      -8
## 482      -8      -8
## 483      -8      -8
## 484      -8      -8
## 485      -8      -8
## 486     450      -8
## 487      -9      -9
## 488       0       0
## 489      -8      -8
## 490      -8      -8
## 491      -9      -9
## 492      -8      -8
## 493      -8      -8
## 494      -8      -8
## 495      -8      -8
## 496      -8      -8
## 497    1800    1400
## 498      -8      -8
## 499    1200      -8
```

```
## 500     590      -8
## 501      -8      -8
## 502    1200      -9
## 503      -8      -8
## 504     300      -8
## 505      -8      -8
## 506      -8      -8
## 507      -8      -8
## 508      -8      -8
## 509      -9      -9
## 510    5150      -9
## 511      -8    2200
## 512      -9      -9
## 513    3500      -9
## 514    1000      -8
## 515    3500      -8
## 516    1500      -8
## 517      -8      -8
## 518      -9      -9
## 519      -9      -9
## 520    1100      -8
## 521    1800      -9
## 522      -8      -8
## 523      -8      -8
## 524      -9      -9
## 525      -8      -8
## 526       0       0
## 527      -8      -8
## 528     300     150
## 529    2000    1800
## 530    1200      -9
## 531     500       0
## 532    5150      -8
## 533       0       0
## 534      -9      -9
## 535      -8      -8
## 536      -8      -8
## 537    2400   17314
## 538     520     160
## 539      -8      -8
## 540      -8      -8
## 541      -8      -8
## 542     600      -8
## 543      -8      -8
## 544      -8   10000
## 545     700      -9
## 546      -9      -9
## 547      -8      -9
## 548      -8     200
## 549      -8      -8
## 550      -8     600
## 551    1200      -8
## 552     500       0
## 553      -9      -8
```

```
## 554      -8      -8
## 555    3000     300
## 556      -9      -9
## 557      -9      -9
## 558      -9      -9
## 559      -9      -9
## 560      -9    2700
## 561      -8      -8
## 562      -8      -9
## 563      -8      -8
## 564      -8      -8
## 565      -8      -8
## 566      -9      -9
## 567     300      -8
## 568     200      -9
## 569    1000      -8
## 570      -8      -8
## 571      -8      -8
## 572      -9      -9
## 573      -8      -8
## 574    2400       0
## 575      -9      -9
## 576      -9      -9
## 577      -8      -8
## 578      -8      -8
## 579      -9      -9
## 580      -8      -8
## 581      -9      -9
## 582      -8      -8
## 583      -9      -9
## 584     800      -9
## 585      -8      -9
## 586      -8      -9
## 587      -9      -9
## 588     100      -8
## 589      -8      -8
## 590      -9      -9
## 591      -9      -9
## 592      -9      -9
## 593      -9      -9
## 594      -9      -9
## 595      -8      -8
## 596     250      -8
## 597    1000    1000
## 598     250      -8
## 599      -9      -9
## 600      -9      -9
## 601    1200       0
## 602       0       0
## 603      -8      -8
## 604      -9      -9
## 605      -8      -8
## 606       0      -8
## 607     600     250
```

```
## 608      -9      -9
## 609     400     660
## 610      -9      -9
## 611     300      -9
## 612    1800       0
## 613      -9      -9
## 614      -9      -9
## 615      -9      -9
## 616    2900    1000
## 617     500       0
## 618      -9      -9
## 619    1000      -8
## 620    1200       0
## 621    2000      -8
## 622      -8      -8
## 623      -9      -8
## 624    1000      -8
## 625     510      -8
## 626      -9      -8
## 627      -8      -8
## 628      -8      -8
## 629      -9      -9
## 630      -8      -8
## 631      -8      -8
## 632     500      -9
## 633      -9      -9
## 634    1000      -8
## 635      -9      -9
## 636       0       0
## 637      -8      -8
## 638    1200    7600
## 639      -8      -8
## 640      -9      -9
## 641      -9      -9
## 642     500      -8
## 643      -9      -9
## 644      -8      -8
## 645     400      -9
## 646      -8      -8
## 647      -9      -9
## 648      -9      -9
## 649    1200       0
## 650      -8      -8
## 651      -9      -9
## 652     480    1800
## 653    1000      -8
## 654     400      -8
## 655      -8      -8
## 656      -9      -9
## 657      -8    3000
## 658      -9      -9
## 659      -8      -8
## 660     500    2000
## 661      -9      -9
```

```
## 662      -8     280
## 663    1200      -8
## 664      -9      -9
## 665      -9      -9
## 666      -8      -8
## 667    1200       0
## 668     500      -8
## 669       0    1500
## 670      -8      -8
## 671      -8      -8
## 672    1000      -8
## 673      -8      -9
## 674      -8      -8
## 675    1100      -8
## 676    2400   10500
## 677      -8      -8
## 678      -9   13000
## 679      -8      -8
## 680    1400      -8
## 681    3500      -8
## 682      -8      -8
## 683     440      -8
## 684    1400      -9
## 685     300      -8
## 686      -9      -9
## 687    2000       0
## 688    1000      -9
## 689       0      -8
## 690     630      -8
## 691      -9      -9
## 692      -9      -9
## 693    3000      -9
## 694      -8      -9
## 695      -8      -8
## 696      -8      -8
## 697    1000     410
## 698      -9    1200
## 699     300      -8
## 700      -8    1300
## 701      -8      -8
## 702      -9      -9
## 703     480      -8
## 704      -9      -9
## 705      -9      -9
## 706      -9      -9
## 707     400     400
## 708      -8      -8
## 709      -9      -9
## 710     200      -8
## 711      -9      -9
## 712      -9      -9
## 713     550      -8
## 714      -9      -9
## 715       0       0
```

```
## 716      -8      -8
## 717    1500      -8
## 718     600      -9
## 719    1200    4000
## 720      -8    2100
## 721    2500       0
## 722      -8      -8
## 723      -8      -8
## 724     700      -8
## 725    3000    2000
## 726      -9      -9
## 727      -8      -8
## 728    1700   17314
## 729      -9      -9
## 730    1200      -8
## 731      -9      -9
## 732      -8      -8
## 733      -8      -8
## 734     800      -9
## 735      -9      -9
## 736      -9      -9
## 737    3000      -8
## 738     600     270
## 739      -8      -8
## 740     900      -8
## 741      -9      -9
## 742      -8    1000
## 743    1400     960
## 744      -8      -8
## 745      -9      -9
## 746      -8      -8
## 747      -9      -9
## 748      -8      -8
## 749      -9      -9
## 750     600      -8
## 751    1000    1500
## 752     200      -8
## 753      -8      -8
## 754    4000     600
## 755      -8      -8
## 756      -8      -8
## 757    1000      -8
## 758      -9      -8
## 759     300      -8
## 760    3600      -8
## 761      -9      -9
## 762      -9      -9
## 763      -8      -9
## 764      -9      -9
## 765      -8      -8
## 766      -9      -9
## 767    1400      -8
## 768      -8      -8
## 769      -8      -8
```

```
## 770      -8      -9
## 771      -8      -8
## 772    1000      -8
## 773      -8      -8
## 774      -8      -8
## 775      -9      -9
## 776      -8      -8
## 777      -9      -9
## 778      -8      -8
## 779    1200      -8
## 780    1000      -8
## 781     500    1000
## 782    1200      -8
## 783      -8      -8
## 784      -8      -8
## 785     600    1000
## 786      -9      -9
## 787      -8      -8
## 788      -8    1100
## 789      -8      -9
## 790    1500      -8
## 791    4000    7000
## 792      -8    2000
## 793    2400      -8
## 794      -8      -8
## 795     800      -8
## 796      -8      -8
## 797      -9      -8
## 798      -8      -8
## 799       0      -8
## 800      -9      -9
## 801      -8      -8
## 802      -8      -8
## 803    2000      -8
## 804      -9      -9
## 805      -9      -9
## 806    1800      -8
## 807    5150     800
## 808      -8      -8
## 809       0      -9
## 810     720      -8
## 811    2500    1200
## 812     500      -8
## 813    1000      -8
## 814      -9      -9
## 815      -8      -8
## 816    2000      -9
## 817      -9      -9
## 818      -8    4700
## 819       0    2000
## 820     430      -8
## 821      -8      -8
## 822      -8    2000
## 823    1200      -9
```

```
## 824    1100    1000
## 825    5150      -8
## 826      -9      -9
## 827      -9      -9
## 828      -9      -9
## 829      -9      -9
## 830      -9      -9
## 831      -9      -9
## 832      -9      -9
## 833      -9      -9
## 834      -9      -9
## 835      -9      -9
## 836      -9      -9
## 837      -9      -9
## 838      -8      -8
## 839      -9      -9
## 840      -9      -9
## 841      -9      -9
## 842      -9      -9
## 843      -9      -9
## 844      -9      -9
## 845      -9      -9
## 846      -9      -9
## 847      -9      -9
## 848      -8      -8
## 849      -8      -8
## 850      -8      -8
## 851      -9      -9
## 852    7000      -8
## 853      -8      -8
## 854    2400   31700
## 855      -8      -8
## 856      -9      -9
## 857      -8   40000
## 858    6700      -9
## 859      -9      -9
## 860      -9      -9
## 861      -9      -9
## 862       0      -8
## 863      -9      -9
## 864      -9      -9
## 865      -9      -9
## 866      -8      -8
## 867      -9      -9
## 868      -8      -8
## 869      -9      -9
## 870      -9      -9
## 871      -9      -9
## 872      -9      -9
## 873      -9      -9
## 874      -9      -9
## 875      -9      -9
## 876      -9      -9
## 877      -8      -9
```

```
## 878      -9      -9
## 879      -9      -9
## 880      -9      -9
## 881      -9      -9
## 882      -9      -9
## 883      -9      -9
## 884   11300     490
## 885      -9      -9
## 886      -8      -8
## 887      -9      -9
## 888      -8      -8
## 889      -9      -9
## 890      -9      -9
## 891      -9      -9
## 892      -9      -9
## 893      -9      -9
## 894      -9      -9
## 895      -9      -9
## 896      -9      -9
## 897    1500    4000
## 898      -9      -9
## 899      -9      -9
## 900      -9      -9
## 901      -9      -9
## 902      -9      -9
## 903      -9      -9
## 904      -9      -9
## 905      -8      -8
## 906    4200      -9
## 907      -9      -9
## 908    1500      -8
## 909      -8      -8
## 910      -9      -9
## 911      -9      -9
## 912      -9      -9
## 913     720      -9
## 914      -9      -9
## 915      -8      -8
## 916      -8      -9
## 917      -9      -9
## 918    2500      -8
## 919      -9      -9
## 920      -8      -9
## 921      -9      -9
## 922     750    3500
## 923      -8      -8
## 924      -8      -8
## 925      -8      -8
## 926      -8      -8
## 927      -9  120000
## 928      -9      -9
## 929      -8      -8
## 930      -9      -9
## 931      -9      -9
```

```
## 932    1800    7000
## 933      -9      -9
## 934   14191   42400
## 935    4300     180
## 936      -8      -8
## 937     400       0
## 938      -9      -9
## 939    3500      -9
## 940      -9      -9
## 941      -9      -9
## 942     600       0
## 943      -9      -9
## 944      -8      -9
## 945      -9    2700
## 946      -9      -9
## 947      -9      -9
## 948      -9      -9
## 949      -9      -9
## 950      -8      -8
## 951      -8      -8
## 952      -9      -9
## 953      -8      -8
## 954       0      -9
## 955      -8      -8
## 956      -8      -8
## 957      -8      -8
## 958   12000      -9
## 959      -8      -9
## 960      -8      -8
## 961      -9      -9
## 962    5400    3000
## 963      -9      -9
## 964      -8      -8
## 965      -8      -8
## 966      -9      -9
## 967      -9      -9
## 968      -8      -8
## 969      -8      -8
## 970      -8   19000
## 971      -8      -8
## 972      -9      -9
## 973     460   11300
## 974      -8      -8
## 975      -8      -8
## 976      -8      -8
## 977      -8      -8
## 978      -9      -9
## 979      -8      -8
## 980      -8      -8
## 981      -8      -8
## 982      -8      -8
## 983      -8      -9
## 984      -9      -9
## 985      -8      -8
```

```
## 986      -9      -9
## 987      -8      -8
## 988    1200      -8
## 989      -8    7900
## 990    4200    2800
## 991      -9      -9
## 992   12000      -8
## 993      -9      -9
## 994      -8      -8
## 995    5000   15000
## 996      -8   26000
## 997      -8      -8
## 998      -8      -8
## 999    2400   16200
## 1000     -8      -8
## 1001   2000   10000
## 1002     -8      -8
## 1003     -8      -8
## 1004     -8      -8
## 1005   1600    5000
## 1006   4200      -8
## 1007     -9      -9
## 1008    340   43700
## 1009    720    1000
## 1010     -8      -8
## 1011     -8      -8
## 1012     -8      -8
## 1013    540      -8
## 1014   2500    7200
## 1015     -8      -8
## 1016   8500    9700
## 1017     -8      -8
## 1018     -8      -8
## 1019     -8      -8
## 1020    800      -8
## 1021   6500    3400
## 1022     -8      -8
## 1023     -8      -8
## 1024     -8      -8
## 1025     -9      -9
## 1026     -8      -8
## 1027     -8      -8
## 1028     -8      -8
## 1029     -9      -9
## 1030     -9      -9
## 1031   2000      -9
## 1032   2400      -8
## 1033     -8      -9
## 1034     -8   13700
## 1035     -9      -9
## 1036      0    6300
## 1037     -8      -8
## 1038     -9    6300
## 1039      0   12100
```

```
## 1040   2100    1900
## 1041     -8  191949
## 1042     -8      -8
## 1043     -9    7500
## 1044   1100    3000
## 1045   4000    2000
## 1046     -9      -9
## 1047     -8      -8
## 1048   9800    7400
## 1049     -8      -8
## 1050   2200    5200
## 1051   3300     600
## 1052     -9      -9
## 1053     -8      -8
## 1054  14191      -9
## 1055     -8      -8
## 1056     -9      -9
## 1057     -9      -9
## 1058      0    1400
## 1059     -8      -8
## 1060     -8      -8
## 1061   5200   16800
## 1062   5000    1300
## 1063   2900    8700
## 1064   2200    1400
## 1065  14191  191949
## 1066   1800      -9
## 1067   4400   11100
## 1068   3600       0
## 1069   2000      -8
## 1070     -8      -8
## 1071     -8      -8
## 1072   1000     800
## 1073     -8      -9
## 1074    810     530
## 1075    600      -8
## 1076     -9      -9
## 1077     -8      -8
## 1078   2100    2900
## 1079   1100      -8
## 1080     -8      -8
## 1081   3000      -8
## 1082     -8      -8
## 1083     -9      -9
## 1084     -9      -9
## 1085     -8      -8
## 1086     -9      -9
## 1087     -9      -9
## 1088     -8      -8
## 1089   7100   55000
## 1090     -9      -9
## 1091     -8      -8
## 1092     -9      -9
## 1093      0   31200
```

```
## 1094      0       0
## 1095     -9   12000
## 1096  11100   39600
## 1097     -8      -8
## 1098   3900      -9
## 1099   7000    9000
## 1100     -8      -8
## 1101   7500  191949
## 1102   2800      -9
## 1103     -8      -8
## 1104     -8      -8
## 1105   3300    9300
## 1106     -8      -8
## 1107     -8      -8
## 1108   8400      -8
## 1109     -9      -9
## 1110     -9      -9
## 1111     -8      -8
## 1112    800       0
## 1113     -8      -8
## 1114     -9      -9
## 1115     -8      -8
## 1116   1300   75000
## 1117   2000      -8
## 1118     -9      -9
## 1119     -8      -8
## 1120     -8      -8
## 1121   1800      -9
## 1122   4100      -8
## 1123      0       0
## 1124     -9      -9
## 1125     -8      -8
## 1126     -8      -8
## 1127     -9      -9
## 1128   6300    2500
## 1129     -8      -8
## 1130     -9      -9
## 1131   6000    5000
## 1132     -9      -9
## 1133     -8      -8
## 1134     -8      -8
## 1135     -8      -8
## 1136  10000   20000
## 1137     -8      -8
## 1138     -8      -8
## 1139     -8      -8
## 1140     -8      -8
## 1141   2400       0
## 1142     -9      -9
## 1143   2900    6100
## 1144   3900      -8
## 1145   5000      -9
## 1146   4700      -9
## 1147     -8      -8
```

```
## 1148     -9      -9
## 1149     -9      -9
## 1150     -9      -9
## 1151     -9      -9
## 1152     -8      -8
## 1153   2400    3600
## 1154     -9      -9
## 1155     -8      -8
## 1156   4000    7500
## 1157     -8      -8
## 1158   6500      -8
## 1159   1200      -8
## 1160     -8      -8
## 1161     -8      -8
## 1162     -8    9000
## 1163     -9      -9
## 1164   5400   13600
## 1165   2500      -8
## 1166     -8      -8
## 1167     -9      -9
## 1168     -8      -8
## 1169    490    2500
## 1170     -8      -8
## 1171   1800      -8
## 1172     -8      -8
## 1173     -8      -8
## 1174     -8      -8
## 1175     -9      -8
## 1176   2500    2700
## 1177     -9    8000
## 1178     -9      -9
## 1179   4200      -8
## 1180     -9      -9
## 1181   1500      -8
## 1182     -8      -9
## 1183     -8      -8
## 1184     -8      -8
## 1185   2000    2000
## 1186     -9      -9
## 1187     -9      -9
## 1188     -9    1200
## 1189     -8      -8
## 1190      0       0
## 1191   5500      -9
## 1192     -8      -8
## 1193     -9      -9
## 1194     -9      -9
## 1195   3300      -8
## 1196   1500       0
## 1197   1400    3300
## 1198   8500      -8
## 1199     -9      -9
## 1200     -9      -9
## 1201    500      -8
```

```
## 1202   1300    1100
## 1203   6900    3800
## 1204     -9      -9
## 1205     -8      -9
## 1206    890    1100
## 1207   4000       0
## 1208      0      -9
## 1209     -8      -8
## 1210     -9      -9
## 1211   3600      -8
## 1212     -9      -9
## 1213   6000      -8
## 1214     -8    6200
## 1215     -9      -9
## 1216     -8      -9
## 1217   2300    2000
## 1218   1200    4700
## 1219   5000      -8
## 1220     -9      -9
## 1221     -9      -9
## 1222     -9      -9
## 1223   6000      -8
## 1224   2800      -8
## 1225     -8      -8
## 1226     -8       0
## 1227     -8      -9
## 1228   5000      -8
## 1229     -8      -8
## 1230     -9      -9
## 1231    750      -8
## 1232   1600       0
## 1233     -8      -8
## 1234   2200   18000
## 1235    600    2600
## 1236   2400      -9
## 1237   1300      -9
## 1238     -8      -8
## 1239     -8    4100
## 1240     -9      -9
## 1241     -9      -9
## 1242     -9    1200
## 1243   2400      -9
## 1244   6200   11200
## 1245   5000   10000
## 1246     -8      -8
## 1247     -9      -9
## 1248    900      -8
## 1249     -9      -9
## 1250    500      -9
## 1251     -8      -9
## 1252   1800      -9
## 1253     -9      -9
## 1254     -8      -8
## 1255   1800      -8
```

```
## 1256     -9      -9
## 1257   1200       0
## 1258   4500      -8
## 1259      0       0
## 1260     -8      -8
## 1261   9000      -8
## 1262     -9      -9
## 1263      0     170
## 1264     -9      -9
## 1265     -9   70000
## 1266   1200      -8
## 1267     -8    4600
## 1268     -8      -8
## 1269   1000    3000
## 1270     -9      -9
## 1271   1000      -8
## 1272     -9      -9
## 1273     -8      -8
## 1274     -9      -9
## 1275     -9      -9
## 1276     -8      -8
## 1277   3000      -8
## 1278     -8      -8
## 1279     -8      -8
## 1280     -9      -8
## 1281     -8      -8
## 1282   5000      -8
## 1283     -9      -9
## 1284     -9      -9
## 1285   1400    4900
## 1286     -8      -8
## 1287   1300      -8
## 1288    360      -9
## 1289     -9      -9
## 1290     -9      -9
## 1291      0    5800
## 1292     -8      -9
## 1293     -9      -9
## 1294    700    1400
## 1295     -9      -9
## 1296   1000      -8
## 1297     -9      -9
## 1298   4300    4100
## 1299     -9      -9
## 1300     -9      -9
## 1301     -9      -9
## 1302   4900    7500
## 1303   5000   10000
## 1304    500      -8
## 1305   2600    3200
## 1306     -9      -9
## 1307      0       0
## 1308     -9      -9
## 1309     -9      -9
```

```
## 1310     -9      -9
## 1311   1000      -8
## 1312     -8      -9
## 1313   6400    2800
## 1314      0       0
## 1315     -9      -9
## 1316     -9      -9
## 1317   2200    9000
## 1318     -8    6500
## 1319   2400    5200
## 1320     -8    2400
## 1321     -8    6500
## 1322     -9      -9
## 1323     -9   15700
## 1324   2100      -8
## 1325     -8      -8
## 1326     -9      -9
## 1327   3600       0
## 1328     -8      -8
## 1329     -8      -8
## 1330    240      -8
## 1331    750    3500
## 1332   4900       0
## 1333   5000      -9
## 1334     -8      -8
## 1335   4000   45000
## 1336     -9      -9
## 1337     -9      -9
## 1338      0    2000
## 1339     -9      -9
## 1340     -8      -8
## 1341     -8      -8
## 1342     -8    3500
## 1343    840    5200
## 1344     -9      -9
## 1345     -8      -8
## 1346   6300   34300
## 1347     -9      -9
## 1348   3600      -8
## 1349   2500      -9
## 1350   3600    6000
## 1351   5500   57000
## 1352   1100    3900
## 1353      0      -9
## 1354   3000      -8
## 1355     -8      -8
## 1356   1000   20000
## 1357     -9    3300
## 1358  14191       0
## 1359     -9      -9
## 1360     -9      -9
## 1361   1000       0
## 1362     -9      -9
## 1363     -9      -9
```

```
## 1364     -8      -8
## 1365   9500      -8
## 1366     -9      -9
## 1367   6000      -8
## 1368     -9      -9
## 1369   3500      -8
## 1370     -8      -8
## 1371     -9      -9
## 1372   1400      -8
## 1373   2400      -8
## 1374   3200      -9
## 1375   1200   10400
## 1376   1200      -8
## 1377     -9  191949
## 1378     -9      -9
## 1379     -9      -9
## 1380    850    1300
## 1381      0       0
## 1382    930    3700
## 1383     -9      -9
## 1384   2600      -8
## 1385    500      -8
## 1386     -9      -9
## 1387   2200      -9
## 1388     -8    3000
## 1389   5000   10000
## 1390   5400      -9
## 1391     -9      -9
## 1392   2200      -8
## 1393     -9      -9
## 1394   4000      -8
## 1395     -9      -9
## 1396     -9      -9
## 1397     -9      -9
## 1398   3000      -9
## 1399   3100      -8
## 1400   1400      -8
## 1401   6200    8800
## 1402  14191   12500
## 1403     -9      -9
## 1404     -9      -9
## 1405     -9      -9
## 1406   1100    1400
## 1407   5000    8000
## 1408     -9      -9
## 1409     -9      -9
## 1410     -9      -9
## 1411   2100    2300
## 1412   2000       0
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## 1414     -9      -9
## 1415   5000    8000
## 1416     -8      -9
## 1417     -9      -9
```

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## 1418     -8      -8
## 1419     -8      -8
## 1420   3500    2500
## 1421     -9      -9
## 1422     -9      -9
## 1423     -9      -9
## 1424   1000    1800
## 1425     -8      -8
## 1426     -8      -8
## 1427   3000      -8
## 1428     -9      -9
## 1429   2000   21300
## 1430   3600      -9
## 1431     -9      -9
## 1432   4000    6000
## 1433   2000      -8
## 1434   3000      -8
## 1435   1200   24000
## 1436     -8    9000
## 1437   3600      -8
## 1438  14191    1200
## 1439   4800      -8
## 1440   1900      -9
## 1441   1200      -9
## 1442   2500      -8
## 1443   7500      -9
## 1444     -8      -8
## 1445   8000      -9
## 1446   1200    1200
## 1447      0      -9
## 1448   3000    5000
## 1449   8000   24000
## 1450   1800    3100
## 1451   1200    2300
## 1452      0    1000
## 1453     -8      -8
## 1454     -8      -8
## 1455     -8      -8
## 1456     -9      -9
## 1457     -9      -9
## 1458    960     230
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## 1461     -9      -9
## 1462   8800   10900
## 1463      0       0
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## 1465     -9   25000
## 1466     -9      -9
## 1467     -9      -9
## 1468     -9      -9
## 1469   9300    6700
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## 1471     -8      -9
```

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## 1473     -9   30400
## 1474   4400  124000
## 1475     -8      -8
## 1476     -9      -9
## 1477     -8   25000
## 1478   1800      -8
## 1479     -9      -9
## 1480     -8    8300
## 1481     -9      -9
## 1482     -9      -9
## 1483     -9      -9
## 1484     -8   23100
## 1485     -8       0
## 1486     -9      -9
## 1487     -9      -9
## 1488     -9      -9
## 1489     -9      -9
## 1490     -9      -9
## 1491     -9      -9
## 1492     -9      -9
## 1493     -9      -9
## 1494     -9      -9
## 1495     -9      -9
## 1496     -9      -9
## 1497     -9      -9
## 1498     -9      -9
## 1499     -9      -9
## 1500     -9      -9
## 1501     -9      -9
## 1502     -9      -9
## 1503     -9      -9
## 1504     -9      -9
## 1505     -9      -9
## 1506     -9      -9
## 1507     -8      -8
## 1508     -9      -9
## 1509     -9      -9
## 1510     -9      -9
## 1511     -9      -9
## 1512     -9      -9
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## 1514     -9      -9
## 1515     -9      -9
## 1516     -9      -9
## 1517     -8      -8
## 1518     -9      -9
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## 1521     -9      -9
## 1522     -8      -8
## 1523     -9      -9
## 1524     -9      -9
## 1525     -9      -9
```

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## 1526     -9      -9
## 1527     -9      -9
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## 1532     -9      -9
## 1533     -9      -9
## 1534     -9      -9
## 1535     -9      -9
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## 1539     -9      -9
## 1540     -9      -9
## 1541     -9      -9
## 1542     -9      -9
## 1543     -8      -8
## 1544      0       0
## 1545   8000   14000
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## 1547   1200    9400
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## 1549     -9      -9
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## 1551   5200   70000
## 1552   5000      -9
## 1553     -9      -9
## 1554   1000      -8
## 1555  18000   15000
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## 1557   7700    9900
## 1558     -9      -9
## 1559     -9      -9
## 1560  18000   30000
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## 1562     -9      -9
## 1563     -9      -9
## 1564     -9      -9
## 1565     -9      -9
## 1566     -9      -9
## 1567     -9      -9
## 1568     -9      -9
## 1569     -9      -9
## 1570     -8      -8
## 1571     -8      -8
## 1572     -8      -8
## 1573     -8      -8
## 1574     -8      -8
## 1575     -8      -8
## 1576     -8  402571
## 1577     -8      -8
## 1578     -8      -8
## 1579     -9      -9
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## 1580     -9      -9
## 1581     -9      -9
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## 1584  11800    3000
## 1585     -8      -8
## 1586   4800      -9
## 1587     -8      -8
## 1588     -8      -8
## 1589     -9      -9
## 1590     -9      -9
## 1591     -8      -8
## 1592     -8      -8
## 1593     -8      -8
## 1594     -8      -8
## 1595     -9      -9
## 1596     -9      -9
## 1597     -9      -9
## 1598     -9      -9
## 1599     -9      -9
## 1600     -9      -9
## 1601     -8   19300
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## 1603     -9      -9
## 1604  12600    6800
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## 1606     -8      -8
## 1607  12000      -9
## 1608     -9      -9
## 1609     -9      -9
## 1610     -8      -8
## 1611     -8      -9
## 1612     -8      -8
## 1613     -8      -8
## 1614     -8      -8
## 1615     -8      -8
## 1616     -8      -8
## 1617     -8      -8
## 1618     -8      -8
## 1619   4500      -9
## 1620     -9      -9
## 1621     -8      -8
## 1622   8800      -9
## 1623     -8      -8
## 1624  10700   23300
## 1625     -8      -8
## 1626  33337  182000
## 1627     -8      -8
## 1628     -8      -8
## 1629     -8      -9
## 1630     -8      -8
## 1631     -9      -9
## 1632   7100    4300
## 1633     -8      -8
```

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## 1634     -8      -8
## 1635     -8      -8
## 1636     -9      -9
## 1637     -8      -8
## 1638   9100   11600
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## 1640     -9      -9
## 1641   4300   19200
## 1642     -8      -8
## 1643   8900      -8
## 1644     -8      -8
## 1645     -9      -9
## 1646     -8      -8
## 1647   4300   13400
## 1648     -8      -8
## 1649     -8      -8
## 1650   6400   35400
## 1651   8800   23000
## 1652   9400   73000
## 1653   5000   48700
## 1654     -8      -8
## 1655     -9      -9
## 1656   7200   26300
## 1657     -8      -8
## 1658     -8      -8
## 1659   7800   12200
## 1660     -8      -8
## 1661     -8      -8
## 1662     -8      -8
## 1663     -8      -8
## 1664     -8   13900
## 1665     -8      -8
## 1666  20000      -9
## 1667     -8      -8
## 1668     -8      -8
## 1669     -8      -8
## 1670     -8      -8
## 1671     -8      -8
## 1672   1200    5400
## 1673     -8      -8
## 1674  17700   45600
## 1675     -8      -8
## 1676   4400  113000
## 1677  20400       0
## 1678   2400   18800
## 1679     -9      -9
## 1680  17600   40100
## 1681   1400    5800
## 1682     -9  189000
## 1683   9200   52000
## 1684   8900   20200
## 1685     -8      -8
## 1686     -9  137000
## 1687      0   93000
```

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## 1688     -9  178000
## 1689  13000  104000
## 1690  15700   13200
## 1691  13700   17900
## 1692     -8      -8
## 1693     -8   12900
## 1694   8100   34500
## 1695    300      -8
## 1696     -8      -8
## 1697     -9      -9
## 1698     -9      -8
## 1699     -8      -8
## 1700     -9      -9
## 1701     -9      -9
## 1702     -8      -8
## 1703   7800   24100
## 1704     -8      -8
## 1705     -8      -8
## 1706   6000  100000
## 1707   1300   11900
## 1708   7000      -8
## 1709   1300   11600
## 1710  14200   22700
## 1711  13800    7900
## 1712    830   19700
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## 1714   4000      -8
## 1715   2400   33500
## 1716  16300   26100
## 1717   8400   64000
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## 1719     -9      -9
## 1720   6600   18900
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## 1722  15100      -9
## 1723     -9      -9
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## 1725  16500  402571
## 1726     -8      -8
## 1727      0   95000
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## 1731     -8      -8
## 1732     -8      -8
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## 1739     -8      -8
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## 1741     -8      -8
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## 1744     -9      -9
## 1745     -8      -8
## 1746   2000      -9
## 1747     -8   37400
## 1748     -8      -8
## 1749     -8      -8
## 1750  12400    8600
## 1751   6500   21300
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## 1753     -8      -8
## 1754   2900    1700
## 1755     -8      -8
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## 1757     -8      -8
## 1758   5500   29800
## 1759     -8      -8
## 1760  20800   62000
## 1761     -8      -8
## 1762   2200      -8
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## 1764   7000      -8
## 1765   3600      -9
## 1766  10800   16700
## 1767  12000      -8
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## 1769     -8      -8
## 1770     -8      -8
## 1771      0   43700
## 1772  20000      -8
## 1773     -9      -9
## 1774     -8   14200
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## 1776     -8      -8
## 1777     -8      -8
## 1778   3900      -9
## 1779     -8      -9
## 1780     -8      -8
## 1781     -8      -8
## 1782     -8      -8
## 1783   3800    5800
## 1784     -8      -8
## 1785  12000  200000
## 1786     -9      -9
## 1787  16000   20000
## 1788     -9      -8
## 1789     -9      -9
## 1790     -9      -9
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## 1792     -9      -9
## 1793     -9      -9
## 1794     -9      -9
## 1795     -8      -8
```

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## 1796     -9      -9
## 1797     -8      -8
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## 1799    980   25600
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## 1801   6100      -8
## 1802   4600   18200
## 1803     -8    8800
## 1804     -8    4000
## 1805   5000      -8
## 1806     -9      -9
## 1807     -8   66000
## 1808     -8      -8
## 1809   9800      -8
## 1810   3200      -9
## 1811     -9      -9
## 1812     -9      -9
## 1813     -9    3200
## 1814     -8      -9
## 1815     -8      -8
## 1816     -9      -9
## 1817     -8      -8
## 1818   1500    2000
## 1819  33337   91000
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## 1821     -8      -9
## 1822     -8      -9
## 1823     -9      -9
## 1824   8000   42700
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## 1831     -9      -9
## 1832     -9      -9
## 1833   4000      -8
## 1834     -9      -9
## 1835     -9      -9
## 1836     -8  402571
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## 1838  10000      -9
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## 1842     -9      -9
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## 1844     -9      -9
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## 1846   2700   12000
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## 1849     -8      -8
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## 1852   2000    4100
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## 1857     -8      -8
## 1858     -9      -9
## 1859     -9      -9
## 1860  11500      -9
## 1861     -8   26700
## 1862     -9      -9
## 1863     -9   47900
## 1864   9800   26200
## 1865     -9      -9
## 1866   4000   10000
## 1867   1000   24100
## 1868  10000   25700
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## 1873   1300   13500
## 1874   1900    2200
## 1875   3000    5000
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## 1877   8000   27900
## 1878     -8      -8
## 1879  10000      -8
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## 1882     -9      -9
## 1883     -8      -8
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## 1885   2400      -8
## 1886    800   36000
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## 1888     -8      -9
## 1889   1800   61000
## 1890     -9      -9
## 1891   1300  113000
## 1892     -9      -9
## 1893     -8      -8
## 1894  17000      -8
## 1895     -8   55000
## 1896   5000    5000
## 1897     -9      -9
## 1898   3800      -8
## 1899     -8      -8
## 1900     -9      -9
## 1901   7000    2400
## 1902     -8      -8
## 1903      0   53000
```

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## 1906     -9      -9
## 1907   8000   20000
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## 1909   6000      -8
## 1910     -9   30000
## 1911   8500   15000
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## 1914     -9      -9
## 1915     -8      -8
## 1916   4500    2200
## 1917     -8      -9
## 1918   2500      -9
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## 1921   7200      -8
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## 1924  33337   42000
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## 1928   5700   18100
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## 1933     -9      -9
## 1934     -9      -9
## 1935     -9      -9
## 1936     -9      -9
## 1937     -9      -9
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## 1940     -9      -9
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## 2009     -9      -9
## 2010     -9      -9
## 2011     -9      -9
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## 2017     -8      -8
## 2018     -9      -9
## 2019     -9      -9
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## 2023     -9      -9
## 2024     -9      -9
## 2025     -9   27200
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## 2035     -9      -9
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## 2047     -9      -9
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## 2117     -9      -9
## 2118  94000      -9
## 2119  12000      -9
```

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## 3182     -8      -8
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## 3190     -9      -9
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## 3194  18000   72000
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## 3196     -9      -9
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## 3199     -9      -9
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## 3228  15900    2700
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## 3230  20600   96000
## 3231     -8      -8
## 3232   3500      -8
## 3233     -8      -8
## 3234     -9   52000
## 3235     -8      -8
## 3236  40100   62000
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## 3238     -9      -9
## 3239  12200  211000
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## 3242     -8      -8
## 3243     -8      -8
## 3244     -8      -8
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## 3248  63000  714000
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## 3250     -9      -9
## 3251  33800  217000
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## 3253     -8      -8
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## 3263     -9      -9
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## 3277     -9      -9
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## 3299     -8      -8
## 3300     -8      -8
## 3301     -8      -8
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## 3305  11100  279000
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## 3307     -9      -9
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## 3317  18300  117000
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## 3320  12100   15200
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## 3322  28000  120000
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## 3331  24900   16600
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## 3358     -9      -9
## 3359  30600   77000
## 3360  12400   81000
## 3361  17900  117000
```

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## 3377     -9      -9
## 3378 146000  169000
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## 3382  23300   10000
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## 3384   5000      -8
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## 3386     -9      -9
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## 3388     -8      -8
## 3389     -9      -9
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## 3395     -9      -9
## 3396     -8      -8
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## 3398     -8      -8
## 3399     -8      -8
## 3400  12000  240000
## 3401     -8      -8
## 3402  33300      -9
## 3403     -8      -8
## 3404  23000  137000
## 3405 259046      -9
## 3406     -8   64000
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## 3409     -9      -9
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## 3411     -8      -8
## 3412     -8      -8
## 3413     -8      -8
## 3414     -9      -9
## 3415     -8      -8
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## 3419     -9      -9
## 3420     -9      -9
## 3421     -8      -8
## 3422  13000   65000
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## 3424     -8      -8
## 3425     -9      -9
## 3426     -8      -8
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## 3432  10700   88000
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## 3434  26800   78000
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## 3436  10700   33400
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## 3447     -8 2548054
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## 3449     -8      -8
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## 3469     -8      -8
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## 3508     -8      -8
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## 3512     -8      -8
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## 3516     -8      -8
## 3517     -8      -8
## 3518     -8   40000
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## 3523     -9  139000
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## 3572     -8      -8
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## 3574   5000      -8
## 3575     -8      -8
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## 3577     -9      -9
```

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## 3592     -9      -9
## 3593  39200      -8
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## 3605  56000  532000
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## 3607     -8      -8
## 3608     -8      -8
## 3609     -9      -9
## 3610     -9  111000
## 3611     -9      -9
## 3612     -9      -9
## 3613     -9      -9
## 3614     -8      -8
## 3615     -8      -8
## 3616     -8      -8
## 3617  24000   94000
## 3618     -8      -8
## 3619     -9      -9
## 3620     -9      -9
## 3621 102000  218000
## 3622     -8      -8
## 3623     -8  138000
## 3624   2000  218000
## 3625  34600  160000
## 3626  60000  130000
## 3627   7100   73000
## 3628     -9      -9
## 3629     -9      -9
## 3630     -8      -8
## 3631     -8      -8
```

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## 3632     -8      -8
## 3633 116000  394000
## 3634  39500  235000
## 3635  29900   60000
## 3636     -8      -8
## 3637  47100  177000
## 3638 259046  100000
## 3639     -8      -8
## 3640     -9 2548054
## 3641  20100  266000
## 3642  57000      -9
## 3643  11100  304000
## 3644  29200      -8
## 3645 259046      -9
## 3646     -8      -8
## 3647  57000  119000
## 3648     -9      -9
## 3649     -8      -8
## 3650     -9      -9
## 3651     -8      -8
## 3652     -8      -8
## 3653     -9      -9
## 3654     -8      -8
## 3655     -8      -8
## 3656  30700   59000
## 3657  54000      -8
## 3658  41600  184000
## 3659  56000      -9
## 3660     -9      -9
## 3661     -8      -8
## 3662     -9      -9
## 3663     -8      -8
## 3664     -9      -9
## 3665  50000      -9
## 3666 259046      -8
## 3667   4800      -8
## 3668     -9      -9
## 3669     -9      -9
## 3670  80000  110000
## 3671     -9      -9
## 3672     -9      -9
## 3673     -8   29500
## 3674     -9      -9
## 3675  99000  449000
## 3676     -9      -9
## 3677  26900      -8
## 3678  85000      -9
## 3679     -8      -8
## 3680     -8      -8
## 3681     -9      -9
## 3682     -8      -8
## 3683     -8      -8
## 3684     -8      -8
## 3685  64000  142000
```

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## 3686     -9      -9
## 3687  34800      -8
## 3688  78000  105000
## 3689     -8      -8
## 3690     -8      -8
## 3691     -8      -8
## 3692  52000      -9
## 3693     -9      -9
## 3694     -8      -8
## 3695     -8      -8
## 3696  78000  284000
## 3697     -9      -9
## 3698  51000  156000
## 3699     -9      -9
## 3700 100000  120000
## 3701     -9      -9
## 3702     -8      -8
## 3703  63000  156000
## 3704     -8      -8
## 3705     -9      -9
## 3706     -8      -8
## 3707     -8      -8
## 3708     -8      -8
## 3709     -9      -9
## 3710     -8      -8
## 3711     -8      -8
## 3712 100000      -8
## 3713     -9      -9
## 3714  42700   66000
## 3715  90000  400000
## 3716     -8      -8
## 3717     -8      -8
## 3718  67000   33700
## 3719     -9      -9
## 3720     -8      -8
## 3721  43500  272000
## 3722     -9      -9
## 3723  81000  504000
## 3724     -8      -8
## 3725     -8      -8
## 3726     -8      -8
## 3727     -8      -8
## 3728     -9      -9
## 3729     -8      -8
## 3730     -9      -9
## 3731     -9      -9
## 3732     -8      -8
## 3733     -8      -8
## 3734     -8      -8
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## 3736     -8      -8
## 3737  24500      -8
## 3738  77000 1090000
## 3739  59000  228000
```

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## 3740     -9      -9
## 3741     -9      -9
## 3742     -8      -8
## 3743     -8      -8
## 3744     -8      -8
## 3745     -8      -8
## 3746     -8      -8
## 3747     -9      -9
## 3748  48000      -8
## 3749     -8      -8
## 3750     -8      -8
## 3751  30000   10000
## 3752     -9      -9
## 3753     -8      -8
## 3754     -9      -9
## 3755     -9      -9
## 3756  97000  413000
## 3757  85000 2000000
## 3758     -8      -8
## 3759     -8      -8
## 3760 154000      -8
## 3761     -9      -9
## 3762  38800      -8
## 3763 143000 1290000
## 3764  45600      -8
## 3765  85000   83000
## 3766 259046   35000
## 3767     -9      -9
## 3768     -8      -8
## 3769     -8      -8
## 3770  75000   68000
## 3771     -9      -9
## 3772     -8      -8
## 3773     -8      -8
## 3774     -9      -9
## 3775     -8      -8
## 3776  82000  189000
## 3777   3900   53000
## 3778     -8      -8
## 3779  18000      -8
## 3780     -9      -9
## 3781     -8      -8
## 3782     -8      -8
## 3783  88000  514000
## 3784     -8      -8
## 3785     -9      -9
## 3786     -9      -9
## 3787     -8      -8
## 3788  50000      -9
## 3789  73000  378000
## 3790     -8      -8
## 3791  24000   15000
## 3792     -8      -8
## 3793     -9      -9
```

```
## 3794  43900       0
## 3795      0       0
## 3796     -8      -8
## 3797     -8      -8
## 3798     -9      -9
## 3799     -9      -9
## 3800     -8      -8
## 3801  74000 1800000
## 3802     -9      -9
## 3803  12700      -8
## 3804     -9      -9
## 3805     -8      -8
## 3806 100000      -8
## 3807  41100      -8
## 3808     -8      -8
## 3809 259046 1620000
## 3810     -9      -9
## 3811 259046      -8
## 3812     -8      -8
## 3813     -8      -8
## 3814     -9      -9
## 3815     -8      -8
## 3816     -9      -9
## 3817     -8      -8
## 3818     -8      -8
## 3819     -8      -8
## 3820     -9      -9
## 3821     -8      -8
## 3822 145000  571000
## 3823 100000   75000
## 3824     -8      -8
## 3825     -9      -9
## 3826     -9      -9
## 3827     -9      -9
## 3828   5300   48000
## 3829     -9      -9
## 3830     -9      -9
## 3831  46700      -8
## 3832     -8      -8
## 3833     -9      -9
## 3834     -9      -9
## 3835     -9   15400
## 3836     -8      -8
## 3837     -9  127000
## 3838     -9      -9
## 3839     -9      -9
## 3840     -9      -9
## 3841     -9      -9
## 3842 259046 1560000
## 3843   2200       0
## 3844   9500   74000
## 3845     -9      -9
## 3846   9900   48300
## 3847     -9      -9
```

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## 3848     -9      -9
## 3849  15000      -9
## 3850     -8      -8
## 3851     -8      -8
## 3852     -9      -9
## 3853     -9      -9
## 3854     -9      -9
## 3855     -9      -9
## 3856     -9      -9
## 3857  20000   12800
## 3858     -9      -9
## 3859     -9      -9
## 3860     -9      -9
## 3861     -9      -9
## 3862     -9      -9
## 3863     -8      -8
## 3864     -9      -9
## 3865     -9      -9
## 3866     -9      -9
## 3867     -9      -9
## 3868     -9      -9
## 3869   2900   28000
## 3870     -9      -9
## 3871   4000      -8
## 3872  16000      -9
## 3873     -9      -9
## 3874     -9      -9
## 3875   1800      -8
## 3876     -8      -8
## 3877  27200      -8
## 3878  88000 1520000
## 3879   2500      -8
## 3880     -9      -9
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## 3882  95000       0
## 3883  22000   35000
## 3884      0       0
## 3885  40000   50000
## 3886  13000   20000
## 3887  25100  258000
## 3888     -9      -9
## 3889     -9      -9
## 3890     -9      -9
## 3891  25500      -8
## 3892     -8      -8
## 3893   6500   38400
## 3894     -9      -9
## 3895     -9      -9
## 3896     -9      -9
## 3897  10000      -9
## 3898  33900      -8
## 3899     -9      -9
## 3900  22000   50000
## 3901     -8      -8
```

```
## 3902     -9      -9
## 3903   4000    5000
## 3904     -8      -9
## 3905  29200    5000
## 3906     -9      -9
## 3907     -9      -9
## 3908   3200      -8
## 3909     -9      -9
## 3910     -9      -9
## 3911   6000   10000
## 3912  65000      -8
## 3913  71000      -9
## 3914      0   10000
## 3915     -8      -8
## 3916     -9      -9
## 3917   6000   19700
## 3918     -9      -9
## 3919  21000  628000
## 3920  16300    2000
## 3921     -9      -9
## 3922     -9      -9
## 3923     -8      -8
## 3924     -9      -9
## 3925      0       0
## 3926  38000      -8
## 3927     -9      -9
## 3928     -8      -8
## 3929   2000    5000
## 3930     -9      -9
## 3931  14100      -9
## 3932     -8      -8
## 3933   3000    5000
## 3934     -9      -9
## 3935  12000   25000
## 3936     -8      -8
## 3937     -8      -8
## 3938     -9      -9
## 3939     -9      -9
## 3940     -9      -9
## 3941     -8      -9
## 3942  12500    8500
## 3943     -9      -9
## 3944  17400   47600
## 3945  17000      -9
## 3946  32700  119000
## 3947     -9      -9
## 3948     -9      -9
## 3949     -8    3500
## 3950   9600  141000
## 3951  11500      -8
## 3952   7500      -8
## 3953  30500  578000
## 3954     -9      -9
## 3955     -9      -9
```

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## 3956  25900   69000
## 3957   5000  122000
## 3958   5400      -9
## 3959  41800   34100
## 3960  45700  104000
## 3961     -9   34100
## 3962     -8      -8
## 3963  16200   15100
## 3964     -8      -8
## 3965     -9      -9
## 3966     -8      -8
## 3967     -8      -8
## 3968  61000      -9
## 3969  40800   54000
## 3970     -9      -9
## 3971   7000       0
## 3972      0       0
## 3973     -8      -8
## 3974  18000   15000
## 3975     -8      -8
## 3976     -9      -9
## 3977     -9      -9
## 3978     -8      -8
## 3979  30000      -9
## 3980  17100  134000
## 3981  67000  270000
## 3982     -9   41100
## 3983     -8      -8
## 3984     -9      -9
## 3985     -9      -9
## 3986     -9      -9
## 3987     -9      -9
## 3988  60000  164000
## 3989     -8      -8
## 3990 123000      -8
## 3991     -9      -9
## 3992     -9      -9
## 3993     -8   28500
## 3994     -8      -8
## 3995  38200  517000
## 3996  21500    2100
## 3997     -9      -9
## 3998  59000   82000
## 3999     -9      -9
## 4000     -8      -8
## 4001     -9      -9
## 4002  36300      -8
## 4003   2000      -8
## 4004  28400   83000
## 4005     -9      -9
## 4006  19000   34100
## 4007  20000      -8
## 4008     -9      -9
## 4009   5700   27400
```

```
## 4010   5000   15000
## 4011     -9      -9
## 4012  71000   53000
## 4013     -8      -8
## 4014     -8      -8
## 4015  16700   11500
## 4016  21700   53000
## 4017     -9      -9
## 4018  26100  129000
## 4019     -9      -9
## 4020     -8      -8
## 4021     -9      -9
## 4022  12000    9600
## 4023   7200      -8
## 4024  32000   58000
## 4025     -9      -9
## 4026  52000  116000
## 4027   5000    7000
## 4028     -9      -9
## 4029  17500  122000
## 4030  27900      -9
## 4031  38000  210000
## 4032     -9      -9
## 4033  91000  782000
## 4034     -9      -9
## 4035     -9      -9
## 4036  20000      -8
## 4037     -9      -9
## 4038  28100  119000
## 4039   5700      -8
## 4040     -9      -9
## 4041     -9      -9
## 4042     -9      -9
## 4043     -8      -8
## 4044     -9      -9
## 4045     -9      -9
## 4046  27600      -8
## 4047     -9      -9
## 4048     -9      -9
## 4049     -9      -9
## 4050     -9      -9
## 4051  25000  164000
## 4052     -9      -9
## 4053   5000    5000
## 4054     -9      -9
## 4055     -9      -9
## 4056     -9      -9
## 4057  25900   94000
## 4058  30800  107000
## 4059     -9      -9
## 4060  50000  208000
## 4061     -9      -9
## 4062     -9      -9
## 4063  42000  348000
```

```
## 4064     -9      -9
## 4065  81000  324000
## 4066     -9      -9
## 4067     -9      -9
## 4068  15000      -8
## 4069     -9      -9
## 4070     -9      -9
## 4071     -9      -9
## 4072     -8      -8
## 4073  15000  257000
## 4074     -9      -9
## 4075     -9      -9
## 4076  18000      -9
## 4077     -9      -9
## 4078     -8      -8
## 4079  12000      -9
## 4080  12000       0
## 4081     -9      -9
## 4082     -9      -9
## 4083     -8  239000
## 4084     -9      -9
## 4085  19300  233000
## 4086     -9      -9
## 4087   3300      -8
## 4088     -9      -9
## 4089  19700      -8
## 4090  65000      -9
## 4091  18000      -9
## 4092  23000      -9
## 4093     -8      -8
## 4094     -9      -8
## 4095     -9      -9
## 4096     -9      -9
## 4097  46000   34000
## 4098  18000       0
## 4099     -9      -9
## 4100     -9      -9
## 4101     -8      -8
## 4102  29000      -8
## 4103  22500   50000
## 4104  17000   25000
## 4105     -9      -9
## 4106  41800  223000
## 4107     -8      -8
## 4108  35000      -8
## 4109   7200      -8
## 4110     -9      -9
## 4111     -8      -8
## 4112     -8      -8
## 4113     -9      -9
## 4114  18000      -8
## 4115     -9      -9
## 4116     -8   29200
## 4117     -9      -9
```

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## 4118   6000      -8
## 4119  89000  211000
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## 4121     -8      -8
## 4122     -8  357000
## 4123  39100   36700
## 4124     -9      -9
## 4125     -9      -9
## 4126  54000      -8
## 4127     -9      -9
## 4128  68000      -9
## 4129   6900   34000
## 4130  44000  782000
## 4131  13000      -8
## 4132     -9      -9
## 4133     -8      -8
## 4134     -9      -9
## 4135     -8   10400
## 4136     -9      -9
## 4137     -9      -9
## 4138     -9      -9
## 4139  31800       0
## 4140     -9      -9
## 4141     -9      -9
## 4142  14700  253000
## 4143  22200      -8
## 4144  12800   67000
## 4145     -8      -8
## 4146     -9      -9
## 4147     -9      -9
## 4148 100000  100000
## 4149     -9      -9
## 4150     -8      -8
## 4151     -9      -9
## 4152  15000      -8
## 4153     -9      -9
## 4154     -9      -9
## 4155     -9      -9
## 4156     -9      -8
## 4157  60000 1400000
## 4158  16600  155000
## 4159     -9      -9
## 4160     -9      -9
## 4161     -8      -8
## 4162     -9      -9
## 4163     -9      -9
## 4164     -9      -9
## 4165     -8      -8
## 4166  56000  395000
## 4167  16900  483000
## 4168     -8      -8
## 4169     -9  166000
## 4170     -9      -9
## 4171  27000  485000
```

```
## 4172    -9      -9
## 4173 19600  625000
## 4174 64000  350000
## 4175 69000  350000
## 4176    -9      -9
## 4177    -9      -9
## 4178    -9      -9
## 4179    -9      -9
## 4180    -8      -8
## 4181 52000   64000
## 4182 84000      -8
## 4183  3100   13800
## 4184 26400      -8
## 4185 22500      -8
## 4186    -9  200000
## 4187 48600  147000
## 4188    -9      -9
## 4189    -9      -9
## 4190    -9      -9
## 4191 52000      -9
## 4192    -9      -9
## 4193    -9      -9
## 4194 22000      -9
## 4195    -9      -9
## 4196    -9      -9
## 4197    -9      -9
## 4198    -9      -9
## 4199    -9      -9
## 4200 41900  165000
## 4201 99000  940000
## 4202    -8      -8
## 4203 20000   80000
## 4204    -8      -8
## 4205    -8      -8
## 4206 17100   82000
## 4207    -9      -9
## 4208    -9      -9
## 4209    -9      -9
## 4210 45000  300000
## 4211 57000 2548054
## 4212    -8      -8
## 4213    -9      -9
## 4214 40000      -8
## 4215 44400       0
## 4216    -8      -8
## 4217    -8      -8
## 4218 45000      -8
## 4219 25300       0
## 4220    -9      -9
## 4221 60000      -9
## 4222    -9      -9
## 4223    -9      -9
## 4224    -9      -9
## 4225    -9      -9
```

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## 4226     -9      -9
## 4227     -9      -9
## 4228     -9      -9
## 4229     -9      -9
## 4230     -9      -9
## 4231     -9      -9
## 4232  30000      -8
## 4233     -8      -8
## 4234     -9      -9
## 4235     -8      -8
## 4236     -9      -9
## 4237 259046      -9
## 4238     -9      -9
## 4239  80000      -9
## 4240     -9      -9
## 4241  49800  930000
## 4242     -8      -8
## 4243     -8      -8
## 4244  66000      -9
## 4245  48400      -8
## 4246     -9      -9
## 4247     -9      -9
## 4248     -9      -9
## 4249  53000       0
## 4250     -9      -9
## 4251     -8  110000
## 4252     -9      -9
## 4253     -9      -9
## 4254     -9      -9
## 4255     -9      -9
## 4256  10000      -8
## 4257     -8      -8
## 4258 154000      -9
## 4259 106000      -9
## 4260     -9      -9
## 4261     -8      -8
## 4262  71000  190000
## 4263     -9      -9
## 4264     -9      -9
## 4265  33700      -9
## 4266  52000   94000
## 4267     -9      -9
## 4268     -9      -9
## 4269     -8      -8
## 4270     -9      -9
## 4271     -9      -9
## 4272     -8  172000
## 4273     -9      -9
## 4274     -9      -9
## 4275     -8  109000
## 4276     -9      -9
## 4277  36000  107000
## 4278     -8      -8
## 4279     -9      -9
```

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## 4280  59000      -8
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## 4282     -9      -9
## 4283     -9      -9
## 4284     -9      -9
## 4285     -9      -9
## 4286     -9      -9
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## 4288     -9      -9
## 4289     -8      -8
## 4290     -9      -9
## 4291     -8 1660000
## 4292     -9      -9
## 4293  74000      -8
## 4294     -9      -9
## 4295  29800  482000
## 4296  75000 1940000
## 4297     -9      -9
## 4298  48300      -8
## 4299  43300   19200
## 4300     -8      -8
## 4301  53000      -8
## 4302  70000  130000
## 4303     -8      -8
## 4304     -9      -9
## 4305     -9      -9
## 4306     -9      -9
## 4307  46200   49300
## 4308     -9      -9
## 4309  10000      -8
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## 4311     -9      -9
## 4312  67000  162000
## 4313  61000      -9
## 4314     -9      -9
## 4315     -8      -8
## 4316  67000      -8
## 4317     -9      -9
## 4318     -9      -9
## 4319     -9      -9
## 4320     -8      -8
## 4321     -8      -8
## 4322     -9      -9
## 4323     -9      -9
## 4324  58000   75000
## 4325  46700  349000
## 4326     -9      -9
## 4327     -8      -8
## 4328     -9      -9
## 4329     -9      -9
## 4330  41600   81000
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## 4332     -9      -9
## 4333     -9      -9
```

```
## 4334  50000  200000
## 4335     -9      -9
## 4336     -9      -9
## 4337     -9      -9
## 4338     -9      -9
## 4339  34600   50000
## 4340  39600  250000
## 4341  45000      -8
## 4342  54000  253000
## 4343  25200      -8
## 4344     -9      -9
## 4345 103000   38900
## 4346   4800    2700
## 4347     -9      -9
## 4348     -9      -9
## 4349 115000  211000
## 4350  98000  140000
## 4351  54000  156000
## 4352     -9      -9
## 4353     -9      -9
## 4354     -9      -9
## 4355     -9      -9
## 4356  71000  406000
## 4357     -9      -9
## 4358     -9      -9
## 4359     -9      -9
## 4360  66000  142000
## 4361  15000      -9
## 4362     -9      -9
## 4363     -9      -8
## 4364  11400       0
## 4365     -9      -9
## 4366  52000   43900
## 4367 120000      -8
## 4368  78000  247000
## 4369     -9      -9
## 4370     -9      -9
## 4371     -9      -9
## 4372     -9      -9
## 4373     -9      -9
## 4374     -9      -9
## 4375     -9      -9
## 4376     -9      -9
## 4377      0       0
## 4378     -9      -9
## 4379     -8      -8
## 4380     -9      -9
## 4381  26900      -8
## 4382 259046      -8
## 4383     -9      -9
## 4384     -9      -9
## 4385     -8      -8
## 4386     -9      -9
## 4387     -9      -9
```

```
## 4388  20600      -8
## 4389     -9 2548054
## 4390     -9      -9
## 4391     -9      -9
## 4392     -9      -9
## 4393     -9      -9
## 4394     -9      -9
## 4395     -9      -9
## 4396 121000  196000
## 4397     -9      -9
## 4398     -8      -8
## 4399 100000      -8
## 4400     -9      -9
## 4401  34500      -8
## 4402  42700      -9
## 4403     -8      -8
## 4404     -9      -9
## 4405     -9      -9
## 4406     -9      -9
## 4407     -8      -8
## 4408 163000  401000
## 4409     -8      -8
## 4410     -9      -9
## 4411     -9      -9
## 4412  12200      -8
## 4413     -9      -9
## 4414  36000  182000
## 4415     -9      -9
## 4416  54000      -9
## 4417     -9      -9
## 4418     -9      -9
## 4419  49200 1450000
## 4420 259046 2548054
## 4421     -9      -9
## 4422     -9      -9
## 4423  75000      -8
## 4424     -8  207000
## 4425     -8      -8
```

```
# A component of OPEX_R coded -8 contributes zero when OPEX_R is observed.
rhfs <- rhfs |>
  mutate(
    across(
      all_of(c("OPPAY", "OPINSUR", "OPELGAS", "OPWTRSWR", "OPREP", "OPMNG",
               "OPPROTH", "OPSECUR", "OPLAND", "OPOTHEX")),
      \(x) if_else(OPEX_R >= 0 & x == -8, 0L, x)
    )
  )


rhfs_before <- rhfs

# Apply only to explicit sentinel cells. This also preserves unclassified -8/-9
# levels in factors when only SOME rows have a known skip reason.
```

```
apply_clean_rule <- function(rule, variable, guard, replacement,
                             expected = c("-8", "-9")) {
  if (!variable %in% names(rhfs)) stop(rule, ": missing column ", variable)
  if (length(guard) != nrow(rhfs)) stop(rule, ": invalid guard length")
  rows <- which(guard %in% TRUE)
  if (!length(rows)) return(invisible(0L))
  old_column <- rhfs[[variable]]
  old <- as.character(old_column[rows])
  if (anyNA(old) || any(!old %in% expected)) {
    bad <- rows[is.na(old) | !old %in% expected]
    stop(rule, ": non-target cell(s) in ", variable, "; CONTROLPUF: ",
         paste(head(rhfs$CONTROLPUF[bad], 20), collapse = ", "))
  }
  if (length(replacement) == 1L) replacement <- rep(replacement, length(rows))
  if (length(replacement) != length(rows) || anyNA(replacement) ||
      any(as.character(replacement) %in% c("-8", "-9"))) {
    stop(rule, ": invalid replacement for ", variable)
  }
  new_column <- old_column
  if (is.factor(new_column)) {
    new_column <- forcats::fct_expand(new_column, unique(as.character(replacement)))
    new_column[rows] <- as.character(replacement)
  } else {
    if (!is.numeric(new_column) || !is.numeric(replacement)) {
      stop(rule, ": replacement must preserve the numeric type of ", variable)
    }
    new_column[rows] <- replacement
  }
  if (!identical(as.character(new_column[rows]), as.character(replacement))) {
    stop(rule, ": assignment failed for ", variable)
  }
  rhfs[[variable]] <<- new_column
  invisible(length(rows))
}

# Structural labels describe the skip, NOT a presumed 'No' or a zero response.
# The factor types and their existing response levels are left intact.
apply_clean_rule("S01_PROPCON", "PROPCON",
  as.character(rhfs$PROPCON) == "-8" & as.character(rhfs$PROPTOWN) == "1",
  "Townhome or row house; condominium item skipped", "-8")
apply_clean_rule("S02_COMPLEX", "COMPLEX",
  as.character(rhfs$COMPLEX) == "-8" & as.character(rhfs$PROPCON) == "1",
  "Condominium; complex item skipped", "-8")
apply_clean_rule("S03_YRNEWBLG_R", "YRNEWBLG_R",
  as.character(rhfs$YRNEWBLG_R) == "-8" & as.character(rhfs$NUMBLD_R) == "1",
  "Only one residential building", "-8")
apply_clean_rule("S04_LEASLNG", "LEASLNG",
  as.character(rhfs$LEASLNG) == "-8" & rhfs$NUMUNITS_R > 1,
  "Multi-unit property; single-unit lease item skipped", "-8")
apply_clean_rule("S05_LNGTRMPRV_MULTI", "LNGTRMPRV",
  as.character(rhfs$LNGTRMPRV) == "-8" & rhfs$NUMUNITS_R > 1,
  "Multi-unit property; prior lease item skipped", "-8")
apply_clean_rule("S06_LNGTRMPRV_LONG", "LNGTRMPRV",
```

```
  as.character(rhfs$LNGTRMPRV) == "-8" & rhfs$NUMUNITS_R == 1 &
    as.character(rhfs$LEASLNG) == "1",
  "Long-term lease; prior long-term item skipped", "-8")
apply_clean_rule("S07_WEEKLNG", "WEEKLNG",
  as.character(rhfs$WEEKLNG) == "-8" & rhfs$NUMUNITS_R == 1,
  "Single-unit route; weekly availability item skipped", "-8")
apply_clean_rule("S08_RNTAPPFEEWAIV", "RNTAPPFEEWAIV",
  as.character(rhfs$RNTAPPFEEWAIV) == "-8" &
    as.character(rhfs$NORNTAPPFEE) == "1" &
    as.character(rhfs$RNTAPPFEE_R) == "1",
  "No application fee to waive", "-8")
apply_clean_rule("S09_NORNTAPPFEE", "NORNTAPPFEE",
  as.character(rhfs$NORNTAPPFEE) == "-9" &
    as.character(rhfs$RNTAPPFEE_R) %in% as.character(2:6),
  "Application fee amount reported", "-9")
apply_clean_rule("S10_YRPROPREHAB_DEV", "YRPROPREHAB_R",
  as.character(rhfs$YRPROPREHAB_R) == "-8" & as.character(rhfs$HOWACQ) == "1",
  "Developed new residential buildings", "-8")
apply_clean_rule("S11_YRPROPREHAB_BUY", "YRPROPREHAB_R",
  as.character(rhfs$YRPROPREHAB_R) == "-8" & as.character(rhfs$HOWACQ) == "3",
  "Purchased without substantial rehabilitation", "-8")

# Q9 and Q10 use a count for multi-unit properties and a yes/no answer for
# single-unit properties. The handful of multi-unit yes/no answers already
# present in the PUF are preserved; no yes/no value is invented for a skip.
apply_clean_rule("S12_RCONTROLAHS", "RCONTROLAHS",
  as.character(rhfs$RCONTROLAHS) == "-9" & rhfs$NUMUNITS_R > 1,
  "Multi-unit route; rent-control yes/no not recorded", "-9")
apply_clean_rule("S13_OFFSTPARKAHS", "OFFSTPARKAHS",
  as.character(rhfs$OFFSTPARKAHS) == "-8" & rhfs$NUMUNITS_R > 1,
  "Multi-unit route; parking yes/no not recorded", "-8")

# The program-benefit questions were omitted on the single-unit survey route.
# These labels do not assert that the property had no program benefits.
for (variable in c("GOVBELOW", "RSUBSIDY", "ELDERLY", "GOVGRANT",
                   "BENEOTH", "STLOWINC")) {
  apply_clean_rule(paste0("S14_", variable), variable,
    as.character(rhfs[[variable]]) == "-8" & rhfs$NUMUNITS_R == 1,
    "Single-unit property; program item skipped", "-8")
}

# A recorded debt makes the 'why no mortgage' question inapplicable.
apply_clean_rule("S15_WHYNOMRT", "WHYNOMRT",
  as.character(rhfs$WHYNOMRT) == "-8" & as.character(rhfs$DEBT) == "1",
  "Mortgage or similar debt present", "-8")

# For no-debt properties, first-debt questions have no defined answer. These
# factor labels keep the difference between 'no debt' and unreported DEBT.
no_debt_factors <- c(
  "DEBT1", "YRMORT_R", "MORTAMT_R", "OUTBAL_R", "GREENMORT",
  "MORTGET", "LOCTYP", "REGPAY", "PMTFREQ", "RPIPRIN",
  "RPIPRPINS", "RPIPMI", "RPIPRPTAX"
)
```

```
for (variable in no_debt_factors) {
  apply_clean_rule(paste0("S16_", variable), variable,
    as.character(rhfs[[variable]]) == "-8" & as.character(rhfs$DEBT) == "2",
    "No mortgage or similar debt", "-8")
}
apply_clean_rule("S17_LOCTYP", "LOCTYP",
  as.character(rhfs$LOCTYP) == "-8" &
    as.character(rhfs$DEBT1) %in% c("1", "2", "4"),
  "First debt not a home equity line or loan", "-8")
for (variable in c("PMTFREQ", "RPIPRIN", "RPIPRPINS", "RPIPMI", "RPIPRPTAX")) {
  apply_clean_rule(paste0("S18_", variable), variable,
    as.character(rhfs[[variable]]) == "-8" & as.character(rhfs$REGPAY) == "2",
    "No regularly required payment", "-8")
}

# A website question routes to properties with at least ten units. The other
# advertising methods route past a respondent who said no advertising.
apply_clean_rule("S19_ADWEBSITE_SMALL", "ADWEBSITE",
  as.character(rhfs$ADWEBSITE) == "-8" & rhfs$NUMUNITS_R < 10,
  "Fewer than ten units; website item skipped", "-8")
apply_clean_rule("S20_ADWEBSITE_NOAD", "ADWEBSITE",
  as.character(rhfs$ADWEBSITE) == "-8" & rhfs$NUMUNITS_R >= 10 &
    as.character(rhfs$ADVERTISE) == "2",
  "Did not advertise; website item skipped", "-8")
for (variable in c("ADONLINE", "ADOTHONLINE", "ADNEWSPAPER",
                   "ADAGENCY", "ADAGENT")) {
  apply_clean_rule(paste0("S21_", variable), variable,
    as.character(rhfs[[variable]]) == "-8" & as.character(rhfs$ADVERTISE) == "2",
    "Did not advertise; method item skipped", "-8")
}
apply_clean_rule("S22_MLS_NOAD", "MLS",
  as.character(rhfs$MLS) == "-8" & as.character(rhfs$ADVERTISE) == "2",
  "Did not advertise; MLS item skipped", "-8")
apply_clean_rule("S23_MLS_NOAGENT", "MLS",
  as.character(rhfs$MLS) == "-8" & as.character(rhfs$ADAGENT) == "2",
  "No real estate agent used", "-8")

# The parking yes/no question was answered 'No' for these single-unit homes.
apply_clean_rule("N01_OFFSTPARK", "OFFSTPARK",
  as.character(rhfs$OFFSTPARK) == "-9" & rhfs$NUMUNITS_R == 1 &
    as.character(rhfs$OFFSTPARKAHS) == "2", 0, "-9")

# All residential units in these seven rows are already accounted for without
# any vacant-for-rent units. The independent rental total also equals rented.
occupancy_components <- c("TRENOC_R", "TOWNOC_R", "TEMPOC_R", "TOTHVA_R")
occupancy_complete <- rowSums(sapply(rhfs[occupancy_components],
                                     function(x) !is.na(x) & x >= 0)) == 4L
occupancy_sum <- rowSums(rhfs[occupancy_components])
vacant_zero_guard <- occupancy_complete & rhfs$NUMUNITS_R >= 0 &
  rhfs$NUMUNITS_R == occupancy_sum & rhfs$TOTRENTVAC >= 0 &
  rhfs$TOTRENTVAC == rhfs$TRENOC_R
vacant_observed <- which(vacant_zero_guard & rhfs$TRENVA_R >= 0)
if (length(vacant_observed) &&
```

```
    mean(rhfs$TRENVA_R[vacant_observed] == 0) < 0.99) {
  bad <- vacant_observed[rhfs$TRENVA_R[vacant_observed] != 0]
  stop("N02_TRENVA_R: observed-case accuracy under 99%; CONTROLPUF: ",
       paste(head(rhfs$CONTROLPUF[bad], 20), collapse = ", "))
}
apply_clean_rule("N02_TRENVA_R", "TRENVA_R",
  as.character(rhfs$TRENVA_R) == "-9" & vacant_zero_guard, 0, "-9")

# Q17 calls each CP* cell marked -8 'Not Applicable' and defines the reported
# total as a sum. Validate that identity on complete observed totals first.
capital_components <- c(
  "CPHEATAMT", "CPRFAMT", "CPWINAMT", "CPDOORAMT", "CPFLRAMT",
  "CPEXTAMT", "CPPLBAMT", "CPELEAMT", "CPKITAMT", "CPBATAMT",
  "CPACCAMT", "CPOTHAMT"
)
capital_matrix <- as.matrix(rhfs[capital_components])
capital_complete <- rowSums(!is.na(capital_matrix) &
  (capital_matrix == -8 | capital_matrix >= 0)) == length(capital_components)
capital_observed <- which(capital_complete & rhfs$CAPEX_R >= 0)
capital_sum <- rowSums(pmax(capital_matrix, 0))
if (length(capital_observed) &&
    mean(abs(rhfs$CAPEX_R[capital_observed] -
             capital_sum[capital_observed]) <= 0.5) < 0.99) {
  bad <- capital_observed[abs(rhfs$CAPEX_R[capital_observed] -
                              capital_sum[capital_observed]) > 0.5]
  stop("N03_CAPEX_R: observed-case sum accuracy under 99%; CONTROLPUF: ",
       paste(head(rhfs$CONTROLPUF[bad], 20), collapse = ", "))
}

# A respondent able to report 2023 finances who explicitly supplied at least
# one zero component, and no positive or unreported component, has a zero sum.
# All--8 rows are excluded: their entire capital question may have been skipped.
capital_zero_guard <- as.character(rhfs$PROPANS) == "1" &
  rhfs$NUMUNITS_R > 0 & rhfs$CAPEX_R == -9 &
  rhfs$CAPEXPU_R == -9 & capital_complete &
  rowSums(capital_matrix == 0) >= 1L & capital_sum == 0
apply_clean_rule("N03_CAPEX_R", "CAPEX_R", capital_zero_guard, 0, "-9")
# Only the guarded -9 cells may change. Rebuild this column from the snapshot
# so every published, observed CAPEX_R total stays exactly as it was.
rhfs$CAPEX_R <- replace(rhfs_before$CAPEX_R,
                        which(capital_zero_guard %in% TRUE), 0)
apply_clean_rule("N04_CAPEXPU_R", "CAPEXPU_R", capital_zero_guard, 0, "-9")

# Once a positive or explicitly reconstructed zero total establishes that the
# capital question was answered, an item marked Not Applicable costs zero.
for (variable in capital_components) {
  apply_clean_rule(paste0("N05_", variable), variable,
    rhfs[[variable]] == -8 & rhfs$CAPEX_R >= 0, 0, "-8")
}

# The existing cleaning made COMMRENT and OPEX_R explicitly zero on one row
# whose published cap rate is still -9. Validate the four-input formula using
# observed rates, including legitimate negative rates. Ratios have 4 decimals.
```

```r
caprate_inputs <- rhfs$TOTCOLL >= 0 & rhfs$COMMRENT >= 0 &
  rhfs$OPEX_R >= 0 & rhfs$MRKTVAL_R > 0
caprate_value <- (rhfs$TOTCOLL + rhfs$COMMRENT - rhfs$OPEX_R) /
  rhfs$MRKTVAL_R
caprate_observed <- which(caprate_inputs &
  !as.character(rhfs$CAPRATE_R) %in% c("-8", "-9"))
caprate_error <- abs(rhfs$CAPRATE_R[caprate_observed] -
                       caprate_value[caprate_observed])
if (length(caprate_observed) && mean(caprate_error <= 0.000051) < 0.99) {
  bad <- caprate_observed[caprate_error > 0.000051]
  stop("N06_CAPRATE_R: observed-case accuracy under 99%; CONTROLPUF: ",
       paste(head(rhfs$CONTROLPUF[bad], 20), collapse = ", "))
}
caprate_rows <- which(caprate_inputs &
  as.character(rhfs$CAPRATE_R) %in% c("-8", "-9"))
apply_clean_rule("N06_CAPRATE_R", "CAPRATE_R",
  seq_len(nrow(rhfs)) %in% caprate_rows,
  round(caprate_value[caprate_rows], 4))


write.csv(rhfs, "RHFS-2024-Cleaned.csv", row.names = FALSE)
```

```r
# Brings back all of the columns that were dropped for convenience at the
# beginning of Cleaning the Dataset
rhfs <- rhfs |>
  left_join(
    rhfs.original |> select(-any_of(names(rhfs)), CONTROLPUF),
    join_by(CONTROLPUF)
  )

# Loads the data frame with srvyr
df <- rhfs |>
  select(-REPWGT0) |> # REPWGT0 is equivalent to WEIGHT
  as_survey_rep(
    repweights = starts_with("REPWGT"),
    weights = WEIGHT,
    type = "successive-difference",
    combined_weights = TRUE,
    # Decided to set to T to follow SDR as outlined by Ash (to match what was
    # done in Census's Methodology)
    mse = TRUE
  )

# Sanity check for if the data was successfully loaded with srvyr
# using Table 5 of Census's RHFS Methodology
df |>
  summarize(
    properties = survey_total(vartype = "se") # Applies weights and calculates se of properties
  ) |>
  mutate( # Dividing by 1000 to match Table 5's scale
    properties = properties / 1000,
    properties_se = properties_se / 1000,
    moe = properties_se * 1.645 # 90% margin of error
```

```
  )
```

```
## # A tibble: 1 x 3
##   properties properties_se   moe
##        <dbl>         <dbl> <dbl>
## 1     18965.          165.  271.
```

```
missing_by_var <- data.frame(
  variable = names(rhfs),
  percent_missing = sapply(
    rhfs,
    \(x) mean(x %in% c(-9))
  )
) |>
  arrange(desc(percent_missing))


# Evidence for MAR (missing at random) in MRKTVAL_R
df |>
  mutate(
    mrktNA = MRKTVAL_R == -9
  ) |>
  summarize(
    survey_mean(mrktNA),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 3
##   NUMCAT_R      coef  `_se`
##   <fct>        <dbl>  <dbl>
## 1 1 Unit      0.0923 0.0263
## 2 2-4 Units   0.0716 0.0224
## 3 5-24 Units  0.154  0.0224
## 4 25-49 Units 0.335  0.0547
## 5 50+ Units   0.418  0.0163
```

```
df |>
  filter(MRKTVAL_R != -9) |>
  summarize(
    mean_mrktval = survey_mean(MRKTVAL_R),
    median_mrktval = survey_median(MRKTVAL_R),
    .by = NUMCAT_R
  )
```

```
## # A tibble: 5 x 5
##   NUMCAT_R    mean_mrktval mean_mrktval_se median_mrktval median_mrktval_se
##   <fct>              <dbl>           <dbl>          <dbl>             <dbl>
## 1 1 Unit           397050.          42218.         250000            33154.
## 2 2-4 Units        569582.          33304.         425000            50491.
## 3 5-24 Units      2178309.         118540.        2000000           469164.
## 4 25-49 Units     5000928.         333704.        4000000           224403.
## 5 50+ Units      33895580.        2865340.       20000000          4987831.
```

```r
# For citing R, RStudio, and the packages that were used in this analysis
cite_packages(
  pkgs = "All",
  omit = NULL,
  citation.style = "apa",
  include.RStudio = TRUE,
  cite.tidyverse = FALSE,
  out.format = "docx",
  out.dir = getwd()
)
```

```
##
## Citation report available at C:/Users/bfeler/Downloads/Theophilus College/Senior Thesis/Senior-Thesis
```

```
## [1] "C:/Users/bfeler/Downloads/Theophilus College/Senior Thesis/Senior-Thesis-WIP/grateful-report.doc
```